

# **RiskPulse: Enterprise AI Workforce Risk Intelligence Platform**

Exhaustive FAANG Technical Interview Handbook

Target Roles: Google L4/L5, Meta E5, OpenAI Applied AI, System Design

*Generated on July 16, 2026*

-#- R-i-s-k-P-u-l-s-e-:- F-A-A-N-G- I-n-t-e-r-v-i-e-w- P-r-e-p-a-r-a-t-i-o-n- H-a-n-d-b-o-o-k-

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-##- 1.- P-r-o-j-e-c-t- O-v-e-r-v-i-e-w- &- R-e-q-u-e-s-t- F-l-o-w-

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-R-i-s-k-P-u-l-s-e- -i-s- -a-n- -A-l---d-r-i-v-e-n- p-r-e-d-i-c-t-i-v-e- -i-n-t-e-l-l-i-g-e-n-c-e- -p-l-a-t-f-o-r-m- -f-o-r- -w-o-r-k-f-o-r-c-e- -r-i-s-k-  
-m-o-n-i-t-o-r-i-n-g-.- -l-t- -i-n-g-e-s-t-s- -h-i-s-t-o-r-i-c-a-l- -c-o-m-p-a-n-y- -l-a-y-o-f-f-s-,- -f-u-n-d-i-n-g- -s-t-a-g-e-s-,- -i-n-d-u-s-t-r-y-  
-g-r-o-u-p-s-,- -a-n-d- -n-e-w-s- -s-e-n-t-i-m-e-n-t-,- -p-i-p-e-l-i-n-e-s- -t-h-i-s- -d-a-t-a- -t-h-r-o-u-g-h- -c-a-l-i-b-r-a-t-e-d- -M-L-  
-c-l-a-s-s-i-f-i-e-r-s-,- -a-n-d- -s-e-r-v-e-s- -p-r-e-d-i-c-t-i-o-n-s- -v-i-a- -a-n- -i-n-t-e-r-a-c-t-i-v-e- -S-t-r-e-a-m-l-i-t- -U-l-.-

-

-###- I-n-f-e-r-r-e-d- A-r-c-h-i-t-e-c-t-u-r-e- &- F-l-o-w-

-`-`-`-

-[U-s-e-r- B-r-o-w-s-e-r-]-

- - - - -|- -(H-T-T-P-S)-

- - - - -v-

-[S-t-r-e-a-m-l-i-t- S-e-r-v-e-r- -(m-a-i-n-.p-y)-]-

- - - - -|-

- - - - -|->- [-D-a-t-a-b-a-s-e- H-e-l-p-e-r- -(d-b-.p-y)-]- ----->- [-P-o-s-t-g-r-e-S-Q-L- D-a-t-a-b-a-s-e-  
-(-R-l-s-k-P-u-l-s-e\_-D-B)-]-

- - - - -|-

- - - - -|->- [-N-e-w-s- S-e-n-t-i-m-e-n-t- H-e-l-p-e-r- -(h-e-l-p-e-r-s-.p-y)-]-

- - - - -|- - - - - -|-

- - - - -|- - - - - ->- [-N-e-w-s- A-P-l- -/- C-a-c-h-e-d- N-e-w-s- A-r-t-i-c-l-e-s]-

- - - - -|-

- - - - ->- [-A-l- F-o-r-e-c-a-s-t- C-o-m-p-o-n-e-n-t- -(a-i\_-p-r-e-d-i-c-t-i-o-n-.p-y)-]-

- - - - - - - - - - -|-

- - - - - - - - - - ->- [-P-r-e-d-i-c-t-i-o-n- S-e-r-v-i-c-e- -(p-r-e-d-i-c-t-\_s-e-r-v-i-c-e-.p-y)-]-

- - - - - - - - - - -|-

- - - - - - - - - - -|->- [-L-o-a-d- C-a-l-i-b-r-a-t-e-d- S-t-a-c-k-i-n-g- -M-o-d-e-l- -(.-p-k-l)-]-

- - - - - - - - - - -|-

- - - - - - - - - - ->- [-S-H-A-P- E-x-p-l-a-i-n-e-r- -(s-h-a-p\_-e-x-p-l-a-i-n-e-r-.p-k-l)-]- ----->- [-L-o-c-a-l-  
-S-H-A-P- -W-a-t-e-r-f-a-l-l- -P-N-G-]-

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-###- M-L- P-i-p-e-l-i-n-e- D-a-t-a- F-l-o-w-

-`-`-`-

-[P-o-s-t-g-r-e-S-Q-L- D-a-t-a-b-a-s-e-]-

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- - - - -|
- - - - -v- -(Q-u-e-r-y- -l-n-g-e-s-t-i-o-n)-

-[P-r-e-p-r-o-c-e-s-s-i-n-g-      &-      -F-e-a-t-u-r-e-      -E-n-g-i-n-e-e-r-i-n-g-      -(p-r-e-p-r-o-c-e-s-s-.p-y-      -/-
-f-e-a-t-u-r-e_-e-n-g-i-n-e-e-r-i-n-g-.p-y)-]

- - - - -|
- - - - -|--->- -l-m-p-u-t-e- &- -E-n-c-o-d-e- -F-e-a-t-u-r-e-s-
- - - - -|
- - - - -v- -(M-a-t-r-i-x- -S-p-l-i-t-s)-

-[B-a-y-e-s-i-a-n- -H-P-O- &- -O-p-t-i-m-i-z-a-t-i-o-n- -(h-p-o-.p-y)-]

- - - - -|
- - - - -|--->- -O-p-t-u-n-a- -T-r-i-a-l-s- -(X-G-B-o-o-s-t- -/- -L-i-g-h-t-G-B-M)-
- - - - -|
- - - - -v- -(V-a-l-i-d-a-t-i-o-n)-

-[E-n-s-e-m-b-l-e-      -B-e-n-c-h-m-a-r-k-i-n-g-      &-      -P-r-o-b-a-b-i-l-i-t-y-      -C-a-l-i-b-r-a-t-i-o-n-      -(b-e-n-c-h-m-a-r-k-.p-y-      -/-
-c-a-l-i-b-r-a-t-i-o-n-.p-y)-]

- - - - -|
- - - - -|--->- -P-l-a-t-t- -S-c-a-l-i-n-g- -/- -l-s-o-t-o-n-i-c- -C-a-l-i-b-r-a-t-i-o-n-
- - - - -|
- - - - -v-

-[O-p-t-i-m-i-z-e-d- -T-h-r-e-s-h-o-l-d- -D-e-c-i-s-i-o-n- -(v-a-l-i-d-a-t-i-o-n-.p-y)-]

- - - - -|
- - - - -|--->- -O-p-t-i-m-i-z-e-d- -T-h-r-e-s-h-o-l-d- -J-S-O-N- -(0-.2-0- -t-h-r-e-s-h-o-l-d- -f-o-r- -F-1-/-R-e-c-a-l-l)-
- - - - -|
- - - - -v-

-[l-n-f-e-r-e-n-c-e- -R-e-g-i-s-t-r-y- -(.-.p-k-l- -M-o-d-e-l-s- +- -S-H-A-P- -E-x-p-l-a-i-n-e-r)-]
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-##-#- -Q-u-e-s-t-i-o-n- -1:- -C-a-n- -y-o-u- -w-a-l-k- -m-e- -t-h-r-o-u-g-h- -t-h-e- -h-i-g-h---l-e-v-e-l- -a-r-c-h-i-t-e-c-t-u-r-e- -o-f-
-R-i-s-k-P-u-l-s-e?-

-*-*-*R-o-u-n-d:-*-*-* -S-y-s-t-e-m- -D-e-s-i-g-n-

-*-*-*D-i-f-f-i-c-u-l-t-y:-*-*-* -E-a-s-y- -| -*-*-*l-m-p-o-r-t-a-n-c-e:-*-*-* -8-/1-0-

-*-*-*C-o-n-c-e-p-t- -T-e-s-t-e-d:-*-*-* -S-y-s-t-e-m- -t-o-p-o-l-o-g-y,- -c-o-m-m-u-n-i-c-a-t-i-o-n- -p-r-o-t-o-c-o-l-s,-
-s-y-n-c-h-r-o-n-o-u-s- -v-s- -a-s-y-n-c-h-r-o-n-o-u-s- -o-p-e-r-a-t-i-o-n-s,-

-*-*-*W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-*-*-* -T-e-s-t-s- -s-y-s-t-e-m- -a-r-c-h-i-t-e-c-t-u-r-e- -l-a-y-o-u-t- -a-n-d- -f-l-o-w.-

-*-*-*I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-*-*-* -R-i-s-k-P-u-l-s-e- -u-s-e-s- -a- -t-i-e-r-e-d- -a-r-c-h-i-t-e-c-t-u-r-e.- -T-h-e-
-p-r-e-s-e-n-t-a-t-i-o-n- -l-a-y-e-r- -i-s- -a- -S-t-r-e-a-m-l-i-t- -U-l,- -c-a-l-l-i-n-g- -a- -P-y-t-h-o-n---b-a-s-e-d- -b-a-c-k-e-n-d.- -T-h-e-

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-b-u-s-i-n-e-s-s- -l-o-g-i-c- -q-u-e-r-i-e-s- -a- -P-o-s-t-g-r-e-S-Q-L- -r-e-l-a-t-i-o-n-a-l- -d-a-t-a-b-a-s-e- -f-o-r- -m-e-t-r-i-c-s-.- -F-o-r- -M-L-  
-p-r-e-d-i-c-t-i-o-n-s-,- -t-h-e- -a-p-p- -i-n-v-o-k-e-s- -a-n- -o-f-f-l-i-n-e---t-r-a-i-n-e-d- -P-r-e-d-i-c-t-i-o-n- -S-e-r-v-i-c-e- -t-h-a-t- -l-o-a-d-s-  
-a- -p-i-c-k-l-e-d- -S-t-a-c-k-i-n-g/-X-G-B-o-o-s-t- -e-n-s-e-m-b-l-e- -a-n-d- -c-o-m-p-u-t-e-s- -l-o-c-a-l- -S-H-A-P- -w-a-t-e-r-f-a-l-l-s-  
-d-y-n-a-m-i-c-a-l-l-y-.- -l-f- -n-e-w-s- -i-s- -r-e-q-u-e-s-t-e-d-,- -t-h-e- -s-y-s-t-e-m- -r-e-a-d-s- -c-a-c-h-e-d- -n-e-w-s- -a-r-t-i-c-l-e-s-  
-w-i-t-h- -p-r-e---c-o-m-p-u-t-e-d- -s-e-n-t-i-m-e-n-t- -s-c-o-r-e-s- -f-r-o-m- -t-h-e- -d-a-t-a-b-a-s-e-.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-:-\*-\* -D-e-s-c-r-i-b-i-n-g- -i-t- -a-s- -a- -m-o-n-o-l-i-t-h-i-c- -s-c-r-i-p-t- -w-i-t-h-o-u-t-  
-s-e-p-a-r-a-t-i-n-g- -t-h-e- -d-a-t-a- -p-i-p-e-l-i-n-e-,- -t-r-a-i-n-i-n-g-,- -a-n-d- -i-n-f-e-r-e-n-c-e- -l-i-f-e-c-y-c-l-e-s-.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-:-\*-\* -T-h-e- -a-r-c-h-i-t-e-c-t-u-r-e- -i-s- -s-p-l-i-t- -i-n-t-o- -a-  
-T-r-a-i-n-i-n-g- -P-i-p-e-l-i-n-e- -(o-f-f-l-i-n-e)-,- -a- -n-e-w-s- -f-e-t-c-h-i-n-g- -b-a-c-k-g-r-o-u-n-d- -w-o-r-k-e-r-  
-(a-s-y-n-c-h-r-o-n-o-u-s- -s-c-h-e-d-u-l-e-r)-,- -a-n-d- -a-n- -O-n-l-i-n-e- -l-n-f-e-r-e-n-c-e- -S-e-r-v-i-c-e-.- -T-h-e- -d-a-t-a-b-a-s-e-  
-a-c-t-s- -a-s- -t-h-e- -s-i-n-g-l-e- -s-o-u-r-c-e- -o-f- -t-r-u-t-h-,- -s-t-o-r-i-n-g- -c-o-m-p-a-n-y- -r-e-c-o-r-d-s-,- -l-a-y-o-f-f- -m-e-t-r-i-c-s-,-  
-n-e-w-s- -t-e-x-t-,- -a-n-d- -p-r-e-d-i-c-t-i-o-n- -l-o-g-s-.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s-:-\*-\* -H-o-w- -d-o- -y-o-u- -h-a-n-d-l-e- -m-o-d-e-l- -c-o-l-d---s-t-a-r-t- -t-i-m-e-s- -w-h-e-n-  
-t-h-e- -S-t-r-e-a-m-l-i-t- -c-o-n-t-a-i-n-e-r- -b-o-o-t-s- -u-p-?-

-\*- \*-D-e-e-p- -D-i-v-e-:-\*-\* -l-f- -t-h-e- -m-o-d-e-l- -s-i-z-e- -g-r-o-w-s- -t-o- -10-G-B+,- -h-o-w- -w-o-u-l-d- -y-o-u- -s-h-i-f-t- -t-o- -a-  
-m-i-c-r-o-s-e-r-v-i-c-e- -d-e-s-i-g-n-?-

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-##- -Q-u-e-s-t-i-o-n- -2-:- -W-h-y- -d-i-d- -y-o-u- -c-h-o-o-s-e- -a- -M-a-c-h-i-n-e- -L-e-a-r-n-i-n-g- -a-p-p-r-o-a-c-h- -i-n-s-t-e-a-d- -o-f-  
-a- -h-e-u-r-i-s-t-i-c-/r-u-l-e---b-a-s-e-d- -s-y-s-t-e-m-?-

-\*- \*-R-o-u-n-d-:-\*-\* -M-L-

-\*- \*-D-i-f-f-i-c-u-l-t-y-:-\*-\* -E-a-s-y- -|- \*-\*-l-m-p-o-r-t-a-n-c-e-:-\*-\* -7-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d-:-\*-\* -R-u-l-e- -s-y-s-t-e-m-s- -v-s- -M-L- -c-l-a-s-s-i-f-i-e-r-s-,- -c-o-m-p-l-e-x-i-t-y- -l-i-m-i-t-s- -o-f-  
-h-e-u-r-i-s-t-i-c-s-.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t-:-\*-\* -T-e-s-t-s- -t-h-e- -b-u-s-i-n-e-s-s- -j-u-s-t-i-f-i-c-a-t-i-o-n- -f-o-r- -M-L-.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-:-\*-\* -A- -r-u-l-e---b-a-s-e-d- -h-e-u-r-i-s-t-i-c- -(e.-g.- -f-l-a-g- -i-f- -c-o-m-p-a-n-y- -s-i-z-e-  
-d-e-c-r-e-a-s-e-s- -o-r- -m-o-n-e-y- -i-s- -l-o-w)- -f-a-i-l-s- -t-o- -c-a-p-t-u-r-e- -n-o-n---l-i-n-e-a-r- -f-e-a-t-u-r-e- -i-n-t-e-r-a-c-t-i-o-n-s-  
-a-n-d- -h-i-g-h---d-i-m-e-n-s-i-o-n-a-l- -c-o-r-r-e-l-a-t-i-o-n-s- -(l-i-k-e- -l-a-y-o-f-f- -v-e-l-o-c-i-t-y- -o-v-e-r- -t-i-m-e- -c-o-m-p-a-r-e-d-  
-t-o- -f-u-n-d-i-n-g- -s-t-a-g-e- -t-r-a-n-s-i-t-i-o-n-s)-.- -A-n- -M-L- -m-o-d-e-l- -l-e-a-r-n-s- -s-o-f-t- -b-o-u-n-d-a-r-i-e-s- -a-n-d-  
-q-u-a-n-t-i-f-i-e-s- -r-i-s-k- -p-r-o-b-a-b-i-l-i-s-t-i-c-a-l-l-y-,- -a-l-l-o-w-i-n-g- -t-h-r-e-s-h-o-l-d- -t-u-n-i-n-g- -f-o-r- -o-p-t-i-m-a-l-  
-r-e-c-a-l-l-.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-:-\*-\* -C-l-a-i-m-i-n-g- -M-L- -i-s- -a-l-w-a-y-s- -b-e-t-t-e-r- -w-i-t-h-o-u-t- -d-i-s-c-u-s-s-i-n-g-  
-r-u-l-e-s'- -s-i-m-p-l-i-c-i-t-y- -o-r- -t-r-a-n-s-p-a-r-e-n-c-y-.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-:-\*-\* -W-e- -u-s-e- -a- -h-y-b-r-i-d- -a-p-p-r-o-a-c-h-:- -t-h-e-  
-S-t-a-c-k-i-n-g- -e-n-s-e-m-b-l-e- -a-c-t-s- -a-s- -t-h-e- -p-r-i-m-a-r-y- -p-r-o-b-a-b-i-l-i-s-t-i-c- -m-o-d-e-l-,- -a-n-d- -w-e- -f-e-e-d- -t-h-e-  
-p-r-o-b-a-b-i-l-i-t-i-e-s- -i-n-t-o- -a- -r-u-l-e-s- -e-n-g-i-n-e- -t-h-a-t- -s-e-t-s- -t-h-e- -f-i-n-a-l- -a-l-e-r-t- -v-e-r-d-i-c-t- -b-a-s-e-d- -o-n-  
-b-u-s-i-n-e-s-s- -r-i-s-k- -a-p-p-e-t-i-t-e-s-.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s-:-\*-\* -W-h-a-t- -w-a-s- -t-h-e- -e-x-a-c-t- -b-a-s-e-l-i-n-e- -m-e-t-r-i-c- -o-f- -t-h-e-  
-r-u-l-e---b-a-s-e-d- -s-y-s-t-e-m-?-

-\*- \*-D-e-e-p- -D-i-v-e-:-\*-\* -U-n-d-e-r- -w-h-a-t- -c-o-n-d-i-t-i-o-n- -w-o-u-l-d- -y-o-u- -r-o-l-l- -b-a-c-k- -f-r-o-m- -t-h-e- -M-L-  
-m-o-d-e-l- -t-o- -a- -r-u-l-e---b-a-s-e-d- -s-y-s-t-e-m- -i-n- -p-r-o-d-u-c-t-i-o-n-?-

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##- Q-u-e-s-t-i-o-n- 3:- H-o-w- d-o- y-o-u- d-e-f-i-n-e- a-n-d- m-e-a-s-u-r-e- b-u-s-i-n-e-s-s- s-u-c-c-e-s-s- f-o-r- R-i-s-k-P-u-l-s-e-?

-\*-\*R-o-u-n-d:-\*- H-i-r-i-n-g- C-o-m-m-i-t-t-e-e-

-\*-\*D-i-f-f-i-c-u-l-t-y:-\*- M-e-d-i-u-m- |- \*-\*-l-m-p-o-r-t-a-n-c-e:-\*- -8/-1-0-

-\*-\*C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- K-P-l- m-a-p-p-i-n-g,- C-o-s-t---B-e-n-e-f-i-t- o-f- f-a-l-s-e- p-o-s-i-t-i-v-e-s- v-s- f-a-l-s-e- n-e-g-a-t-i-v-e-s.-

-\*-\*W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- C-h-e-c-k-s- i-f- t-h-e- e-n-g-i-n-e-e-r- a-l-i-g-n-s- t-e-c-h-n-i-c-a-l- w-o-r-k-t-o- b-u-s-i-n-e-s-s- o-u-t-c-o-m-e-s.-

-\*-\*l-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- B-u-s-i-n-e-s-s- s-u-c-c-e-s-s- i-s- m-e-a-s-u-r-e-d- b-y- t-h-e- R-e-c-a-l-l- o-f- l-a-y-o-f-f- e-v-e-n-t-s- (-c-a-t-c-h-i-n-g- a-t- l-e-a-s-t- 5-0%- o-f- u-p-c-o-m-i-n-g- l-a-y-o-f-f-s)- a-n-d- t-h-e- P-r-e-c-i-s-i-o-n- (-m-i-n-i-m-i-z-i-n-g- f-a-l-s-e- a-l-a-r-m-s)-. A- f-a-l-s-e- a-l-a-r-m- c-a-u-s-e-s- r-e-p-u-t-a-t-i-o-n-a-l- r-i-s-k- o-r- w-a-s-t-e- o-f- a-n-a-l-y-s-t- t-i-m-e,- w-h-i-l-e- a- m-i-s-s-e-d- l-a-y-o-f-f- r-e-p-r-e-s-e-n-t-s- a- c-r-i-t-i-c-a-l- b-l-i-n-d-s-p-o-t.-

-\*-\*C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- O-n-l-y- m-e-n-t-i-o-n-i-n-g- A-U-C---R-O-C- o-r- F-1- w-i-t-h-o-u-t- t-r-a-n-s-l-a-t-i-n-g- t-h-e-m- t-o- f-i-n-a-n-c-i-a-l- o-r- o-p-e-r-a-t-i-o-n-a-l- K-P-l-s.-

-\*-\*B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- d-e-f-i-n-e- a- C-o-s-t---U-t-i-l-i-t-y- f-u-n-c-t-i-o-n:- C-o-s-t- == C\_-f-n- \*- F-N- +- C\_-f-p- \*- F-P,- B-y- o-p-t-i-m-i-z-a-t-i-o-n,- w-e- s-h-o-w- t-h-a-t- t-h-e- t-u-n-e-d- M-L- m-o-d-e-l- r-e-d-u-c-e-s- o-p-e-r-a-t-i-o-n-a-l- r-i-s-k- e-x-p-o-s-u-r-e- b-y- 3-4%- c-o-m-p-a-r-e-d- t-o- t-h-e- r-u-l-e- b-a-s-e-l-i-n-e.-

-\*-\*F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- W-h-a-t- i-s- t-h-e- c-o-s-t- r-a-t-i-o- o-f- a- F-a-l-s-e- N-e-g-a-t-i-v-e- t-o- a- F-a-l-s-e- P-o-s-i-t-i-v-e-?

-\*-\*D-e-e-p- D-i-v-e:-\*- H-o-w- w-o-u-l-d- y-o-u- d-e-s-i-g-n- a- s-h-a-d-o-w- d-e-p-l-o-y-m-e-n-t- t-o- v-a-l-i-d-a-t-e- t-h-e- c-o-s-t- m-o-d-e-l- i-n- r-e-a-l---t-i-m-e-?

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##- Q-u-e-s-t-i-o-n- 4:- W-h-a-t- i-s- t-h-e- r-e-q-u-e-s-t- f-l-o-w- w-h-e-n- a- u-s-e-r- c-l-i-c-k-s- o-n- a- c-o-m-p-a-n-y- i-n- t-h-e- S-t-r-e-a-m-l-i-t- U-l-?

-\*-\*R-o-u-n-d:-\*- B-a-c-k-e-n-d-

-\*-\*D-i-f-f-i-c-u-l-t-y:-\*- M-e-d-i-u-m- |- \*-\*-l-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

-\*-\*C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- S-y-n-c-h-r-o-n-o-u-s- d-a-t-a-b-a-s-e- q-u-e-r-i-e-s,- c-a-c-h-i-n-g- l-a-y-e-r-s,- m-o-d-e-l- l-a-t-e-n-c-y.-

-\*-\*W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- V-e-r-i-f-i-e-s- d-a-t-a- a-n-d- c-o-n-t-r-o-l- f-l-o-w- u-n-d-e-r-s-t-a-n-d-i-n-g.-

-\*-\*l-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- 1.- U-s-e-r- s-e-l-e-c-t-s- a- c-o-m-p-a-n-y- n-a-m-e.- 2.- S-t-r-e-a-m-l-i-t- q-u-e-r-i-e-s- p-o-s-t-g-r-e-s- t-o- f-e-t-c-h- c-o-m-p-a-n-y- o-v-e-r-v-i-e-w- m-e-t-a-d-a-t-a- a-n-d- l-a-y-o-f-f- h-i-s-t-o-r-y.- 3.- S-i-m-u-l-t-a-n-e-o-u-s-l-y,- i-t- q-u-e-r-i-e-s- n-e-w-s- a-r-t-i-c-l-e-s- f-o-r- t-h-e- c-o-m-p-a-n-y.- 4.- S-t-r-e-a-m-l-i-t- s-e-n-d-s- t-h-e- f-e-a-t-u-r-e-s- t-o- t-h-e- P-r-e-d-i-c-t-i-o-n-S-e-r-v-i-c-e.- 5.- P-r-e-d-i-c-t-i-o-n-S-e-r-v-i-c-e- p-r-o-c-e-s-s-e-s- t-h-e- v-a-l-u-e-s,- l-o-a-d-s- t-h-e- p-r-e-p-r-o-c-e-s-s-o-r- a-n-d- c-a-l-i-b-r-a-t-e-d- S-t-a-c-k-i-n-g- m-o-d-e-l,- o-u-t-p-u-t-s- a- p-r-o-b-a-b-i-l-i-t-y,- a-n-d- g-e-n-e-r-a-t-e-s- a- S-H-A-P- w-a-t-e-r-f-a-l-l- p-l-o-t.- 6.- S-t-r-e-a-m-l-i-t- r-e-n-d-e-r-s- P-l-o-t-l-y- c-h-a-r-t-s- a-n-d- t-h-e- S-H-A-P- p-l-o-t.-

-\*-\*C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- F-o-r-g-e-t-t-i-n-g- t-h-a-t- m-o-d-e-l- r-e-l-o-a-d-i-n-g- m-u-s-t- b-e- c-a-c-h-e-d- u-s-i-n-g- -`@-s-t.-c-a-c-h-e\_-r-e-s-o-u-r-c-e-`.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*-\* -T-h-e- -d-a-t-a-b-a-s-e- -r-e-a-d-s- -u-s-e-  
 -`-@-s-t-.c-a-c-h-e-\_d-a-t-a-(t-t-l=-6-0-0)-`- -a-n-d- -t-h-e- -m-o-d-e-l- -l-o-a-d-e-r- -u-s-e-s- -`-@-s-t-.c-a-c-h-e-\_r-e-s-o-u-r-c-e-`-.-  
 -F-e-a-t-u-r-e-s- -a-r-e- -c-o-n-s-t-r-u-c-t-e-d- -i-n- -S-Q-L- -t-o- -m-i-n-i-m-i-z-e- -p-y-t-h-o-n- -p-r-o-c-e-s-s-i-n-g- -o-v-e-r-h-e-a-d,-  
 -a-n-d- -p-r-e-d-i-c-t-i-o-n-s- -a-r-e- -l-o-g-g-e-d- -a-s-y-n-c-h-r-o-n-o-u-s-l-y- -t-o- -a- -p-o-s-t-g-r-e-s- -p-r-e-d-i-c-t-i-o-n- -h-i-s-t-o-r-y-  
 -t-a-b-l-e.-

-\* -\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*-\* -W-h-a-t- -h-a-p-p-e-n-s- -i-f- -t-h-e- -P-o-s-t-g-r-e-S-Q-L- -c-o-n-n-e-c-t-i-o-n- -p-o-o-l-  
 -i-s- -e-x-h-a-u-s-t-e-d- -d-u-r-i-n-g- -p-e-a-k- -r-e-q-u-e-s-t-s?-

-\* -\*-D-e-e-p- -D-i-v-e:-\*-\* -H-o-w- -w-o-u-l-d- -y-o-u- -r-e-d-e-s-i-g-n- -t-h-e- -S-H-A-P- -r-e-n-d-e-r-i-n-g- -t-o- -a-v-o-i-d-  
 -b-l-o-c-k-i-n-g- -t-h-e- -m-a-i-n- -U-I- -t-h-r-e-a-d?-

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-#-#-#- -Q-u-e-s-t-i-o-n- -5:- -E-x-p-l-a-i-n- -y-o-u-r- -m-o-d-e-l- -d-e-p-l-o-y-m-e-n-t- -a-n-d- -r-e-g-i-s-t-r-y- -w-o-r-k-f-l-o-w.-

-\* -\*-R-o-u-n-d:-\*-\* -S-y-s-t-e-m- -D-e-s-i-g-n-

-\* -\*-D-i-f-f-i-c-u-l-t-y:-\*-\* -M-e-d-i-u-m- -|- \*-\*-l-m-p-o-r-t-a-n-c-e:-\*-\* -8-/1-0-

-\* -\*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*-\* -M-o-d-e-l- -s-e-r-i-a-l-i-z-a-t-i-o-n,- -v-e-r-s-i-o-n-i-n-g,- -r-o-l-l-b-a-c-k,-  
 -c-o-n-t-a-i-n-e-r-i-z-a-t-i-o-n.-

-\* -\*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*-\* -T-e-s-t-s- -M-L-O-p-s- -k-n-o-w-l-e-d-g-e.-

-\* -\*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*-\* -W-e- -r-u-n- -a-n- -o-f-f-l-i-n-e- -t-r-a-i-n-i-n-g- -p-i-p-e-l-i-n-e.- -T-h-e- -b-e-s-t-  
 -p-r-e-p-r-o-c-e-s-s-o-r,- -S-t-a-c-k-i-n-g- -c-l-a-s-s-i-f-i-e-r,- -a-n-d- -S-H-A-P- -e-x-p-l-a-i-n-e-r- -a-r-e- -s-e-r-i-a-l-i-z-e-d- -t-o-  
 -`-p-k-l`- -f-i-l-e-s- -a-n-d- -s-t-o-r-e-d- -w-i-t-h- -t-h-e-i-r- -m-e-t-a-d-a-t-a- -(t-r-a-i-n-i-n-g- -m-e-t-r-i-c-s,- -t-h-r-e-s-h-o-l-d)- -i-n- -a-  
 -v-e-r-s-i-o-n-e-d- -f-o-l-d-e-r.- -T-h-e- -p-r-o-d-u-c-t-i-o-n- -s-e-r-v-e-r- -p-u-l-l-s- -t-h-e- -l-a-t-e-s-t- -t-a-g- -d-u-r-i-n-g- -C-I/-C-D-  
 -d-e-p-l-o-y.-

-\* -\*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*-\* -U-n-p-i-c-k-l-i-n-g- -m-o-d-e-l-s- -a-c-r-o-s-s- -d-i-f-f-e-r-e-n-t- -P-y-t-h-o-n-  
 -e-n-v-i-r-o-n-m-e-n-t-s- -w-i-t-h-o-u-t- -s-a-f-e-t-y- -g-u-a-r-d-s.-

-\* -\*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*-\* -W-e- -u-s-e- -a- -m-o-d-e-l- -r-e-g-i-s-t-r-y- -p-a-t-t-e-r-n- -(e.-g.-  
 -M-L-f-l-o-w/-l-o-c-a-l- -S-3- -v-e-r-s-i-o-n-i-n-g)-.- -W-e- -b-u-n-d-l-e- -m-o-d-e-l-s- -i-n- -a- -c-o-n-t-a-i-n-e-r- -a-n-d- -r-u-n- -a-  
 -s-t-a-r-t-u-p- -c-h-e-c-k- -t-o- -v-e-r-i-f-y- -t-h-a-t- -t-h-e- -u-n-p-i-c-k-l-e-d- -m-o-d-e-l- -o-u-t-p-u-t-s- -m-a-t-c-h- -b-a-s-e-l-i-n-e- -t-e-s-t-  
 -m-a-t-r-i-c-e-s.-

-\* -\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*-\* -W-h-a-t- -c-h-e-c-k-s- -w-o-u-l-d- -y-o-u- -p-u-t- -i-n- -y-o-u-r- -C-I/-C-D- -t-e-s-t-  
 -s-u-i-t-e- -b-e-f-o-r-e- -d-e-p-l-o-y-i-n-g- -a- -n-e-w- -m-o-d-e-l- -p-i-c-k-l-e?-

-\* -\*-D-e-e-p- -D-i-v-e:-\*-\* -H-o-w- -d-o- -y-o-u- -d-e-t-e-c-t- -s-i-l-e-n-t- -d-a-t-a- -c-o-r-r-u-p-t-i-o-n- -d-u-r-i-n-g- -m-o-d-e-l-  
 -d-e-s-e-r-i-a-l-i-z-a-t-i-o-n?-

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-#-#-#- -Q-u-e-s-t-i-o-n- -6:- -D-B- -Q-u-e-s-t-i-o-n- -6:- -H-o-w- -w-o-u-l-d- -y-o-u- -o-p-t-i-m-i-z-e- -P-o-s-t-g-r-e-S-Q-L- -f-o-r-  
 -h-i-g-h---v-o-l-u-m-e- -h-i-s-t-o-r-i-c-a-l- -a-g-g-r-e-g-a-t-i-o-n-s- -i-n- -R-i-s-k-P-u-l-s-e?-

-\* -\*-R-o-u-n-d:-\*-\* -B-a-c-k-e-n-d-

-\* -\*-D-i-f-f-i-c-u-l-t-y:-\*-\* -M-e-d-i-u-m- -|- \*-\*-l-m-p-o-r-t-a-n-c-e:-\*-\* -8-/1-0-

-\* -\*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*-\* -l-n-d-e-x-e-s,- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w-s,- -W-i-n-d-o-w- -F-u-n-c-t-i-o-n-s,-  
 -P-o-s-t-g-r-e-S-Q-L- -q-u-e-r-y- -p-l-a-n-n-e-r.-

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- *-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-*- -T-e-s-t-s- -d-a-t-a-b-a-s-e- -t-u-n-i-n-g- -a-n-d- -r-e-l-a-t-i-o-n-a-l- -d-e-s-i-g-n-.
- *- -l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-*- -T-o- -o-p-t-i-m-i-z-e- -r-o-l-l-i-n-g- -a-g-g-r-e-g-a-t-i-o-n-s- -(e.-g.-,-
-c-u-m-u-l-a-t-i-v-e- -l-a-y-o-f-f-s,- -r-o-l-l-i-n-g- -90---d-a-y- -l-a-y-o-f-f- -c-o-u-n-t-s-),- -w-e- -s-h-o-u-l-d- -a-v-o-i-d- -s-c-a-n-n-i-n-g-
-t-h-e- -r-a-w- -l-a-y-o-f-f-s`- -t-a-b-l-e- -r-e-p-e-a-t-e-d-l-y.- -l-n-s-t-e-a-d,- -c-r-e-a-t-e- -c-o-m-p-o-s-i-t-e- -i-n-d-e-x-e-s- -o-n-
-`-(c-o-m-p-a-n-y_-i-d,- -l-a-y-o-f-f_-d-a-t-e)-`- -o-r- -s-e-t- -u-p- -a- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w- -t-h-a-t- -r-e-f-r-e-s-h-e-s-
-p-e-r-i-o-d-i-c-a-l-l-y- -t-o- -s-t-o-r-e- -p-r-e---a-g-g-r-e-g-a-t-e-d- -m-e-t-r-i-c-s.-
- *- -C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-*- -R-e-l-y-i-n-g- -o-n- -O-R-M- -q-u-e-r-i-e-s- -t-h-a-t- -p-e-r-f-o-r-m- -N+-1- -d-a-t-a-b-a-s-e-
-h-i-t-s.-
- *- -B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-*- -W-e- -w-r-i-t-e- -r-a-w- -S-Q-L- -q-u-e-r-i-e-s- -u-s-i-n-g-
-w-i-n-d-o-w- -f-u-n-c-t-i-o-n-s- -(`S-U-M(-) -O-V-E-R- -(P-A-R-T-I-T-I-O-N- -B-Y- -.-.- -O-R-D-E-R- -B-Y- -.-.-)-`),- -W-e- -s-e-t-
-u-p- -d-a-t-a-b-a-s-e- -p-a-r-t-i-t-i-o-n-i-n-g- -o-n- -t-h-e- -l-a-y-o-f-f-s`- -t-a-b-l-e- -b-y- -y-e-a-r,- -a-n-d- -c-a-c-h-e- -q-u-e-r-y-
-r-e-s-u-l-t-s- -u-s-i-n-g- -R-e-d-i-s.-
- *- -F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-*- -H-o-w- -d-o- -y-o-u- -m-a-n-a-g-e- -d-a-t-a-b-a-s-e- -m-i-g-r-a-t-i-o-n-s- -w-h-e-n-
-m-o-d-i-f-y-i-n-g- -t-h-e- -s-c-h-e-m-a- -o-f- -c-o-m-p-a-n-i-e-s?-
- *- -D-e-e-p- -D-i-v-e:-*- -E-x-p-l-a-i-n- -t-h-e- -d-i-f-f-e-r-e-n-c-e- -b-e-t-w-e-e-n- -a- -i-n-d-e-x-e-s- -s-c-a-n- -a-n-d- -a-
-s-e-q-u-e-n-t-i-a-l- -s-c-a-n- -u-n-d-e-r- -P-o-s-t-g-r-e-S-Q-L.-
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- #-#- -Q-u-e-s-t-i-o-n- -7:- -D-B- -Q-u-e-s-t-i-o-n- -7:- -H-o-w- -w-o-u-l-d- -y-o-u- -o-p-t-i-m-i-z-e- -P-o-s-t-g-r-e-S-Q-L- -f-o-r-
-h-i-g-h---v-o-l-u-m-e- -h-i-s-t-o-r-i-c-a-l- -a-g-g-r-e-g-a-t-i-o-n-s- -i-n- -R-i-s-k-P-u-l-s-e-?
- *- -R-o-u-n-d:-*- -B-a-c-k-e-n-d-
- *- -D-i-f-f-i-c-u-l-t-y:-*- -M-e-d-i-u-m- -|- *- -l-m-p-o-r-t-a-n-c-e:-*- -8-/10-
- *- -C-o-n-c-e-p-t- -T-e-s-t-e-d:-*- -l-n-d-e-x-e-s,- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w-s,- -W-i-n-d-o-w- -F-u-n-c-t-i-o-n-s,-
-P-o-s-t-g-r-e-S-Q-L- -q-u-e-r-y- -p-l-a-n-n-e-r.-
- *- -W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-*- -T-e-s-t-s- -d-a-t-a-b-a-s-e- -t-u-n-i-n-g- -a-n-d- -r-e-l-a-t-i-o-n-a-l- -d-e-s-i-g-n.-
- *- -l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-*- -T-o- -o-p-t-i-m-i-z-e- -r-o-l-l-i-n-g- -a-g-g-r-e-g-a-t-i-o-n-s- -(e.-g.-,-
-c-u-m-u-l-a-t-i-v-e- -l-a-y-o-f-f-s,- -r-o-l-l-i-n-g- -90---d-a-y- -l-a-y-o-f-f- -c-o-u-n-t-s-),- -w-e- -s-h-o-u-l-d- -a-v-o-i-d- -s-c-a-n-n-i-n-g-
-t-h-e- -r-a-w- -l-a-y-o-f-f-s`- -t-a-b-l-e- -r-e-p-e-a-t-e-d-l-y.- -l-n-s-t-e-a-d,- -c-r-e-a-t-e- -c-o-m-p-o-s-i-t-e- -i-n-d-e-x-e-s- -o-n-
-`-(c-o-m-p-a-n-y_-i-d,- -l-a-y-o-f-f_-d-a-t-e)-`- -o-r- -s-e-t- -u-p- -a- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w- -t-h-a-t- -r-e-f-r-e-s-h-e-s-
-p-e-r-i-o-d-i-c-a-l-l-y- -t-o- -s-t-o-r-e- -p-r-e---a-g-g-r-e-g-a-t-e-d- -m-e-t-r-i-c-s.-
- *- -C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-*- -R-e-l-y-i-n-g- -o-n- -O-R-M- -q-u-e-r-i-e-s- -t-h-a-t- -p-e-r-f-o-r-m- -N+-1- -d-a-t-a-b-a-s-e-
-h-i-t-s.-
- *- -B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-*- -W-e- -w-r-i-t-e- -r-a-w- -S-Q-L- -q-u-e-r-i-e-s- -u-s-i-n-g-
-w-i-n-d-o-w- -f-u-n-c-t-i-o-n-s- -(`S-U-M(-) -O-V-E-R- -(P-A-R-T-I-T-I-O-N- -B-Y- -.-.- -O-R-D-E-R- -B-Y- -.-.-)-`),- -W-e- -s-e-t-
-u-p- -d-a-t-a-b-a-s-e- -p-a-r-t-i-t-i-o-n-i-n-g- -o-n- -t-h-e- -l-a-y-o-f-f-s`- -t-a-b-l-e- -b-y- -y-e-a-r,- -a-n-d- -c-a-c-h-e- -q-u-e-r-y-
-r-e-s-u-l-t-s- -u-s-i-n-g- -R-e-d-i-s.-
- *- -F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-*- -H-o-w- -d-o- -y-o-u- -m-a-n-a-g-e- -d-a-t-a-b-a-s-e- -m-i-g-r-a-t-i-o-n-s- -w-h-e-n-
-m-o-d-i-f-y-i-n-g- -t-h-e- -s-c-h-e-m-a- -o-f- -c-o-m-p-a-n-i-e-s?-
- *- -D-e-e-p- -D-i-v-e:-*- -E-x-p-l-a-i-n- -t-h-e- -d-i-f-f-e-r-e-n-c-e- -b-e-t-w-e-e-n- -a- -i-n-d-e-x-e-s- -s-c-a-n- -a-n-d- -a-
-s-e-q-u-e-n-t-i-a-l- -s-c-a-n- -u-n-d-e-r- -P-o-s-t-g-r-e-S-Q-L.-
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##### -Q-u-e-s-t-i-o-n- -8:- -D-B- -Q-u-e-s-t-i-o-n- -8:- -H-o-w- -w-o-u-l-d- -y-o-u- -o-p-t-i-m-i-z-e- -P-o-s-t-g-r-e-S-Q-L- -f-o-r-  
-h-i-g-h---v-o-l-u-m-e- -h-i-s-t-o-r-i-c-a-l- -a-g-g-r-e-g-a-t-i-o-n-s- -i-n- -R-i-s-k-P-u-l-s-e-?-

\*- \*-R-o-u-n-d:-\*- -B-a-c-k-e-n-d-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -l-n-d-e-x-e-s,- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w-s,- -W-i-n-d-o-w- -F-u-n-c-t-i-o-n-s,-  
-P-o-s-t-g-r-e-S-Q-L- -q-u-e-r-y- -p-l-a-n-n-e-r.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -d-a-t-a-b-a-s-e- -t-u-n-i-n-g- -a-n-d- -r-e-l-a-t-i-o-n-a-l- -d-e-s-i-g-n.-

\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-o- -o-p-t-i-m-i-z-e- -r-o-l-l-i-n-g- -a-g-g-r-e-g-a-t-i-o-n-s- -(e.-g.-,-  
-c-u-m-u-l-a-t-i-v-e- -l-a-y-o-f-f-s,- -r-o-l-l-i-n-g- -9-0---d-a-y- -l-a-y-o-f-f- -c-o-u-n-t-s-),- -w-e- -s-h-o-u-l-d- -a-v-o-i-d- -s-c-a-n-n-i-n-g-  
-t-h-e- -r-a-w- -`l-a-y-o-f-f-s`- -t-a-b-l-e- -r-e-p-e-a-t-e-d-l-y,- -l-n-s-t-e-a-d,- -c-r-e-a-t-e- -c-o-m-p-o-s-i-t-e- -i-n-d-e-x-e-s- -o-n-  
-`(c-o-m-p-a-n-y\_-i-d,- -l-a-y-o-f-f\_-d-a-t-e)`- -o-r- -s-e-t- -u-p- -a- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w- -t-h-a-t- -r-e-f-r-e-s-h-e-s-  
-p-e-r-i-o-d-i-c-a-l-l-y- -t-o- -s-t-o-r-e- -p-r-e---a-g-g-r-e-g-a-t-e-d- -m-e-t-r-i-c-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -R-e-l-y-i-n-g- -o-n- -O-R-M- -q-u-e-r-i-e-s- -t-h-a-t- -p-e-r-f-o-r-m- -N++1- -d-a-t-a-b-a-s-e-  
-h-i-t-s.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -w-r-i-t-e- -r-a-w- -S-Q-L- -q-u-e-r-i-e-s- -u-s-i-n-g-  
-w-i-n-d-o-w- -f-u-n-c-t-i-o-n-s- -(`S-U-M(-) -O-V-E-R- -(P-A-R-T-I-T-I-O-N- -B-Y- -...- -O-R-D-E-R- -B-Y- -...-)`),- -W-e- -s-e-t-  
-u-p- -d-a-t-a-b-a-s-e- -p-a-r-t-i-t-i-o-n-i-n-g- -o-n- -t-h-e- -`l-a-y-o-f-f-s`- -t-a-b-l-e- -b-y- -y-e-a-r,- -a-n-d- -c-a-c-h-e- -q-u-e-r-y-  
-r-e-s-u-l-t-s- -u-s-i-n-g- -R-e-d-i-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -d-o- -y-o-u- -m-a-n-a-g-e- -d-a-t-a-b-a-s-e- -m-i-g-r-a-t-i-o-n-s- -w-h-e-n-  
-m-o-d-i-f-y-i-n-g- -t-h-e- -s-c-h-e-m-a- -o-f- -c-o-m-p-a-n-i-e-s-?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -t-h-e- -d-i-f-f-e-r-e-n-c-e- -b-e-t-w-e-e-n- -a- -i-n-d-e-x-e-s- -s-c-a-n- -a-n-d- -a-  
-s-e-q-u-e-n-t-i-a-l- -s-c-a-n- -u-n-d-e-r- -P-o-s-t-g-r-e-S-Q-L.-

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##### -Q-u-e-s-t-i-o-n- -9:- -D-B- -Q-u-e-s-t-i-o-n- -9:- -H-o-w- -w-o-u-l-d- -y-o-u- -o-p-t-i-m-i-z-e- -P-o-s-t-g-r-e-S-Q-L- -f-o-r-  
-h-i-g-h---v-o-l-u-m-e- -h-i-s-t-o-r-i-c-a-l- -a-g-g-r-e-g-a-t-i-o-n-s- -i-n- -R-i-s-k-P-u-l-s-e-?-

\*- \*-R-o-u-n-d:-\*- -B-a-c-k-e-n-d-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -l-n-d-e-x-e-s,- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w-s,- -W-i-n-d-o-w- -F-u-n-c-t-i-o-n-s,-  
-P-o-s-t-g-r-e-S-Q-L- -q-u-e-r-y- -p-l-a-n-n-e-r.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -d-a-t-a-b-a-s-e- -t-u-n-i-n-g- -a-n-d- -r-e-l-a-t-i-o-n-a-l- -d-e-s-i-g-n.-

\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-o- -o-p-t-i-m-i-z-e- -r-o-l-l-i-n-g- -a-g-g-r-e-g-a-t-i-o-n-s- -(e.-g.-,-  
-c-u-m-u-l-a-t-i-v-e- -l-a-y-o-f-f-s,- -r-o-l-l-i-n-g- -9-0---d-a-y- -l-a-y-o-f-f- -c-o-u-n-t-s-),- -w-e- -s-h-o-u-l-d- -a-v-o-i-d- -s-c-a-n-n-i-n-g-  
-t-h-e- -r-a-w- -`l-a-y-o-f-f-s`- -t-a-b-l-e- -r-e-p-e-a-t-e-d-l-y,- -l-n-s-t-e-a-d,- -c-r-e-a-t-e- -c-o-m-p-o-s-i-t-e- -i-n-d-e-x-e-s- -o-n-  
-`(c-o-m-p-a-n-y\_-i-d,- -l-a-y-o-f-f\_-d-a-t-e)`- -o-r- -s-e-t- -u-p- -a- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w- -t-h-a-t- -r-e-f-r-e-s-h-e-s-  
-p-e-r-i-o-d-i-c-a-l-l-y- -t-o- -s-t-o-r-e- -p-r-e---a-g-g-r-e-g-a-t-e-d- -m-e-t-r-i-c-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -R-e-l-y-i-n-g- -o-n- -O-R-M- -q-u-e-r-i-e-s- -t-h-a-t- -p-e-r-f-o-r-m- -N++1- -d-a-t-a-b-a-s-e-  
-h-i-t-s.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -w-r-i-t-e- -r-a-w- -S-Q-L- -q-u-e-r-i-e-s- -u-s-i-n-g-  
-w-i-n-d-o-w- -f-u-n-c-t-i-o-n-s- -(`S-U-M(-) -O-V-E-R- -(P-A-R-T-I-T-I-O-N- -B-Y- -...- -O-R-D-E-R- -B-Y- -...-)`),- -W-e- -s-e-t-  
-u-p- -d-a-t-a-b-a-s-e- -p-a-r-t-i-t-i-o-n-i-n-g- -o-n- -t-h-e- -`l-a-y-o-f-f-s`- -t-a-b-l-e- -b-y- -y-e-a-r,- -a-n-d- -c-a-c-h-e- -q-u-e-r-y-

-r-e-s-u-l-t-s- -u-s-i-n-g- -R-e-d-i-s-.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -d-o- -y-o-u- -m-a-n-a-g-e- -d-a-t-a-b-a-s-e- -m-i-g-r-a-t-i-o-n-s- -w-h-e-n-  
-m-o-d-i-f-y-i-n-g- -t-h-e- -s-c-h-e-m-a- -o-f- -c-o-m-p-a-n-i-e-s?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -t-h-e- -d-i-f-f-e-r-e-n-c-e- -b-e-t-w-e-e-n- -a- -i-n-d-e-x-e-s- -s-c-a-n- -a-n-d- -a-  
-s-e-q-u-e-n-t-i-a-l- -s-c-a-n- -u-n-d-e-r- -P-o-s-t-g-r-e-S-Q-L-.-

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-###- -Q-u-e-s-t-i-o-n- -1-0:- -D-B- -Q-u-e-s-t-i-o-n- -1-0:- -H-o-w- -w-o-u-l-d- -y-o-u- -o-p-t-i-m-i-z-e- -P-o-s-t-g-r-e-S-Q-L- -f-o-r-  
-h-i-g-h---v-o-l-u-m-e- -h-i-s-t-o-r-i-c-a-l- -a-g-g-r-e-g-a-t-i-o-n-s- -i-n- -R-i-s-k-P-u-l-s-e?-

-\*- \*-R-o-u-n-d:-\*- -B-a-c-k-e-n-d-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -| -\*- \*-l-m-p-o-r-t-a-n-c-e:-\*- -8/-1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -l-n-d-e-x-e-s,- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w-s,- -W-i-n-d-o-w- -F-u-n-c-t-i-o-n-s,-  
-P-o-s-t-g-r-e-S-Q-L- -q-u-e-r-y- -p-l-a-n-n-e-r.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -d-a-t-a-b-a-s-e- -t-u-n-i-n-g- -a-n-d- -r-e-l-a-t-i-o-n-a-l- -d-e-s-i-g-n.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-o- -o-p-t-i-m-i-z-e- -r-o-l-l-i-n-g- -a-g-g-r-e-g-a-t-i-o-n-s- -(e.g.-,-  
-c-u-m-u-l-a-t-i-v-e- -l-a-y-o-f-f-s,- -r-o-l-l-i-n-g- -9-0---d-a-y- -l-a-y-o-f-f- -c-o-u-n-t-s-),- -w-e- -s-h-o-u-l-d- -a-v-o-i-d- -s-c-a-n-n-i-n-g-  
-t-h-e- -r-a-w- -l-a-y-o-f-f-s-` -t-a-b-l-e- -r-e-p-e-a-t-e-d-l-y,- -l-n-s-t-e-a-d,- -c-r-e-a-t-e- -c-o-m-p-o-s-i-t-e- -i-n-d-e-x-e-s- -o-n-  
-`-(c-o-m-p-a-n-y\_-i-d,- -l-a-y-o-f-f\_-d-a-t-e)-` -o-r- -s-e-t- -u-p- -a- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w- -t-h-a-t- -r-e-f-r-e-s-h-e-s-  
-p-e-r-i-o-d-i-c-a-l-l-y- -t-o- -s-t-o-r-e- -p-r-e---a-g-g-r-e-g-a-t-e-d- -m-e-t-r-i-c-s.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -R-e-l-y-i-n-g- -o-n- -O-R-M- -q-u-e-r-i-e-s- -t-h-a-t- -p-e-r-f-o-r-m- -N-+-1- -d-a-t-a-b-a-s-e-  
-h-i-t-s.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -w-r-i-t-e- -r-a-w- -S-Q-L- -q-u-e-r-i-e-s- -u-s-i-n-g-  
-w-i-n-d-o-w- -f-u-n-c-t-i-o-n-s- -(S-U-M(-) -O-V-E-R- -(P-A-R-T-I-T-I-O-N- -B-Y- -.-.- -O-R-D-E-R- -B-Y- -.-.-)-),- -W-e- -s-e-t-  
-u-p- -d-a-t-a-b-a-s-e- -p-a-r-t-i-t-i-o-n-i-n-g- -o-n- -t-h-e- -l-a-y-o-f-f-s-` -t-a-b-l-e- -b-y- -y-e-a-r,- -a-n-d- -c-a-c-h-e- -q-u-e-r-y-  
-r-e-s-u-l-t-s- -u-s-i-n-g- -R-e-d-i-s-.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -d-o- -y-o-u- -m-a-n-a-g-e- -d-a-t-a-b-a-s-e- -m-i-g-r-a-t-i-o-n-s- -w-h-e-n-  
-m-o-d-i-f-y-i-n-g- -t-h-e- -s-c-h-e-m-a- -o-f- -c-o-m-p-a-n-i-e-s?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -t-h-e- -d-i-f-f-e-r-e-n-c-e- -b-e-t-w-e-e-n- -a- -i-n-d-e-x-e-s- -s-c-a-n- -a-n-d- -a-  
-s-e-q-u-e-n-t-i-a-l- -s-c-a-n- -u-n-d-e-r- -P-o-s-t-g-r-e-S-Q-L-.-

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-###- -Q-u-e-s-t-i-o-n- -1-1:- -D-B- -Q-u-e-s-t-i-o-n- -1-1:- -H-o-w- -w-o-u-l-d- -y-o-u- -o-p-t-i-m-i-z-e- -P-o-s-t-g-r-e-S-Q-L- -f-o-r-  
-h-i-g-h---v-o-l-u-m-e- -h-i-s-t-o-r-i-c-a-l- -a-g-g-r-e-g-a-t-i-o-n-s- -i-n- -R-i-s-k-P-u-l-s-e?-

-\*- \*-R-o-u-n-d:-\*- -B-a-c-k-e-n-d-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -| -\*- \*-l-m-p-o-r-t-a-n-c-e:-\*- -8/-1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -l-n-d-e-x-e-s,- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w-s,- -W-i-n-d-o-w- -F-u-n-c-t-i-o-n-s,-  
-P-o-s-t-g-r-e-S-Q-L- -q-u-e-r-y- -p-l-a-n-n-e-r.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -d-a-t-a-b-a-s-e- -t-u-n-i-n-g- -a-n-d- -r-e-l-a-t-i-o-n-a-l- -d-e-s-i-g-n.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-o- -o-p-t-i-m-i-z-e- -r-o-l-l-i-n-g- -a-g-g-r-e-g-a-t-i-o-n-s- -(e.g.-,-

-c-u-m-u-l-a-t-i-v-e- -l-a-y-o-f-f-s-, -r-o-l-l-i-n-g- -9-0---d-a-y- -l-a-y-o-f-f- -c-o-u-n-t-s-),- -w-e- -s-h-o-u-l-d- -a-v-o-i-d- -s-c-a-n-n-i-n-g-  
-t-h-e- -r-a-w- -`-l-a-y-o-f-f-s-`- -t-a-b-l-e- -r-e-p-e-a-t-e-d-l-y-, -l-n-s-t-e-a-d-, -c-r-e-a-t-e- -c-o-m-p-o-s-i-t-e- -i-n-d-e-x-e-s- -o-n-  
-`-(c-o-m-p-a-n-y\_-i-d-, -l-a-y-o-f-f\_-d-a-t-e)-`- -o-r- -s-e-t- -u-p- -a- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w- -t-h-a-t- -r-e-f-r-e-s-h-e-s-  
-p-e-r-i-o-d-i-c-a-l-l-y- -t-o- -s-t-o-r-e- -p-r-e---a-g-g-r-e-g-a-t-e-d- -m-e-t-r-i-c-s-.-

-\*-\*C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*-\* -R-e-l-y-i-n-g- -o-n- -O-R-M- -q-u-e-r-i-e-s- -t-h-a-t- -p-e-r-f-o-r-m- -N++1- -d-a-t-a-b-a-s-e-  
-h-i-t-s-.-

-\*-\*B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*-\* -W-e- -w-r-i-t-e- -r-a-w- -S-Q-L- -q-u-e-r-i-e-s- -u-s-i-n-g-  
-w-i-n-d-o-w- -f-u-n-c-t-i-o-n-s- -(`S-U-M(-) -O-V-E-R- -(P-A-R-T-I-T-I-O-N- -B-Y- -. . . -O-R-D-E-R- -B-Y- -. . .)-`)-, -W-e- -s-e-t-  
-u-p- -d-a-t-a-b-a-s-e- -p-a-r-t-i-t-i-o-n-i-n-g- -o-n- -t-h-e- -`-l-a-y-o-f-f-s-`- -t-a-b-l-e- -b-y- -y-e-a-r-, -a-n-d- -c-a-c-h-e- -q-u-e-r-y-  
-r-e-s-u-l-t-s- -u-s-i-n-g- -R-e-d-i-s-.-

-\*-\*F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*-\* -H-o-w- -d-o- -y-o-u- -m-a-n-a-g-e- -d-a-t-a-b-a-s-e- -m-i-g-r-a-t-i-o-n-s- -w-h-e-n-  
-m-o-d-i-f-y-i-n-g- -t-h-e- -s-c-h-e-m-a- -o-f- -c-o-m-p-a-n-i-e-s-?-

-\*-\*D-e-e-p- -D-i-v-e:-\*-\* -E-x-p-l-a-i-n- -t-h-e- -d-i-f-f-e-r-e-n-c-e- -b-e-t-w-e-e-n- -a- -i-n-d-e-x-e-s- -s-c-a-n- -a-n-d- -a-  
-s-e-q-u-e-n-t-i-a-l- -s-c-a-n- -u-n-d-e-r- -P-o-s-t-g-r-e-S-Q-L-.-

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-##-#- -Q-u-e-s-t-i-o-n- -1-2:- -D-B- -Q-u-e-s-t-i-o-n- -1-2:- -H-o-w- -w-o-u-l-d- -y-o-u- -o-p-t-i-m-i-z-e- -P-o-s-t-g-r-e-S-Q-L- -f-o-r-  
-h-i-g-h---v-o-l-u-m-e- -h-i-s-t-o-r-i-c-a-l- -a-g-g-r-e-g-a-t-i-o-n-s- -i-n- -R-i-s-k-P-u-l-s-e-?-

-\*-\*R-o-u-n-d:-\*-\* -B-a-c-k-e-n-d-

-\*-\*D-i-f-f-i-c-u-l-t-y:-\*-\* -M-e-d-i-u-m- -| -\*-\*l-m-p-o-r-t-a-n-c-e:-\*-\* -8/-1-0-

-\*-\*C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*-\* -l-n-d-e-x-e-s-, -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w-s-, -W-i-n-d-o-w- -F-u-n-c-t-i-o-n-s-,  
-P-o-s-t-g-r-e-S-Q-L- -q-u-e-r-y- -p-l-a-n-n-e-r-.-

-\*-\*W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*-\* -T-e-s-t-s- -d-a-t-a-b-a-s-e- -t-u-n-i-n-g- -a-n-d- -r-e-l-a-t-i-o-n-a-l- -d-e-s-i-g-n-.-

-\*-\*l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*-\* -T-o- -o-p-t-i-m-i-z-e- -r-o-l-l-i-n-g- -a-g-g-r-e-g-a-t-i-o-n-s- -(e.g.-, -  
-c-u-m-u-l-a-t-i-v-e- -l-a-y-o-f-f-s-, -r-o-l-l-i-n-g- -9-0---d-a-y- -l-a-y-o-f-f- -c-o-u-n-t-s-),- -w-e- -s-h-o-u-l-d- -a-v-o-i-d- -s-c-a-n-n-i-n-g-  
-t-h-e- -r-a-w- -`-l-a-y-o-f-f-s-`- -t-a-b-l-e- -r-e-p-e-a-t-e-d-l-y-, -l-n-s-t-e-a-d-, -c-r-e-a-t-e- -c-o-m-p-o-s-i-t-e- -i-n-d-e-x-e-s- -o-n-  
-`-(c-o-m-p-a-n-y\_-i-d-, -l-a-y-o-f-f\_-d-a-t-e)-`- -o-r- -s-e-t- -u-p- -a- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w- -t-h-a-t- -r-e-f-r-e-s-h-e-s-  
-p-e-r-i-o-d-i-c-a-l-l-y- -t-o- -s-t-o-r-e- -p-r-e---a-g-g-r-e-g-a-t-e-d- -m-e-t-r-i-c-s-.-

-\*-\*C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*-\* -R-e-l-y-i-n-g- -o-n- -O-R-M- -q-u-e-r-i-e-s- -t-h-a-t- -p-e-r-f-o-r-m- -N++1- -d-a-t-a-b-a-s-e-  
-h-i-t-s-.-

-\*-\*B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*-\* -W-e- -w-r-i-t-e- -r-a-w- -S-Q-L- -q-u-e-r-i-e-s- -u-s-i-n-g-  
-w-i-n-d-o-w- -f-u-n-c-t-i-o-n-s- -(`S-U-M(-) -O-V-E-R- -(P-A-R-T-I-T-I-O-N- -B-Y- -. . . -O-R-D-E-R- -B-Y- -. . .)-`)-, -W-e- -s-e-t-  
-u-p- -d-a-t-a-b-a-s-e- -p-a-r-t-i-t-i-o-n-i-n-g- -o-n- -t-h-e- -`-l-a-y-o-f-f-s-`- -t-a-b-l-e- -b-y- -y-e-a-r-, -a-n-d- -c-a-c-h-e- -q-u-e-r-y-  
-r-e-s-u-l-t-s- -u-s-i-n-g- -R-e-d-i-s-.-

-\*-\*F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*-\* -H-o-w- -d-o- -y-o-u- -m-a-n-a-g-e- -d-a-t-a-b-a-s-e- -m-i-g-r-a-t-i-o-n-s- -w-h-e-n-  
-m-o-d-i-f-y-i-n-g- -t-h-e- -s-c-h-e-m-a- -o-f- -c-o-m-p-a-n-i-e-s-?-

-\*-\*D-e-e-p- -D-i-v-e:-\*-\* -E-x-p-l-a-i-n- -t-h-e- -d-i-f-f-e-r-e-n-c-e- -b-e-t-w-e-e-n- -a- -i-n-d-e-x-e-s- -s-c-a-n- -a-n-d- -a-  
-s-e-q-u-e-n-t-i-a-l- -s-c-a-n- -u-n-d-e-r- -P-o-s-t-g-r-e-S-Q-L-.-

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-##-#- -Q-u-e-s-t-i-o-n- -1-3:- -D-B- -Q-u-e-s-t-i-o-n- -1-3:- -H-o-w- -w-o-u-l-d- -y-o-u- -o-p-t-i-m-i-z-e- -P-o-s-t-g-r-e-S-Q-L- -f-o-r-

-h-i-g-h---v-o-l-u-m-e- -h-i-s-t-o-r-i-c-a-l- -a-g-g-r-e-g-a-t-i-o-n-s- -i-n- -R-i-s-k-P-u-l-s-e-?-

-\*- \*-R-o-u-n-d:-\*- -B-a-c-k-e-n-d-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -I-n-d-e-x-e-s,- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w-s,- -W-i-n-d-o-w- -F-u-n-c-t-i-o-n-s,-  
-P-o-s-t-g-r-e-S-Q-L- -q-u-e-r-y- -p-l-a-n-n-e-r,-

-\*- \*-W-h-y- -I-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -d-a-t-a-b-a-s-e- -t-u-n-i-n-g- -a-n-d- -r-e-l-a-t-i-o-n-a-l- -d-e-s-i-g-n,-

-\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-o- -o-p-t-i-m-i-z-e- -r-o-l-l-i-n-g- -a-g-g-r-e-g-a-t-i-o-n-s- -(e.g.,-  
-c-u-m-u-l-a-t-i-v-e- -l-a-y-o-f-f-s,- -r-o-l-l-i-n-g- -9-0---d-a-y- -l-a-y-o-f-f- -c-o-u-n-t-s),- -w-e- -s-h-o-u-l-d- -a-v-o-i-d- -s-c-a-n-n-i-n-g-  
-t-h-e- -r-a-w- -l-a-y-o-f-f-s- -t-a-b-l-e- -r-e-p-e-a-t-e-d-l-y,- -l-n-s-t-e-a-d,- -c-r-e-a-t-e- -c-o-m-p-o-s-i-t-e- -i-n-d-e-x-e-s- -o-n-  
-(-c-o-m-p-a-n-y\_-i-d,- -l-a-y-o-f-f\_-d-a-t-e)- -o-r- -s-e-t- -u-p- -a- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w- -t-h-a-t- -r-e-f-r-e-s-h-e-s-  
-p-e-r-i-o-d-i-c-a-l-l-y- -t-o- -s-t-o-r-e- -p-r-e---a-g-g-r-e-g-a-t-e-d- -m-e-t-r-i-c-s,-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -R-e-l-y-i-n-g- -o-n- -O-R-M- -q-u-e-r-i-e-s- -t-h-a-t- -p-e-r-f-o-r-m- -N++1- -d-a-t-a-b-a-s-e-  
-h-i-t-s,-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -w-r-i-t-e- -r-a-w- -S-Q-L- -q-u-e-r-i-e-s- -u-s-i-n-g-  
-w-i-n-d-o-w- -f-u-n-c-t-i-o-n-s- (-`S-U-M(-) -O-V-E-R- -(P-A-R-T-I-T-I-O-N- -B-Y- -.-.- -O-R-D-E-R- -B-Y- -.-.-)-`),- -W-e- -s-e-t-  
-u-p- -d-a-t-a-b-a-s-e- -p-a-r-t-i-t-i-o-n-i-n-g- -o-n- -t-h-e- -l-a-y-o-f-f-s- -t-a-b-l-e- -b-y- -y-e-a-r,- -a-n-d- -c-a-c-h-e- -q-u-e-r-y-  
-r-e-s-u-l-t-s- -u-s-i-n-g- -R-e-d-i-s,-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -d-o- -y-o-u- -m-a-n-a-g-e- -d-a-t-a-b-a-s-e- -m-i-g-r-a-t-i-o-n-s- -w-h-e-n-  
-m-o-d-i-f-y-i-n-g- -t-h-e- -s-c-h-e-m-a- -o-f- -c-o-m-p-a-n-i-e-s-?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -t-h-e- -d-i-f-f-e-r-e-n-c-e- -b-e-t-w-e-e-n- -a- -i-n-d-e-x-e-s- -s-c-a-n- -a-n-d- -a-  
-s-e-q-u-e-n-t-i-a-l- -s-c-a-n- -u-n-d-e-r- -P-o-s-t-g-r-e-S-Q-L,-

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-###- -Q-u-e-s-t-i-o-n- -1-4:- -D-B- -Q-u-e-s-t-i-o-n- -1-4:- -H-o-w- -w-o-u-l-d- -y-o-u- -o-p-t-i-m-i-z-e- -P-o-s-t-g-r-e-S-Q-L- -f-o-r-  
-h-i-g-h---v-o-l-u-m-e- -h-i-s-t-o-r-i-c-a-l- -a-g-g-r-e-g-a-t-i-o-n-s- -i-n- -R-i-s-k-P-u-l-s-e-?-

-\*- \*-R-o-u-n-d:-\*- -B-a-c-k-e-n-d-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -I-n-d-e-x-e-s,- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w-s,- -W-i-n-d-o-w- -F-u-n-c-t-i-o-n-s,-  
-P-o-s-t-g-r-e-S-Q-L- -q-u-e-r-y- -p-l-a-n-n-e-r,-

-\*- \*-W-h-y- -I-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -d-a-t-a-b-a-s-e- -t-u-n-i-n-g- -a-n-d- -r-e-l-a-t-i-o-n-a-l- -d-e-s-i-g-n,-

-\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-o- -o-p-t-i-m-i-z-e- -r-o-l-l-i-n-g- -a-g-g-r-e-g-a-t-i-o-n-s- -(e.g.,-  
-c-u-m-u-l-a-t-i-v-e- -l-a-y-o-f-f-s,- -r-o-l-l-i-n-g- -9-0---d-a-y- -l-a-y-o-f-f- -c-o-u-n-t-s),- -w-e- -s-h-o-u-l-d- -a-v-o-i-d- -s-c-a-n-n-i-n-g-  
-t-h-e- -r-a-w- -l-a-y-o-f-f-s- -t-a-b-l-e- -r-e-p-e-a-t-e-d-l-y,- -l-n-s-t-e-a-d,- -c-r-e-a-t-e- -c-o-m-p-o-s-i-t-e- -i-n-d-e-x-e-s- -o-n-  
-(-c-o-m-p-a-n-y\_-i-d,- -l-a-y-o-f-f\_-d-a-t-e)- -o-r- -s-e-t- -u-p- -a- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w- -t-h-a-t- -r-e-f-r-e-s-h-e-s-  
-p-e-r-i-o-d-i-c-a-l-l-y- -t-o- -s-t-o-r-e- -p-r-e---a-g-g-r-e-g-a-t-e-d- -m-e-t-r-i-c-s,-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -R-e-l-y-i-n-g- -o-n- -O-R-M- -q-u-e-r-i-e-s- -t-h-a-t- -p-e-r-f-o-r-m- -N++1- -d-a-t-a-b-a-s-e-  
-h-i-t-s,-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -w-r-i-t-e- -r-a-w- -S-Q-L- -q-u-e-r-i-e-s- -u-s-i-n-g-  
-w-i-n-d-o-w- -f-u-n-c-t-i-o-n-s- (-`S-U-M(-) -O-V-E-R- -(P-A-R-T-I-T-I-O-N- -B-Y- -.-.- -O-R-D-E-R- -B-Y- -.-.-)-`),- -W-e- -s-e-t-  
-u-p- -d-a-t-a-b-a-s-e- -p-a-r-t-i-t-i-o-n-i-n-g- -o-n- -t-h-e- -l-a-y-o-f-f-s- -t-a-b-l-e- -b-y- -y-e-a-r,- -a-n-d- -c-a-c-h-e- -q-u-e-r-y-  
-r-e-s-u-l-t-s- -u-s-i-n-g- -R-e-d-i-s,-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -d-o- -y-o-u- -m-a-n-a-g-e- -d-a-t-a-b-a-s-e- -m-i-g-r-a-t-i-o-n-s- -w-h-e-n-

-m-o-d-i-f-y-i-n-g- t-h-e- s-c-h-e-m-a- o-f- c-o-m-p-a-n-i-e-s-?

\*- \*-D-e-e-p- D-i-v-e-;-\*- -E-x-p-l-a-i-n- t-h-e- d-i-f-f-e-r-e-n-c-e- b-e-t-w-e-e-n- a- i-n-d-e-x-e-s- s-c-a-n- a-n-d- a-s-e-q-u-e-n-t-i-a-l- s-c-a-n- u-n-d-e-r- P-o-s-t-g-r-e-S-Q-L-.-

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-##-#- Q-u-e-s-t-i-o-n- 1-5-:- D-B- Q-u-e-s-t-i-o-n- 1-5-:- H-o-w- w-o-u-l-d- y-o-u- o-p-t-i-m-i-z-e- P-o-s-t-g-r-e-S-Q-L- f-o-r- h-i-g-h---v-o-l-u-m-e- h-i-s-t-o-r-i-c-a-l- a-g-g-r-e-g-a-t-i-o-n-s- i-n- R-i-s-k-P-u-l-s-e-?

\*- \*-R-o-u-n-d-;-\*- -B-a-c-k-e-n-d-

\*- \*-D-i-f-f-i-c-u-l-t-y-;-\*- -M-e-d-i-u-m- -|- \*-\*-l-m-p-o-r-t-a-n-c-e-;-\*- -8-/1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d-;-\*- -l-n-d-e-x-e-s-,- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w-s-,- -W-i-n-d-o-w- -F-u-n-c-t-i-o-n-s-,- -P-o-s-t-g-r-e-S-Q-L- -q-u-e-r-y- -p-l-a-n-n-e-r-.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- l-t-;-\*- -T-e-s-t-s- -d-a-t-a-b-a-s-e- t-u-n-i-n-g- -a-n-d- -r-e-l-a-t-i-o-n-a-l- -d-e-s-i-g-n-.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-;-\*- -T-o- -o-p-t-i-m-i-z-e- -r-o-l-l-i-n-g- -a-g-g-r-e-g-a-t-i-o-n-s- -(-e-.g.-.-,- -c-u-m-u-l-a-t-i-v-e- -l-a-y-o-f-f-s-,- -r-o-l-l-i-n-g- -9-0---d-a-y- -l-a-y-o-f-f- -c-o-u-n-t-s-),- -w-e- -s-h-o-u-l-d- -a-v-o-i-d- -s-c-a-n-n-i-n-g- -t-h-e- -r-a-w- -`-l-a-y-o-f-f-s-`- -t-a-b-l-e- -r-e-p-e-a-t-e-d-l-y-.- -l-n-s-t-e-a-d-,- -c-r-e-a-t-e- -c-o-m-p-o-s-i-t-e- i-n-d-e-x-e-s- -o-n- -`-(c-o-m-p-a-n-y\_-i-d-,- -l-a-y-o-f-f\_-d-a-t-e)-`- -o-r- -s-e-t- -u-p- -a- -M-a-t-e-r-i-a-l-i-z-e-d- -V-i-e-w- -t-h-a-t- -r-e-f-r-e-s-h-e-s- -p-e-r-i-o-d-i-c-a-l-l-y- -t-o- -s-t-o-r-e- -p-r-e---a-g-g-r-e-g-a-t-e-d- -m-e-t-r-i-c-s-.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-;-\*- -R-e-l-y-i-n-g- -o-n- -O-R-M- -q-u-e-r-i-e-s- -t-h-a-t- -p-e-r-f-o-r-m- -N-t-1- -d-a-t-a-b-a-s-e- -h-i-t-s-.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-;-\*- -W-e- -w-r-i-t-e- -r-a-w- -S-Q-L- -q-u-e-r-i-e-s- -u-s-i-n-g- -w-i-n-d-o-w- -f-u-n-c-t-i-o-n-s- -(`-S-U-M(-) -O-V-E-R- -(P-A-R-T-I-T-I-O-N- -B-Y- -.-.- -O-R-D-E-R- -B-Y- -.-.-)-`)-.- -W-e- -s-e-t- -u-p- -d-a-t-a-b-a-s-e- -p-a-r-t-i-t-i-o-n-i-n-g- -o-n- -t-h-e- -`-l-a-y-o-f-f-s-`- -t-a-b-l-e- -b-y- -y-e-a-r-,- -a-n-d- -c-a-c-h-e- -q-u-e-r-y- -r-e-s-u-l-t-s- -u-s-i-n-g- -R-e-d-i-s-.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s-;-\*- -H-o-w- -d-o- y-o-u- -m-a-n-a-g-e- -d-a-t-a-b-a-s-e- -m-i-g-r-a-t-i-o-n-s- -w-h-e-n- -m-o-d-i-f-y-i-n-g- t-h-e- s-c-h-e-m-a- o-f- c-o-m-p-a-n-i-e-s-?

\*- \*-D-e-e-p- D-i-v-e-;-\*- -E-x-p-l-a-i-n- t-h-e- d-i-f-f-e-r-e-n-c-e- b-e-t-w-e-e-n- a- i-n-d-e-x-e-s- s-c-a-n- a-n-d- a-s-e-q-u-e-n-t-i-a-l- s-c-a-n- u-n-d-e-r- P-o-s-t-g-r-e-S-Q-L-.-

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-##-#- Q-u-e-s-t-i-o-n- 1-6-:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- 1-6-:- H-o-w- -d-o- y-o-u- -p-r-e-v-e-n-t- -t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -f-e-a-t-u-r-e-?

\*- \*-R-o-u-n-d-;-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y-;-\*- -H-a-r-d- -|- \*-\*-l-m-p-o-r-t-a-n-c-e-;-\*- -9-/1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d-;-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e-,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g-,- -d-a-t-a-i-m-p-u-t-a-t-i-o-n-.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- l-t-;-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n- -t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a-.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-;-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e-.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y- -c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y-.- -F-o-r- -t-r-a-i-n-i-n-g-,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s-

-b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t-, -n-o-t- -t-h-e- -f-u-t-u-r-e- -o-n-e-s.-

-\* -\*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e- -e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

-\* -\*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e.- -E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p-, -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d- -u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s- -t-i-m-e-s-t-a-m-p.-

-\* -\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f'- -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s?-

-\* -\*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'l-n-d-u-s-t-r-y'- -c-a-t-e-g-o-r-y-, -h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g?-

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-###- -Q-u-e-s-t-i-o-n- -1-7:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- -1-7:- -H-o-w- -d-o- -y-o-u- -p-r-e-v-e-n-t- -t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f'- -f-e-a-t-u-r-e?-

-\* -\*-R-o-u-n-d:-\*- -M-L-

-\* -\*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -| -\*- -l-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-

-\* -\*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e-, -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g-, -d-a-t-a- -i-m-p-u-t-a-t-i-o-n.-

-\* -\*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n- -t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a.-

-\* -\*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f'- -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y- -c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g-, -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s- -b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t-, -n-o-t- -t-h-e- -f-u-t-u-r-e- -o-n-e-s.-

-\* -\*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e- -e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

-\* -\*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e.- -E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p-, -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d- -u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s- -t-i-m-e-s-t-a-m-p.-

-\* -\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f'- -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s?-

-\* -\*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'l-n-d-u-s-t-r-y'- -c-a-t-e-g-o-r-y-, -h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g?-

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-###- -Q-u-e-s-t-i-o-n- -1-8:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- -1-8:- -H-o-w- -d-o- -y-o-u- -p-r-e-v-e-n-t- -t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f'- -f-e-a-t-u-r-e?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- |- \*-I-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g,- -d-a-t-a-i-m-p-u-t-a-t-i-o-n.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n-t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-' -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y- -c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s- -b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t,- -n-o-t- -t-h-e- -f-u-t-u-r-e- -o-n-e-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e- -e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e.- -E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p,- -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d- -u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s- -t-i-m-e-s-t-a-m-p.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-' -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'l-n-d-u-s-t-r-y-' -c-a-t-e-g-o-r-y,- -h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g?-

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-#-#-#- -Q-u-e-s-t-i-o-n- -1-9:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- -1-9:- -H-o-w- -d-o- -y-o-u- -p-r-e-v-e-n-t- -t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-' -f-e-a-t-u-r-e?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- |- \*-I-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g,- -d-a-t-a-i-m-p-u-t-a-t-i-o-n.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n-t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-' -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y- -c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s- -b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t,- -n-o-t- -t-h-e- -f-u-t-u-r-e- -o-n-e-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e- -e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e.- -E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p,- -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d- -u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s- -t-i-m-e-s-t-a-m-p.-

Q: What value do you assign to Data Science last year for company with no prior layoffs?

A: Dependent on how the industry categorizes how does it affect the trend based on models compared to target engagement?

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### Question 20: Feature Engineering Question 20: How do you prevent target leakage when engineering the Data Science last year for company with no prior layoffs?

A: Round off ML

A: Difficult to harden the model - 9/10

A: Concise Tested Target leakage, chronological sorting, data imputation.

A: Why interviewers ask this? Tests understanding of data leakage in time-series and tabular data.

A: Ideal FANG Answer: Target leakage happens if we compute Data Science last year for company with no prior layoffs using information from the future. We must strictly compute this feature chronologically. For training, it must be the data before the current event and the immedately preceding event, not the future ones.

A: Common Mistakes: Calculating rolling averages or intervals over the entire dataset before splitting into train/test.

A: Better Prediction Grade Answer: We use a point-in-time feature store. Every training sample is indexed by a time stamp, and features are computed using only records that have a time stamp strictly less than the sample's time stamp.

Q: What value do you assign to Data Science last year for company with no prior layoffs?

A: Dependent on how the industry categorizes how does it affect the trend based on models compared to target engagement?

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### Question 21: Feature Engineering Question 21: How do you prevent target leakage when engineering the Data Science last year for company with no prior layoffs?

A: Round off ML

A: Difficult to harden the model - 9/10

A: Concise Tested Target leakage, chronological sorting, data imputation.

A: Why interviewers ask this? Tests understanding of data leakage in time-series and tabular data.

A: Ideal FANG Answer: Target leakage happens if we compute Data Science last year for company with no prior layoffs using information from the future.

-S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f'- -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y-  
-c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s-  
-b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t,- -n-o-t- -t-h-e-  
-f-u-t-u-r-e- -o-n-e-s.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e-  
-e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e.-  
-E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p,- -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d-  
-u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s-  
-t-i-m-e-s-t-a-m-p.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t-  
-L-a-y-o-f-f'- -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'l-n-d-u-s-t-r-y'- -c-a-t-e-g-o-r-y,-  
-h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g?-  
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-#-#- -Q-u-e-s-t-i-o-n- -2-2:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- -2-2:- -H-o-w- -d-o- -y-o-u- -p-r-e-v-e-n-t-  
-t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f'- -f-e-a-t-u-r-e?-  
-\*- \*-R-o-u-n-d:-\*- -M-L-

-\*- \*-R-o-u-n-d:-\*- -M-L-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*-\*-l-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g,- -d-a-t-a-  
-i-m-p-u-t-a-t-i-o-n.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n-  
-t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'D-a-y-s-  
-S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f'- -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y-  
-c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s-  
-b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t,- -n-o-t- -t-h-e-  
-f-u-t-u-r-e- -o-n-e-s.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e-  
-e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e.-  
-E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p,- -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d-  
-u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s-  
-t-i-m-e-s-t-a-m-p.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t-  
-L-a-y-o-f-f'- -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'l-n-d-u-s-t-r-y'- -c-a-t-e-g-o-r-y,-  
-h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g?-  
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### -Q-u-e-s-t-i-o-n- 2-3:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- 2-3:- -H-o-w- -d-o- -y-o-u- -p-r-e-v-e-n-t-  
-t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f'- -f-e-a-t-u-r-e-?

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -| -\*- -l-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g,- -d-a-t-a-  
-i-m-p-u-t-a-t-i-o-n.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n-  
-t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'D-a-y-s-  
-S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f'- -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y-  
-c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s-  
-b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t,- -n-o-t- -t-h-e-  
-f-u-t-u-r-e- -o-n-e-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e-  
-e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e.-  
-E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p,- -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d-  
-u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s-  
-t-i-m-e-s-t-a-m-p.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t-  
-L-a-y-o-f-f'- -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s-?

\*- \*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'l-n-d-u-s-t-r-y'- -c-a-t-e-g-o-r-y,-  
-h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g-?

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### -Q-u-e-s-t-i-o-n- 2-4:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- 2-4:- -H-o-w- -d-o- -y-o-u- -p-r-e-v-e-n-t-  
-t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f'- -f-e-a-t-u-r-e-?

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -| -\*- -l-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g,- -d-a-t-a-  
-i-m-p-u-t-a-t-i-o-n.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n-  
-t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'D-a-y-s-  
-S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f'- -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y-  
-c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s-  
-b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t,- -n-o-t- -t-h-e-  
-f-u-t-u-r-e- -o-n-e-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e-  
-e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e.-  
-E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p,- -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d-

-u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s-  
-t-i-m-e-s-t-a-m-p-.-

-\* -\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t-  
-L-a-y-o-f-f-'- -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s-?-

-\* -\*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'l-n-d-u-s-t-r-y-'- -c-a-t-e-g-o-r-y-,-  
-h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g-?-  
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-##-#- -Q-u-e-s-t-i-o-n- -2-5:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- -2-5:- -H-o-w- -d-o- -y-o-u- -p-r-e-v-e-n-t-  
-t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -f-e-a-t-u-r-e-?-  
-\* -\*-R-o-u-n-d:-\*- -M-L-

-\* -\*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -| -\*- -l-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-  
-\* -\*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g,- -d-a-t-a-  
-i-m-p-u-t-a-t-i-o-n.-

-\* -\*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n-  
-t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a.-

-\* -\*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'D-a-y-s-  
-S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y-  
-c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s-  
-b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t,- -n-o-t- -t-h-e-  
-f-u-t-u-r-e- -o-n-e-s.-

-\* -\*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e-  
-e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

-\* -\*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e.-  
-E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p,- -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d-  
-u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s-  
-t-i-m-e-s-t-a-m-p.-

-\* -\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t-  
-L-a-y-o-f-f-'- -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s-?-  
-\* -\*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'l-n-d-u-s-t-r-y-'- -c-a-t-e-g-o-r-y-,-  
-h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g-?-  
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-##-#- -Q-u-e-s-t-i-o-n- -2-6:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- -2-6:- -H-o-w- -d-o- -y-o-u- -p-r-e-v-e-n-t-  
-t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -f-e-a-t-u-r-e-?-  
-\* -\*-R-o-u-n-d:-\*- -M-L-

-\* -\*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -| -\*- -l-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-  
-\* -\*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g,- -d-a-t-a-  
-i-m-p-u-t-a-t-i-o-n.-

-\* -\*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n-

-t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a.-

-\* -\*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'-D-a-y-s-  
-S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y-  
-c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s-  
-b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t,- -n-o-t- -t-h-e-  
-f-u-t-u-r-e- -o-n-e-s.-

-\* -\*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e-  
-e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

-\* -\*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e.-  
-E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p,- -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d-  
-u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s-  
-t-i-m-e-s-t-a-m-p.-

-\* -\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'-D-a-y-s- -S-i-n-c-e- -L-a-s-t-  
-L-a-y-o-f-f-'- -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s?-

-\* -\*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'-l-n-d-u-s-t-r-y-'- -c-a-t-e-g-o-r-y,-  
-h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g?-  
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##- -Q-u-e-s-t-i-o-n- -2-7:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- -2-7:- -H-o-w- -d-o- -y-o-u- -p-r-e-v-e-n-t-  
-t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'-D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -f-e-a-t-u-r-e?-  
-\* -\*-R-o-u-n-d:-\*- -M-L-

-\* -\*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- -\*-I-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-

-\* -\*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g,- -d-a-t-a-  
-i-m-p-u-t-a-t-i-o-n.-

-\* -\*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n-  
-t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a.-

-\* -\*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'-D-a-y-s-  
-S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y-  
-c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s-  
-b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t,- -n-o-t- -t-h-e-  
-f-u-t-u-r-e- -o-n-e-s.-

-\* -\*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e-  
-e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

-\* -\*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e.-  
-E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p,- -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d-  
-u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s-  
-t-i-m-e-s-t-a-m-p.-

-\* -\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'-D-a-y-s- -S-i-n-c-e- -L-a-s-t-  
-L-a-y-o-f-f-'- -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s?-

-\* -\*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'-l-n-d-u-s-t-r-y-'- -c-a-t-e-g-o-r-y,-  
-h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g?-  
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### -Q-u-e-s-t-i-o-n- 2-8:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- 2-8:- -H-o-w- -d-o- -y-o-u- -p-r-e-v-e-n-t-  
-t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'-D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -f-e-a-t-u-r-e-?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g,- -d-a-t-a-  
-i-m-p-u-t-a-t-i-o-n.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n-  
-t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'-D-a-y-s-  
-S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y-  
-c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s-  
-b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t,- -n-o-t- -t-h-e-  
-f-u-t-u-r-e- -o-n-e-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e-  
-e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e.-  
-E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p,- -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d-  
-u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s-  
-t-i-m-e-s-t-a-m-p.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'-D-a-y-s- -S-i-n-c-e- -L-a-s-t-  
-L-a-y-o-f-f-'- -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s-?

\*- \*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'-l-n-d-u-s-t-r-y-'- -c-a-t-e-g-o-r-y,-  
-h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g-?

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### -Q-u-e-s-t-i-o-n- 2-9:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- 2-9:- -H-o-w- -d-o- -y-o-u- -p-r-e-v-e-n-t-  
-t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'-D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -f-e-a-t-u-r-e-?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g,- -d-a-t-a-  
-i-m-p-u-t-a-t-i-o-n.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n-  
-t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'-D-a-y-s-  
-S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y-  
-c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s-  
-b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t,- -n-o-t- -t-h-e-  
-f-u-t-u-r-e- -o-n-e-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e-  
-e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e-,-  
-E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p,- -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d-  
-u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s-  
-t-i-m-e-s-t-a-m-p.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t-  
-L-a-y-o-f-f-'- -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'l-n-d-u-s-t-r-y-'- -c-a-t-e-g-o-r-y,-  
-h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g?-  
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###- -Q-u-e-s-t-i-o-n- -3-0:- -F-e-a-t-u-r-e- -E-n-g-i-n-e-e-r-i-n-g- -Q-u-e-s-t-i-o-n- -3-0:- -H-o-w- -d-o- -y-o-u- -p-r-e-v-e-n-t-  
-t-a-r-g-e-t- -l-e-a-k-a-g-e- -w-h-e-n- -e-n-g-i-n-e-e-r-i-n-g- -t-h-e- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -f-e-a-t-u-r-e?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-o-r-t-i-n-g,- -d-a-t-a-  
-i-m-p-u-t-a-t-i-o-n.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -d-a-t-a- -l-e-a-k-a-g-e- -i-n-  
-t-i-m-e---s-e-r-i-e-s- -a-n-d- -t-a-b-u-l-a-r- -d-a-t-a.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -T-a-r-g-e-t- -l-e-a-k-a-g-e- -h-a-p-p-e-n-s- -i-f- -w-e- -c-o-m-p-u-t-e- -'D-a-y-s-  
-S-i-n-c-e- -L-a-s-t- -L-a-y-o-f-f-'- -u-s-i-n-g- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -t-h-e- -f-u-t-u-r-e.- -W-e- -m-u-s-t- -s-t-r-i-c-t-l-y-  
-c-o-m-p-u-t-e- -t-h-i-s- -f-e-a-t-u-r-e- -c-h-r-o-n-o-l-o-g-i-c-a-l-l-y.- -F-o-r- -t-r-a-i-n-i-n-g,- -i-t- -m-u-s-t- -b-e- -t-h-e- -d-a-y-s-  
-b-e-t-w-e-e-n- -t-h-e- -c-u-r-r-e-n-t- -e-v-e-n-t- -a-n-d- -t-h-e- -i-m-m-e-d-i-a-t-e-l-y- -p-r-e-c-e-d-i-n-g- -e-v-e-n-t,- -n-o-t- -t-h-e-  
-f-u-t-u-r-e- -o-n-e-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -C-a-l-c-u-l-a-t-i-n-g- -r-o-l-l-i-n-g- -a-v-e-r-a-g-e-s- -o-r- -i-n-t-e-r-v-a-l-s- -o-v-e-r- -t-h-e-  
-e-n-t-i-r-e- -d-a-t-a-s-e-t- -b-e-f-o-r-e- -s-p-l-i-t-t-i-n-g- -i-n-t-o- -t-r-a-i-n-/t-e-s-t.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- -W-e- -u-s-e- -a- -p-o-i-n-t---i-n---t-i-m-e- -f-e-a-t-u-r-e- -s-t-o-r-e-,-  
-E-v-e-r-y- -t-r-a-i-n-i-n-g- -s-a-m-p-l-e- -i-s- -i-n-d-e-x-e-d- -b-y- -a- -t-i-m-e-s-t-a-m-p,- -a-n-d- -f-e-a-t-u-r-e-s- -a-r-e- -c-o-m-p-u-t-e-d-  
-u-s-i-n-g- -o-n-l-y- -r-e-c-o-r-d-s- -t-h-a-t- -h-a-v-e- -a- -t-i-m-e-s-t-a-m-p- -s-t-r-i-c-t-l-y- -l-e-s-s- -t-h-a-n- -t-h-e- -s-a-m-p-l-e-'s-  
-t-i-m-e-s-t-a-m-p.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -v-a-l-u-e- -d-o- -y-o-u- -a-s-s-i-g-n- -t-o- -'D-a-y-s- -S-i-n-c-e- -L-a-s-t-  
-L-a-y-o-f-f-'- -f-o-r- -a- -c-o-m-p-a-n-y- -w-i-t-h- -n-o- -p-r-i-o-r- -l-a-y-o-f-f-s?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -l-f- -y-o-u- -u-s-e- -o-n-e---h-o-t- -e-n-c-o-d-i-n-g- -f-o-r- -t-h-e- -'l-n-d-u-s-t-r-y-'- -c-a-t-e-g-o-r-y,-  
-h-o-w- -d-o-e-s- -i-t- -a-f-f-e-c-t- -T-r-e-e---b-a-s-e-d- -m-o-d-e-l-s- -c-o-m-p-a-r-e-d- -t-o- -t-a-r-g-e-t- -e-n-c-o-d-i-n-g?-  
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###- -Q-u-e-s-t-i-o-n- -3-1:- -M-o-d-e-l- -S-e-l-e-c-t-i-o-n- -Q-u-e-s-t-i-o-n- -3-1:- -W-h-y- -d-i-d- -y-o-u- -c-h-o-o-s-e- -a-  
-S-t-a-c-k-i-n-g- -E-n-s-e-m-b-l-e- -o-v-e-r- -a- -s-i-n-g-l-e- -t-u-n-e-d- -L-i-g-h-t-G-B-M- -m-o-d-e-l?-  
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\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -E-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g,- -s-t-a-c-k-i-n-g,- -b-i-a-s---v-a-r-i-a-n-c-e- -t-r-a-d-e-o-f-f.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- -T-e-s-t-s- u-n-d-e-r-s-t-a-n-d-i-n-g- o-f- e-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g- a-n-d- v-a-r-i-a-n-c-e- r-e-d-u-c-t-i-o-n-.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- -S-t-a-c-k-i-n-g- c-o-m-b-i-n-e-s- m-o-d-e-l-s- w-i-t-h- d-i-f-f-e-r-e-n-t- i-n-d-u-c-t-i-v-e- b-i-a-s-e-s- -(e-.g.- R-a-n-d-o-m- F-o-r-e-s-t,- L-i-g-h-t-G-B-M,- E-x-t-r-a-T-r-e-e-s-)-. L-i-g-h-t-G-B-M- f-o-c-u-s-e-s- o-n- g-r-a-d-i-e-n-t- b-o-o-s-t-i-n-g- o-n- t-r-e-e-s,- w-h-i-l-e- R-a-n-d-o-m- F-o-r-e-s-t- u-s-e-s- b-a-g-g-i-n-g.- S-t-a-c-k-i-n-g- t-r-a-i-n-s- a- m-e-t-a--l-e-a-r-n-e-r- -(e-.g.- L-o-g-i-s-t-i-c- R-e-g-r-e-s-s-i-o-n)- o-n- t-h-e-i-r- o-u-t--o-f--f-o-l-d- p-r-e-d-i-c-t-i-o-n-s,- w-h-i-c-h- h-e-l-p-s- s-m-o-o-t-h- o-u-t- i-n-d-i-v-i-d-u-a-l- e-r-r-o-r-s- a-n-d- l-o-w-e-r-s- v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- -T-h-i-n-k-i-n-g- S-t-a-c-k-i-n-g- a-l-w-a-y-s- i-m-p-r-o-v-e-s- p-e-r-f-o-r-m-a-n-c-e- w-i-t-h-o-u-t- n-o-t-i-n-g- t-h-e- c-o-s-t- i-n- c-o-m-p-l-e-x-i-t-y- a-n-d- t-r-a-i-n-i-n-g- t-i-m-e.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n--G-r-a-d-e- A-n-s-w-e-r:-\*- -W-e- e-v-a-l-u-a-t-e- S-t-a-c-k-i-n-g- v-e-r-s-u-s- L-i-g-h-t-G-B-M.- S-t-a-c-k-i-n-g- y-i-e-l-d-s- a- +-1-.5%- R-O-C--A-U-C- g-a-i-n.- l-n- p-r-o-d-u-c-t-i-o-n,- w-e- e-v-a-l-u-a-t-e- i-f- t-h-i-s- g-a-i-n- o-u-t-w-e-i-g-h-s- t-h-e- e-x-t-r-a- i-n-f-e-r-e-n-c-e- l-a-t-e-n-c-y- o-f- r-u-n-n-i-n-g- m-u-l-t-i-p-l-e- b-a-s-e- e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- m-o-d-e-l- d-i-d- y-o-u- u-s-e- a-s- t-h-e- m-e-t-a--l-e-a-r-n-e-r- a-n-d- w-h-y?-

\*- \*-D-e-e-p- D-i-v-e:-\*- -E-x-p-l-a-i-n- h-o-w- m-u-l-t-i---s-t-a-g-e- s-t-a-c-k-i-n-g- c-a-n- l-e-a-d- t-o- t-a-r-g-e-t- l-e-a-k-a-g-e- i-f- c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- i-s- n-o-t- c-o-n-f-i-g-u-r-e-d- p-r-o-p-e-r-l-y.-

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###- Q-u-e-s-t-i-o-n- 3-2:- M-o-d-e-l- S-e-l-e-c-t-i-o-n- Q-u-e-s-t-i-o-n- 3-2:- W-h-y- d-i-d- y-o-u- c-h-o-o-s-e- a- S-t-a-c-k-i-n-g- E-n-s-e-m-b-l-e- o-v-e-r- a- s-i-n-g-l-e- t-u-n-e-d- L-i-g-h-t-G-B-M- m-o-d-e-l?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- |- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- -E-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g,- s-t-a-c-k-i-n-g,- b-i-a-s--v-a-r-i-a-n-c-e- t-r-a-d-e-o-f-f.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- -T-e-s-t-s- u-n-d-e-r-s-t-a-n-d-i-n-g- o-f- e-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g- a-n-d- v-a-r-i-a-n-c-e- r-e-d-u-c-t-i-o-n-.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- -S-t-a-c-k-i-n-g- c-o-m-b-i-n-e-s- m-o-d-e-l-s- w-i-t-h- d-i-f-f-e-r-e-n-t- i-n-d-u-c-t-i-v-e- b-i-a-s-e-s- -(e-.g.- R-a-n-d-o-m- F-o-r-e-s-t,- L-i-g-h-t-G-B-M,- E-x-t-r-a-T-r-e-e-s-)-. L-i-g-h-t-G-B-M- f-o-c-u-s-e-s- o-n- g-r-a-d-i-e-n-t- b-o-o-s-t-i-n-g- o-n- t-r-e-e-s,- w-h-i-l-e- R-a-n-d-o-m- F-o-r-e-s-t- u-s-e-s- b-a-g-g-i-n-g.- S-t-a-c-k-i-n-g- t-r-a-i-n-s- a- m-e-t-a--l-e-a-r-n-e-r- -(e-.g.- L-o-g-i-s-t-i-c- R-e-g-r-e-s-s-i-o-n)- o-n- t-h-e-i-r- o-u-t--o-f--f-o-l-d- p-r-e-d-i-c-t-i-o-n-s,- w-h-i-c-h- h-e-l-p-s- s-m-o-o-t-h- o-u-t- i-n-d-i-v-i-d-u-a-l- e-r-r-o-r-s- a-n-d- l-o-w-e-r-s- v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- -T-h-i-n-k-i-n-g- S-t-a-c-k-i-n-g- a-l-w-a-y-s- i-m-p-r-o-v-e-s- p-e-r-f-o-r-m-a-n-c-e- w-i-t-h-o-u-t- n-o-t-i-n-g- t-h-e- c-o-s-t- i-n- c-o-m-p-l-e-x-i-t-y- a-n-d- t-r-a-i-n-i-n-g- t-i-m-e.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n--G-r-a-d-e- A-n-s-w-e-r:-\*- -W-e- e-v-a-l-u-a-t-e- S-t-a-c-k-i-n-g- v-e-r-s-u-s- L-i-g-h-t-G-B-M.- S-t-a-c-k-i-n-g- y-i-e-l-d-s- a- +-1-.5%- R-O-C--A-U-C- g-a-i-n.- l-n- p-r-o-d-u-c-t-i-o-n,- w-e- e-v-a-l-u-a-t-e- i-f- t-h-i-s- g-a-i-n- o-u-t-w-e-i-g-h-s- t-h-e- e-x-t-r-a- i-n-f-e-r-e-n-c-e- l-a-t-e-n-c-y- o-f- r-u-n-n-i-n-g- m-u-l-t-i-p-l-e- b-a-s-e- e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- m-o-d-e-l- d-i-d- y-o-u- u-s-e- a-s- t-h-e- m-e-t-a--l-e-a-r-n-e-r- a-n-d- w-h-y?-

\*- \*-D-e-e-p- D-i-v-e:-\*- -E-x-p-l-a-i-n- h-o-w- m-u-l-t-i---s-t-a-g-e- s-t-a-c-k-i-n-g- c-a-n- l-e-a-d- t-o- t-a-r-g-e-t- l-e-a-k-a-g-e- i-f- c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- i-s- n-o-t- c-o-n-f-i-g-u-r-e-d- p-r-o-p-e-r-l-y.-

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##- Q-u-e-s-t-i-o-n- 3-3:- M-o-d-e-l- S-e-l-e-c-t-i-o-n- Q-u-e-s-t-i-o-n- 3-3:- W-h-y- d-i-d- y-o-u- c-h-o-o-s-e- a-  
S-t-a-c-k-i-n-g- E-n-s-e-m-b-l-e- o-v-e-r- a- s-i-n-g-l-e- t-u-n-e-d- L-i-g-h-t-G-B-M- m-o-d-e-l-?

\*- \*-R-o-u-n-d:-\*- M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- |- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8/-1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- E-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g-, -s-t-a-c-k-i-n-g-, -b-i-a-s---v-a-r-i-a-n-c-e- t-r-a-d-e-o-f-f.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- T-e-s-t-s- u-n-d-e-r-s-t-a-n-d-i-n-g- o-f- e-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g- a-n-d-  
v-a-r-i-a-n-c-e- r-e-d-u-c-t-i-o-n.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- S-t-a-c-k-i-n-g- c-o-m-b-i-n-e-s- m-o-d-e-l-s- w-i-t-h- d-i-f-f-e-r-e-n-t-  
i-n-d-u-c-t-i-v-e- b-i-a-s-e-s- -(e.g.- R-a-n-d-o-m- F-o-r-e-s-t-, L-i-g-h-t-G-B-M-, E-x-t-r-a-T-r-e-e-s)-. L-i-g-h-t-G-B-M-  
f-o-c-u-s-e-s- o-n- g-r-a-d-i-e-n-t- b-o-o-s-t-i-n-g- o-n- t-r-e-e-s-, w-h-i-l-e- R-a-n-d-o-m- F-o-r-e-s-t- u-s-e-s- b-a-g-g-i-n-g.-  
S-t-a-c-k-i-n-g- t-r-a-i-n-s- a- m-e-t-a---l-e-a-r-n-e-r- -(e.g.- L-o-g-i-s-t-i-c- R-e-g-r-e-s-s-i-o-n)- o-n- t-h-e-i-r-  
o-u-t---o-f---f-o-l-d- p-r-e-d-i-c-t-i-o-n-s-, w-h-i-c-h- h-e-l-p-s- s-m-o-o-t-h- o-u-t- i-n-d-i-v-i-d-u-a-l- e-r-r-o-r-s- a-n-d- l-o-w-e-r-s-  
v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- T-h-i-n-k-i-n-g- S-t-a-c-k-i-n-g- a-l-w-a-y-s- i-m-p-r-o-v-e-s- p-e-r-f-o-r-m-a-n-c-e-  
w-i-t-h-o-u-t- n-o-t-i-n-g- t-h-e- c-o-s-t- i-n- c-o-m-p-l-e-x-i-t-y- a-n-d- t-r-a-i-n-i-n-g- t-i-m-e.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- e-v-a-l-u-a-t-e- S-t-a-c-k-i-n-g- v-e-r-s-u-s-  
L-i-g-h-t-G-B-M.- S-t-a-c-k-i-n-g- y-i-e-l-d-s- a- +1-.5%- R-O-C---A-U-C- g-a-i-n-. l-n- p-r-o-d-u-c-t-i-o-n-, w-e-  
e-v-a-l-u-a-t-e- i-f- t-h-i-s- g-a-i-n- o-u-t-w-e-i-g-h-s- t-h-e- e-x-t-r-a- i-n-f-e-r-e-n-c-e- l-a-t-e-n-c-y- o-f- r-u-n-n-i-n-g-  
m-u-l-t-i-p-l-e- b-a-s-e- e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- W-h-a-t- m-o-d-e-l- d-i-d- y-o-u- u-s-e- a-s- t-h-e- m-e-t-a---l-e-a-r-n-e-r- a-n-d-  
w-h-y-?

\*- \*-D-e-e-p- D-i-v-e:-\*- E-x-p-l-a-i-n- h-o-w- m-u-l-t-i---s-t-a-g-e- s-t-a-c-k-i-n-g- c-a-n- l-e-a-d- t-o- t-a-r-g-e-t-  
l-e-a-k-a-g-e- i-f- c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- i-s- n-o-t- c-o-n-f-i-g-u-r-e-d- p-r-o-p-e-r-l-y.-

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##- Q-u-e-s-t-i-o-n- 3-4:- M-o-d-e-l- S-e-l-e-c-t-i-o-n- Q-u-e-s-t-i-o-n- 3-4:- W-h-y- d-i-d- y-o-u- c-h-o-o-s-e- a-  
S-t-a-c-k-i-n-g- E-n-s-e-m-b-l-e- o-v-e-r- a- s-i-n-g-l-e- t-u-n-e-d- L-i-g-h-t-G-B-M- m-o-d-e-l-?

\*- \*-R-o-u-n-d:-\*- M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- |- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8/-1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- E-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g-, -s-t-a-c-k-i-n-g-, -b-i-a-s---v-a-r-i-a-n-c-e- t-r-a-d-e-o-f-f.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- T-e-s-t-s- u-n-d-e-r-s-t-a-n-d-i-n-g- o-f- e-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g- a-n-d-  
v-a-r-i-a-n-c-e- r-e-d-u-c-t-i-o-n.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- S-t-a-c-k-i-n-g- c-o-m-b-i-n-e-s- m-o-d-e-l-s- w-i-t-h- d-i-f-f-e-r-e-n-t-  
i-n-d-u-c-t-i-v-e- b-i-a-s-e-s- -(e.g.- R-a-n-d-o-m- F-o-r-e-s-t-, L-i-g-h-t-G-B-M-, E-x-t-r-a-T-r-e-e-s)-. L-i-g-h-t-G-B-M-  
f-o-c-u-s-e-s- o-n- g-r-a-d-i-e-n-t- b-o-o-s-t-i-n-g- o-n- t-r-e-e-s-, w-h-i-l-e- R-a-n-d-o-m- F-o-r-e-s-t- u-s-e-s- b-a-g-g-i-n-g.-  
S-t-a-c-k-i-n-g- t-r-a-i-n-s- a- m-e-t-a---l-e-a-r-n-e-r- -(e.g.- L-o-g-i-s-t-i-c- R-e-g-r-e-s-s-i-o-n)- o-n- t-h-e-i-r-  
o-u-t---o-f---f-o-l-d- p-r-e-d-i-c-t-i-o-n-s-, w-h-i-c-h- h-e-l-p-s- s-m-o-o-t-h- o-u-t- i-n-d-i-v-i-d-u-a-l- e-r-r-o-r-s- a-n-d- l-o-w-e-r-s-  
v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -T-h-i-n-k-i-n-g- -S-t-a-c-k-i-n-g- -a-l-w-a-y-s- -i-m-p-r-o-v-e-s- -p-e-r-f-o-r-m-a-n-c-e-  
-w-i-t-h-o-u-t- -n-o-t-i-n-g- -t-h-e- -c-o-s-t- -i-n- -c-o-m-p-l-e-x-i-t-y- -a-n-d- -t-r-a-i-n-i-n-g- -t-i-m-e.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -e-v-a-l-u-a-t-e- -S-t-a-c-k-i-n-g- -v-e-r-s-u-s-  
-L-i-g-h-t-G-B-M.- -S-t-a-c-k-i-n-g- -y-i-e-l-d-s- -a- -+1-.5%- -R-O-C---A-U-C- -g-a-i-n.- -l-n- -p-r-o-d-u-c-t-i-o-n.- -w-e-  
-e-v-a-l-u-a-t-e- -i-f- -t-h-i-s- -g-a-i-n- -o-u-t-w-e-i-g-h-s- -t-h-e- -e-x-t-r-a- -i-n-f-e-r-e-n-c-e- -l-a-t-e-n-c-y- -o-f- -r-u-n-n-i-n-g-  
-m-u-l-t-i-p-l-e- -b-a-s-e- -e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -m-o-d-e-l- -d-i-d- -y-o-u- -u-s-e- -a-s- -t-h-e- -m-e-t-a---l-e-a-r-n-e-r- -a-n-d-  
-w-h-y?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -h-o-w- -m-u-l-t-i---s-t-a-g-e- -s-t-a-c-k-i-n-g- -c-a-n- -l-e-a-d- -t-o- -t-a-r-g-e-t-  
-l-e-a-k-a-g-e- -i-f- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -i-s- -n-o-t- -c-o-n-f-i-g-u-r-e-d- -p-r-o-p-e-r-l-y.-

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###- -Q-u-e-s-t-i-o-n- -3-5:- -M-o-d-e-l- -S-e-l-e-c-t-i-o-n- -Q-u-e-s-t-i-o-n- -3-5:- -W-h-y- -d-i-d- -y-o-u- -c-h-o-o-s-e- -a-  
-S-t-a-c-k-i-n-g- -E-n-s-e-m-b-l-e- -o-v-e-r- -a- -s-i-n-g-l-e- -t-u-n-e-d- -L-i-g-h-t-G-B-M- -m-o-d-e-l?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*-l-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -E-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g,- -s-t-a-c-k-i-n-g,- -b-i-a-s---v-a-r-i-a-n-c-e- -t-r-a-d-e-o-f-f.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -e-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g- -a-n-d-  
-v-a-r-i-a-n-c-e- -r-e-d-u-c-t-i-o-n.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -S-t-a-c-k-i-n-g- -c-o-m-b-i-n-e-s- -m-o-d-e-l-s- -w-i-t-h- -d-i-f-f-e-r-e-n-t-  
-i-n-d-u-c-t-i-v-e- -b-i-a-s-e-s- -(e.g.- -R-a-n-d-o-m- -F-o-r-e-s-t,- -L-i-g-h-t-G-B-M,- -E-x-t-r-a-T-r-e-e-s).- -L-i-g-h-t-G-B-M-  
-f-o-c-u-s-e-s- -o-n- -g-r-a-d-i-e-n-t- -b-o-o-s-t-i-n-g- -o-n- -t-r-e-e-s,- -w-h-i-l-e- -R-a-n-d-o-m- -F-o-r-e-s-t- -u-s-e-s- -b-a-g-g-i-n-g.-  
-S-t-a-c-k-i-n-g- -t-r-a-i-n-s- -a- -m-e-t-a---l-e-a-r-n-e-r- -(e.g.- -L-o-g-i-s-t-i-c- -R-e-g-r-e-s-s-i-o-n)- -o-n- -t-h-e-i-r-  
-o-u-t---o-f--f-o-l-d- -p-r-e-d-i-c-t-i-o-n-s,- -w-h-i-c-h- -h-e-l-p-s- -s-m-o-o-t-h- -o-u-t- -i-n-d-i-v-i-d-u-a-l- -e-r-r-o-r-s- -a-n-d- -l-o-w-e-r-s-  
-v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -T-h-i-n-k-i-n-g- -S-t-a-c-k-i-n-g- -a-l-w-a-y-s- -i-m-p-r-o-v-e-s- -p-e-r-f-o-r-m-a-n-c-e-  
-w-i-t-h-o-u-t- -n-o-t-i-n-g- -t-h-e- -c-o-s-t- -i-n- -c-o-m-p-l-e-x-i-t-y- -a-n-d- -t-r-a-i-n-i-n-g- -t-i-m-e.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -e-v-a-l-u-a-t-e- -S-t-a-c-k-i-n-g- -v-e-r-s-u-s-  
-L-i-g-h-t-G-B-M.- -S-t-a-c-k-i-n-g- -y-i-e-l-d-s- -a- -+1-.5%- -R-O-C---A-U-C- -g-a-i-n.- -l-n- -p-r-o-d-u-c-t-i-o-n.- -w-e-  
-e-v-a-l-u-a-t-e- -i-f- -t-h-i-s- -g-a-i-n- -o-u-t-w-e-i-g-h-s- -t-h-e- -e-x-t-r-a- -i-n-f-e-r-e-n-c-e- -l-a-t-e-n-c-y- -o-f- -r-u-n-n-i-n-g-  
-m-u-l-t-i-p-l-e- -b-a-s-e- -e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -m-o-d-e-l- -d-i-d- -y-o-u- -u-s-e- -a-s- -t-h-e- -m-e-t-a---l-e-a-r-n-e-r- -a-n-d-  
-w-h-y?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -h-o-w- -m-u-l-t-i---s-t-a-g-e- -s-t-a-c-k-i-n-g- -c-a-n- -l-e-a-d- -t-o- -t-a-r-g-e-t-  
-l-e-a-k-a-g-e- -i-f- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -i-s- -n-o-t- -c-o-n-f-i-g-u-r-e-d- -p-r-o-p-e-r-l-y.-

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###- -Q-u-e-s-t-i-o-n- -3-6:- -M-o-d-e-l- -S-e-l-e-c-t-i-o-n- -Q-u-e-s-t-i-o-n- -3-6:- -W-h-y- -d-i-d- -y-o-u- -c-h-o-o-s-e- -a-  
-S-t-a-c-k-i-n-g- -E-n-s-e-m-b-l-e- -o-v-e-r- -a- -s-i-n-g-l-e- -t-u-n-e-d- -L-i-g-h-t-G-B-M- -m-o-d-e-l?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -E-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g,- -s-t-a-c-k-i-n-g,- -b-i-a-s---v-a-r-i-a-n-c-e- -t-r-a-d-e-o-f-f.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -e-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g- -a-n-d- -v-a-r-i-a-n-c-e- -r-e-d-u-c-t-i-o-n.-

\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -S-t-a-c-k-i-n-g- -c-o-m-b-i-n-e-s- -m-o-d-e-l-s- -w-i-t-h- -d-i-f-f-e-r-e-n-t- -i-n-d-u-c-t-i-v-e- -b-i-a-s-e-s- -(e.g.- -R-a-n-d-o-m- -F-o-r-e-s-t,- -L-i-g-h-t-G-B-M,- -E-x-t-r-a-T-r-e-e-s)-. -L-i-g-h-t-G-B-M- -f-o-c-u-s-e-s- -o-n- -g-r-a-d-i-e-n-t- -b-o-o-s-t-i-n-g- -o-n- -t-r-e-e-s,- -w-h-i-l-e- -R-a-n-d-o-m- -F-o-r-e-s-t- -u-s-e-s- -b-a-g-g-i-n-g.- -S-t-a-c-k-i-n-g- -t-r-a-i-n-s- -a- -m-e-t-a---l-e-a-r-n-e-r- -(e.g.- -L-o-g-i-s-t-i-c- -R-e-g-r-e-s-s-i-o-n)- -o-n- -t-h-e-i-r- -o-u-t---o-f---f-o-l-d- -p-r-e-d-i-c-t-i-o-n-s,- -w-h-i-c-h- -h-e-l-p-s- -s-m-o-o-t-h- -o-u-t- -i-n-d-i-v-i-d-u-a-l- -e-r-r-o-r-s- -a-n-d- -l-o-w-e-r-s- -v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -T-h-i-n-k-i-n-g- -S-t-a-c-k-i-n-g- -a-l-w-a-y-s- -i-m-p-r-o-v-e-s- -p-e-r-f-o-r-m-a-n-c-e- -w-i-t-h-o-u-t- -n-o-t-i-n-g- -t-h-e- -c-o-s-t- -i-n- -c-o-m-p-l-e-x-i-t-y- -a-n-d- -t-r-a-i-n-i-n-g- -t-i-m-e.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -e-v-a-l-u-a-t-e- -S-t-a-c-k-i-n-g- -v-e-r-s-u-s- -L-i-g-h-t-G-B-M.- -S-t-a-c-k-i-n-g- -y-i-e-l-d-s- -a- +-1-.5%- -R-O-C---A-U-C- -g-a-i-n.- -l-n- -p-r-o-d-u-c-t-i-o-n,- -w-e- -e-v-a-l-u-a-t-e- -i-f- -t-h-i-s- -g-a-i-n- -o-u-t-w-e-i-g-h-s- -t-h-e- -e-x-t-r-a- -i-n-f-e-r-e-n-c-e- -l-a-t-e-n-c-y- -o-f- -r-u-n-n-i-n-g- -m-u-l-t-i-p-l-e- -b-a-s-e- -e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -m-o-d-e-l- -d-i-d- -y-o-u- -u-s-e- -a-s- -t-h-e- -m-e-t-a---l-e-a-r-n-e-r- -a-n-d- -w-h-y?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -h-o-w- -m-u-l-t-i---s-t-a-g-e- -s-t-a-c-k-i-n-g- -c-a-n- -l-e-a-d- -t-o- -t-a-r-g-e-t- -l-e-a-k-a-g-e- -i-f- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -i-s- -n-o-t- -c-o-n-f-i-g-u-r-e-d- -p-r-o-p-e-r-l-y.-

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##- -Q-u-e-s-t-i-o-n- -3-7:- -M-o-d-e-l- -S-e-l-e-c-t-i-o-n- -Q-u-e-s-t-i-o-n- -3-7:- -W-h-y- -d-i-d- -y-o-u- -c-h-o-o-s-e- -a- -S-t-a-c-k-i-n-g- -E-n-s-e-m-b-l-e- -o-v-e-r- -a- -s-i-n-g-l-e- -t-u-n-e-d- -L-i-g-h-t-G-B-M- -m-o-d-e-l?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -E-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g,- -s-t-a-c-k-i-n-g,- -b-i-a-s---v-a-r-i-a-n-c-e- -t-r-a-d-e-o-f-f.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -e-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g- -a-n-d- -v-a-r-i-a-n-c-e- -r-e-d-u-c-t-i-o-n.-

\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -S-t-a-c-k-i-n-g- -c-o-m-b-i-n-e-s- -m-o-d-e-l-s- -w-i-t-h- -d-i-f-f-e-r-e-n-t- -i-n-d-u-c-t-i-v-e- -b-i-a-s-e-s- -(e.g.- -R-a-n-d-o-m- -F-o-r-e-s-t,- -L-i-g-h-t-G-B-M,- -E-x-t-r-a-T-r-e-e-s)-. -L-i-g-h-t-G-B-M- -f-o-c-u-s-e-s- -o-n- -g-r-a-d-i-e-n-t- -b-o-o-s-t-i-n-g- -o-n- -t-r-e-e-s,- -w-h-i-l-e- -R-a-n-d-o-m- -F-o-r-e-s-t- -u-s-e-s- -b-a-g-g-i-n-g.- -S-t-a-c-k-i-n-g- -t-r-a-i-n-s- -a- -m-e-t-a---l-e-a-r-n-e-r- -(e.g.- -L-o-g-i-s-t-i-c- -R-e-g-r-e-s-s-i-o-n)- -o-n- -t-h-e-i-r- -o-u-t---o-f---f-o-l-d- -p-r-e-d-i-c-t-i-o-n-s,- -w-h-i-c-h- -h-e-l-p-s- -s-m-o-o-t-h- -o-u-t- -i-n-d-i-v-i-d-u-a-l- -e-r-r-o-r-s- -a-n-d- -l-o-w-e-r-s- -v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -T-h-i-n-k-i-n-g- -S-t-a-c-k-i-n-g- -a-l-w-a-y-s- -i-m-p-r-o-v-e-s- -p-e-r-f-o-r-m-a-n-c-e- -w-i-t-h-o-u-t- -n-o-t-i-n-g- -t-h-e- -c-o-s-t- -i-n- -c-o-m-p-l-e-x-i-t-y- -a-n-d- -t-r-a-i-n-i-n-g- -t-i-m-e.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -e-v-a-l-u-a-t-e- -S-t-a-c-k-i-n-g- -v-e-r-s-u-s- -L-i-g-h-t-G-B-M.- -S-t-a-c-k-i-n-g- -y-i-e-l-d-s- -a- +-1-.5%- -R-O-C---A-U-C- -g-a-i-n.- -l-n- -p-r-o-d-u-c-t-i-o-n,- -w-e- -e-v-a-l-u-a-t-e- -i-f- -t-h-i-s- -g-a-i-n- -o-u-t-w-e-i-g-h-s- -t-h-e- -e-x-t-r-a- -i-n-f-e-r-e-n-c-e- -l-a-t-e-n-c-y- -o-f- -r-u-n-n-i-n-g- -m-u-l-t-i-p-l-e- -b-a-s-e- -e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -m-o-d-e-l- -d-i-d- -y-o-u- -u-s-e- -a-s- -t-h-e- -m-e-t-a---l-e-a-r-n-e-r- -a-n-d-

-w-h-y-?-

-\*- \*-D-e-e-p- -D-i-v-e-:--\*- -E-x-p-l-a-i-n- -h-o-w- -m-u-l-t-i---s-t-a-g-e- -s-t-a-c-k-i-n-g- -c-a-n- -l-e-a-d- -t-o- -t-a-r-g-e-t-  
-l-e-a-k-a-g-e- -i-f- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -i-s- -n-o-t- -c-o-n-f-i-g-u-r-e-d- -p-r-o-p-e-r-l-y.-

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-###- -Q-u-e-s-t-i-o-n- -3-8-:- -M-o-d-e-l- -S-e-l-e-c-t-i-o-n- -Q-u-e-s-t-i-o-n- -3-8-:- -W-h-y- -d-i-d- -y-o-u- -c-h-o-o-s-e- -a-  
-S-t-a-c-k-i-n-g- -E-n-s-e-m-b-l-e- -o-v-e-r- -a- -s-i-n-g-l-e- -t-u-n-e-d- -L-i-g-h-t-G-B-M- -m-o-d-e-l-?-

-\*- \*-R-o-u-n-d-:--\*- -M-L-

-\*- \*-D-i-f-f-i-c-u-l-t-y-:--\*- -H-a-r-d- -| -\*- \*-l-m-p-o-r-t-a-n-c-e-:--\*- -8-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d-:--\*- -E-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g-, -s-t-a-c-k-i-n-g-, -b-i-a-s---v-a-r-i-a-n-c-e- -t-r-a-d-e-o-f-f.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t-:--\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -e-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g- -a-n-d-  
-v-a-r-i-a-n-c-e- -r-e-d-u-c-t-i-o-n.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-:--\*- -S-t-a-c-k-i-n-g- -c-o-m-b-i-n-e-s- -m-o-d-e-l-s- -w-i-t-h- -d-i-f-f-e-r-e-n-t-  
-i-n-d-u-c-t-i-v-e- -b-i-a-s-e-s- -(e.-g.- -R-a-n-d-o-m- -F-o-r-e-s-t-, -L-i-g-h-t-G-B-M-, -E-x-t-r-a-T-r-e-e-s)-. -L-i-g-h-t-G-B-M-  
-f-o-c-u-s-e-s- -o-n- -g-r-a-d-i-e-n-t- -b-o-o-s-t-i-n-g- -o-n- -t-r-e-e-s-, -w-h-i-l-e- -R-a-n-d-o-m- -F-o-r-e-s-t- -u-s-e-s- -b-a-g-g-i-n-g.-  
-S-t-a-c-k-i-n-g- -t-r-a-i-n-s- -a- -m-e-t-a---l-e-a-r-n-e-r- -(e.-g.- -L-o-g-i-s-t-i-c- -R-e-g-r-e-s-s-i-o-n)- -o-n- -t-h-e-i-r-  
-o-u-t---o-f---f-o-l-d- -p-r-e-d-i-c-t-i-o-n-s-, -w-h-i-c-h- -h-e-l-p-s- -s-m-o-o-t-h- -o-u-t- -i-n-d-i-v-i-d-u-a-l- -e-r-r-o-r-s- -a-n-d- -l-o-w-e-r-s-  
-v-a-r-i-a-n-c-e.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-:--\*- -T-h-i-n-k-i-n-g- -S-t-a-c-k-i-n-g- -a-l-w-a-y-s- -i-m-p-r-o-v-e-s- -p-e-r-f-o-r-m-a-n-c-e-  
-w-i-t-h-o-u-t- -n-o-t-i-n-g- -t-h-e- -c-o-s-t- -i-n- -c-o-m-p-l-e-x-i-t-y- -a-n-d- -t-r-a-i-n-i-n-g- -t-i-m-e.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-:--\*- -W-e- -e-v-a-l-u-a-t-e- -S-t-a-c-k-i-n-g- -v-e-r-s-u-s-  
-L-i-g-h-t-G-B-M.- -S-t-a-c-k-i-n-g- -y-i-e-l-d-s- -a- +-1-.5%- -R-O-C---A-U-C- -g-a-i-n.- -l-n- -p-r-o-d-u-c-t-i-o-n-, -w-e-  
-e-v-a-l-u-a-t-e- -i-f- -t-h-i-s- -g-a-i-n- -o-u-t-w-e-i-g-h-s- -t-h-e- -e-x-t-r-a- -i-n-f-e-r-e-n-c-e- -l-a-t-e-n-c-y- -o-f- -r-u-n-n-i-n-g-  
-m-u-l-t-i-p-l-e- -b-a-s-e- -e-s-t-i-m-a-t-o-r-s.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s-:--\*- -W-h-a-t- -m-o-d-e-l- -d-i-d- -y-o-u- -u-s-e- -a-s- -t-h-e- -m-e-t-a---l-e-a-r-n-e-r- -a-n-d-  
-w-h-y-?-

-\*- \*-D-e-e-p- -D-i-v-e-:--\*- -E-x-p-l-a-i-n- -h-o-w- -m-u-l-t-i---s-t-a-g-e- -s-t-a-c-k-i-n-g- -c-a-n- -l-e-a-d- -t-o- -t-a-r-g-e-t-  
-l-e-a-k-a-g-e- -i-f- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -i-s- -n-o-t- -c-o-n-f-i-g-u-r-e-d- -p-r-o-p-e-r-l-y.-

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-###- -Q-u-e-s-t-i-o-n- -3-9-:- -M-o-d-e-l- -S-e-l-e-c-t-i-o-n- -Q-u-e-s-t-i-o-n- -3-9-:- -W-h-y- -d-i-d- -y-o-u- -c-h-o-o-s-e- -a-  
-S-t-a-c-k-i-n-g- -E-n-s-e-m-b-l-e- -o-v-e-r- -a- -s-i-n-g-l-e- -t-u-n-e-d- -L-i-g-h-t-G-B-M- -m-o-d-e-l-?-

-\*- \*-R-o-u-n-d-:--\*- -M-L-

-\*- \*-D-i-f-f-i-c-u-l-t-y-:--\*- -H-a-r-d- -| -\*- \*-l-m-p-o-r-t-a-n-c-e-:--\*- -8-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d-:--\*- -E-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g-, -s-t-a-c-k-i-n-g-, -b-i-a-s---v-a-r-i-a-n-c-e- -t-r-a-d-e-o-f-f.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t-:--\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -e-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g- -a-n-d-  
-v-a-r-i-a-n-c-e- -r-e-d-u-c-t-i-o-n.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-:--\*- -S-t-a-c-k-i-n-g- -c-o-m-b-i-n-e-s- -m-o-d-e-l-s- -w-i-t-h- -d-i-f-f-e-r-e-n-t-  
-i-n-d-u-c-t-i-v-e- -b-i-a-s-e-s- -(e.-g.- -R-a-n-d-o-m- -F-o-r-e-s-t-, -L-i-g-h-t-G-B-M-, -E-x-t-r-a-T-r-e-e-s)-. -L-i-g-h-t-G-B-M-  
-f-o-c-u-s-e-s- -o-n- -g-r-a-d-i-e-n-t- -b-o-o-s-t-i-n-g- -o-n- -t-r-e-e-s-, -w-h-i-l-e- -R-a-n-d-o-m- -F-o-r-e-s-t- -u-s-e-s- -b-a-g-g-i-n-g.-

-S-t-a-c-k-i-n-g- t-r-a-i-n-s- a- m-e-t-a---l-e-a-r-n-e-r- -(e-.g.- L-o-g-i-s-t-i-c- R-e-g-r-e-s-s-i-o-n)- o-n- t-h-e-i-r- o-u-t---o-f---f-o-l-d- p-r-e-d-i-c-t-i-o-n-s,- w-h-i-c-h- h-e-l-p-s- s-m-o-o-t-h- o-u-t- i-n-d-i-v-i-d-u-a-l- e-r-r-o-r-s- a-n-d- l-o-w-e-r-s- v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- T-h-i-n-k-i-n-g- S-t-a-c-k-i-n-g- a-l-w-a-y-s- i-m-p-r-o-v-e-s- p-e-r-f-o-r-m-a-n-c-e- w-i-t-h-o-u-t- n-o-t-i-n-g- t-h-e- c-o-s-t- i-n- c-o-m-p-l-e-x-i-t-y- a-n-d- t-r-a-i-n-i-n-g- t-i-m-e.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- e-v-a-l-u-a-t-e- S-t-a-c-k-i-n-g- v-e-r-s-u-s- L-i-g-h-t-G-B-M.- S-t-a-c-k-i-n-g- y-i-e-l-d-s- a- +1-.5%- R-O-C---A-U-C- g-a-i-n.- l-n- p-r-o-d-u-c-t-i-o-n,- w-e- e-v-a-l-u-a-t-e- i-f- t-h-i-s- g-a-i-n- o-u-t-w-e-i-g-h-s- t-h-e- e-x-t-r-a- i-n-f-e-r-e-n-c-e- l-a-t-e-n-c-y- o-f- r-u-n-n-i-n-g- m-u-l-t-i-p-l-e- b-a-s-e- e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- W-h-a-t- m-o-d-e-l- d-i-d- y-o-u- u-s-e- a-s- t-h-e- m-e-t-a---l-e-a-r-n-e-r- a-n-d- w-h-y?-

\*- \*-D-e-e-p- D-i-v-e:-\*- E-x-p-l-a-i-n- h-o-w- m-u-l-t-i---s-t-a-g-e- s-t-a-c-k-i-n-g- c-a-n- l-e-a-d- t-o- t-a-r-g-e-t- l-e-a-k-a-g-e- i-f- c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- i-s- n-o-t- c-o-n-f-i-g-u-r-e-d- p-r-o-p-e-r-l-y.-

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###- Q-u-e-s-t-i-o-n- 4-0:- M-o-d-e-l- S-e-l-e-c-t-i-o-n- Q-u-e-s-t-i-o-n- 4-0:- W-h-y- d-i-d- y-o-u- c-h-o-o-s-e- a- S-t-a-c-k-i-n-g- E-n-s-e-m-b-l-e- o-v-e-r- a- s-i-n-g-l-e- t-u-n-e-d- L-i-g-h-t-G-B-M- m-o-d-e-l?-

\*- \*-R-o-u-n-d:-\*- M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- |- \*-l-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- E-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g,- s-t-a-c-k-i-n-g,- b-i-a-s---v-a-r-i-a-n-c-e- t-r-a-d-e-o-f-f.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- T-e-s-t-s- u-n-d-e-r-s-t-a-n-d-i-n-g- o-f- e-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g- a-n-d- v-a-r-i-a-n-c-e- r-e-d-u-c-t-i-o-n.-

\*- \*-l-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- S-t-a-c-k-i-n-g- c-o-m-b-i-n-e-s- m-o-d-e-l-s- w-i-t-h- d-i-f-f-e-r-e-n-t- i-n-d-u-c-t-i-v-e- b-i-a-s-e-s- -(e-.g.- R-a-n-d-o-m- F-o-r-e-s-t,- L-i-g-h-t-G-B-M,- E-x-t-r-a-T-r-e-e-s)- L-i-g-h-t-G-B-M- f-o-c-u-s-e-s- o-n- g-r-a-d-i-e-n-t- b-o-o-s-t-i-n-g- o-n- t-r-e-e-s,- w-h-i-l-e- R-a-n-d-o-m- F-o-r-e-s-t- u-s-e-s- b-a-g-g-i-n-g.- S-t-a-c-k-i-n-g- t-r-a-i-n-s- a- m-e-t-a---l-e-a-r-n-e-r- -(e-.g.- L-o-g-i-s-t-i-c- R-e-g-r-e-s-s-i-o-n)- o-n- t-h-e-i-r- o-u-t---o-f---f-o-l-d- p-r-e-d-i-c-t-i-o-n-s,- w-h-i-c-h- h-e-l-p-s- s-m-o-o-t-h- o-u-t- i-n-d-i-v-i-d-u-a-l- e-r-r-o-r-s- a-n-d- l-o-w-e-r-s- v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- T-h-i-n-k-i-n-g- S-t-a-c-k-i-n-g- a-l-w-a-y-s- i-m-p-r-o-v-e-s- p-e-r-f-o-r-m-a-n-c-e- w-i-t-h-o-u-t- n-o-t-i-n-g- t-h-e- c-o-s-t- i-n- c-o-m-p-l-e-x-i-t-y- a-n-d- t-r-a-i-n-i-n-g- t-i-m-e.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- e-v-a-l-u-a-t-e- S-t-a-c-k-i-n-g- v-e-r-s-u-s- L-i-g-h-t-G-B-M.- S-t-a-c-k-i-n-g- y-i-e-l-d-s- a- +1-.5%- R-O-C---A-U-C- g-a-i-n.- l-n- p-r-o-d-u-c-t-i-o-n,- w-e- e-v-a-l-u-a-t-e- i-f- t-h-i-s- g-a-i-n- o-u-t-w-e-i-g-h-s- t-h-e- e-x-t-r-a- i-n-f-e-r-e-n-c-e- l-a-t-e-n-c-y- o-f- r-u-n-n-i-n-g- m-u-l-t-i-p-l-e- b-a-s-e- e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- W-h-a-t- m-o-d-e-l- d-i-d- y-o-u- u-s-e- a-s- t-h-e- m-e-t-a---l-e-a-r-n-e-r- a-n-d- w-h-y?-

\*- \*-D-e-e-p- D-i-v-e:-\*- E-x-p-l-a-i-n- h-o-w- m-u-l-t-i---s-t-a-g-e- s-t-a-c-k-i-n-g- c-a-n- l-e-a-d- t-o- t-a-r-g-e-t- l-e-a-k-a-g-e- i-f- c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- i-s- n-o-t- c-o-n-f-i-g-u-r-e-d- p-r-o-p-e-r-l-y.-

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###- Q-u-e-s-t-i-o-n- 4-1:- M-o-d-e-l- S-e-l-e-c-t-i-o-n- Q-u-e-s-t-i-o-n- 4-1:- W-h-y- d-i-d- y-o-u- c-h-o-o-s-e- a-

-S-t-a-c-k-i-n-g- -E-n-s-e-m-b-l-e- -o-v-e-r- -a- -s-i-n-g-l-e- -t-u-n-e-d- -L-i-g-h-t-G-B-M- -m-o-d-e-l-?-

-\*- \*-R-o-u-n-d-:-\*- -M-L-

-\*- \*-D-i-f-f-i-c-u-l-t-y-:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e-:-\*- -8-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d-:-\*- -E-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g-, -s-t-a-c-k-i-n-g-, -b-i-a-s---v-a-r-i-a-n-c-e- -t-r-a-d-e-o-f-f.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t-:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -e-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g- -a-n-d- -v-a-r-i-a-n-c-e- -r-e-d-u-c-t-i-o-n.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-:-\*- -S-t-a-c-k-i-n-g- -c-o-m-b-i-n-e-s- -m-o-d-e-l-s- -w-i-t-h- -d-i-f-f-e-r-e-n-t- -i-n-d-u-c-t-i-v-e- -b-i-a-s-e-s- -(e-.g.- -R-a-n-d-o-m- -F-o-r-e-s-t-, -L-i-g-h-t-G-B-M-, -E-x-t-r-a-T-r-e-e-s)-. -L-i-g-h-t-G-B-M- -f-o-c-u-s-e-s- -o-n- -g-r-a-d-i-e-n-t- -b-o-o-s-t-i-n-g- -o-n- -t-r-e-e-s-, -w-h-i-l-e- -R-a-n-d-o-m- -F-o-r-e-s-t- -u-s-e-s- -b-a-g-g-i-n-g.- -S-t-a-c-k-i-n-g- -t-r-a-i-n-s- -a- -m-e-t-a---l-e-a-r-n-e-r- -(e-.g.- -L-o-g-i-s-t-i-c- -R-e-g-r-e-s-s-i-o-n)- -o-n- -t-h-e-i-r- -o-u-t---o-f---f-o-l-d- -p-r-e-d-i-c-t-i-o-n-s-, -w-h-i-c-h- -h-e-l-p-s- -s-m-o-o-t-h- -o-u-t- -i-n-d-i-v-i-d-u-a-l- -e-r-r-o-r-s- -a-n-d- -l-o-w-e-r-s- -v-a-r-i-a-n-c-e.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-:-\*- -T-h-i-n-k-i-n-g- -S-t-a-c-k-i-n-g- -a-l-w-a-y-s- -i-m-p-r-o-v-e-s- -p-e-r-f-o-r-m-a-n-c-e- -w-i-t-h-o-u-t- -n-o-t-i-n-g- -t-h-e- -c-o-s-t- -i-n- -c-o-m-p-l-e-x-i-t-y- -a-n-d- -t-r-a-i-n-i-n-g- -t-i-m-e.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-:-\*- -W-e- -e-v-a-l-u-a-t-e- -S-t-a-c-k-i-n-g- -v-e-r-s-u-s- -L-i-g-h-t-G-B-M.- -S-t-a-c-k-i-n-g- -y-i-e-l-d-s- -a- +-1-.5%- -R-O-C---A-U-C- -g-a-i-n.- -l-n- -p-r-o-d-u-c-t-i-o-n-, -w-e- -e-v-a-l-u-a-t-e- -i-f- -t-h-i-s- -g-a-i-n- -o-u-t-w-e-i-g-h-s- -t-h-e- -e-x-t-r-a- -i-n-f-e-r-e-n-c-e- -l-a-t-e-n-c-y- -o-f- -r-u-n-n-i-n-g- -m-u-l-t-i-p-l-e- -b-a-s-e- -e-s-t-i-m-a-t-o-r-s.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s-:-\*- -W-h-a-t- -m-o-d-e-l- -d-i-d- -y-o-u- -u-s-e- -a-s- -t-h-e- -m-e-t-a---l-e-a-r-n-e-r- -a-n-d- -w-h-y-?-

-\*- \*-D-e-e-p- -D-i-v-e-:-\*- -E-x-p-l-a-i-n- -h-o-w- -m-u-l-t-i---s-t-a-g-e- -s-t-a-c-k-i-n-g- -c-a-n- -l-e-a-d- -t-o- -t-a-r-g-e-t- -l-e-a-k-a-g-e- -i-f- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -i-s- -n-o-t- -c-o-n-f-i-g-u-r-e-d- -p-r-o-p-e-r-l-y.-

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-##-#- -Q-u-e-s-t-i-o-n- -4-2-:- -M-o-d-e-l- -S-e-l-e-c-t-i-o-n- -Q-u-e-s-t-i-o-n- -4-2-:- -W-h-y- -d-i-d- -y-o-u- -c-h-o-o-s-e- -a- -S-t-a-c-k-i-n-g- -E-n-s-e-m-b-l-e- -o-v-e-r- -a- -s-i-n-g-l-e- -t-u-n-e-d- -L-i-g-h-t-G-B-M- -m-o-d-e-l-?-

-\*- \*-R-o-u-n-d-:-\*- -M-L-

-\*- \*-D-i-f-f-i-c-u-l-t-y-:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e-:-\*- -8-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d-:-\*- -E-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g-, -s-t-a-c-k-i-n-g-, -b-i-a-s---v-a-r-i-a-n-c-e- -t-r-a-d-e-o-f-f.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t-:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -e-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g- -a-n-d- -v-a-r-i-a-n-c-e- -r-e-d-u-c-t-i-o-n.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-:-\*- -S-t-a-c-k-i-n-g- -c-o-m-b-i-n-e-s- -m-o-d-e-l-s- -w-i-t-h- -d-i-f-f-e-r-e-n-t- -i-n-d-u-c-t-i-v-e- -b-i-a-s-e-s- -(e-.g.- -R-a-n-d-o-m- -F-o-r-e-s-t-, -L-i-g-h-t-G-B-M-, -E-x-t-r-a-T-r-e-e-s)-. -L-i-g-h-t-G-B-M- -f-o-c-u-s-e-s- -o-n- -g-r-a-d-i-e-n-t- -b-o-o-s-t-i-n-g- -o-n- -t-r-e-e-s-, -w-h-i-l-e- -R-a-n-d-o-m- -F-o-r-e-s-t- -u-s-e-s- -b-a-g-g-i-n-g.- -S-t-a-c-k-i-n-g- -t-r-a-i-n-s- -a- -m-e-t-a---l-e-a-r-n-e-r- -(e-.g.- -L-o-g-i-s-t-i-c- -R-e-g-r-e-s-s-i-o-n)- -o-n- -t-h-e-i-r- -o-u-t---o-f---f-o-l-d- -p-r-e-d-i-c-t-i-o-n-s-, -w-h-i-c-h- -h-e-l-p-s- -s-m-o-o-t-h- -o-u-t- -i-n-d-i-v-i-d-u-a-l- -e-r-r-o-r-s- -a-n-d- -l-o-w-e-r-s- -v-a-r-i-a-n-c-e.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-:-\*- -T-h-i-n-k-i-n-g- -S-t-a-c-k-i-n-g- -a-l-w-a-y-s- -i-m-p-r-o-v-e-s- -p-e-r-f-o-r-m-a-n-c-e- -w-i-t-h-o-u-t- -n-o-t-i-n-g- -t-h-e- -c-o-s-t- -i-n- -c-o-m-p-l-e-x-i-t-y- -a-n-d- -t-r-a-i-n-i-n-g- -t-i-m-e.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-:-\*- -W-e- -e-v-a-l-u-a-t-e- -S-t-a-c-k-i-n-g- -v-e-r-s-u-s- -L-i-g-h-t-G-B-M.- -S-t-a-c-k-i-n-g- -y-i-e-l-d-s- -a- +-1-.5%- -R-O-C---A-U-C- -g-a-i-n.- -l-n- -p-r-o-d-u-c-t-i-o-n-, -w-e- -e-v-a-l-u-a-t-e- -i-f- -t-h-i-s- -g-a-i-n- -o-u-t-w-e-i-g-h-s- -t-h-e- -e-x-t-r-a- -i-n-f-e-r-e-n-c-e- -l-a-t-e-n-c-y- -o-f- -r-u-n-n-i-n-g-

-m-u-l-t-i-p-l-e- -b-a-s-e- -e-s-t-i-m-a-t-o-r-s.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -m-o-d-e-l- -d-i-d- -y-o-u- -u-s-e- -a-s- -t-h-e- -m-e-t-a---l-e-a-r-n-e-r- -a-n-d- -w-h-y?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -h-o-w- -m-u-l-t-i---s-t-a-g-e- -s-t-a-c-k-i-n-g- -c-a-n- -l-e-a-d- -t-o- -t-a-r-g-e-t- -l-e-a-k-a-g-e- -i-f- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -i-s- -n-o-t- -c-o-n-f-i-g-u-r-e-d- -p-r-o-p-e-r-l-y.-

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### -Q-u-e-s-t-i-o-n- -4-3:- -M-o-d-e-l- -S-e-l-e-c-t-i-o-n- -Q-u-e-s-t-i-o-n- -4-3:- -W-h-y- -d-i-d- -y-o-u- -c-h-o-o-s-e- -a- -S-t-a-c-k-i-n-g- -E-n-s-e-m-b-l-e- -o-v-e-r- -a- -s-i-n-g-l-e- -t-u-n-e-d- -L-i-g-h-t-G-B-M- -m-o-d-e-l?-

-\*- \*-R-o-u-n-d:-\*- -M-L-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -I-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -E-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g,- -s-t-a-c-k-i-n-g,- -b-i-a-s---v-a-r-i-a-n-c-e- -t-r-a-d-e-o-f-f.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -e-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g- -a-n-d- -v-a-r-i-a-n-c-e- -r-e-d-u-c-t-i-o-n.-

-\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -S-t-a-c-k-i-n-g- -c-o-m-b-i-n-e-s- -m-o-d-e-l-s- -w-i-t-h- -d-i-f-f-e-r-e-n-t- -i-n-d-u-c-t-i-v-e- -b-i-a-s-e-s- -(e.-g.- -R-a-n-d-o-m- -F-o-r-e-s-t,- -L-i-g-h-t-G-B-M,- -E-x-t-r-a-T-r-e-e-s)-. -L-i-g-h-t-G-B-M- -f-o-c-u-s-e-s- -o-n- -g-r-a-d-i-e-n-t- -b-o-o-s-t-i-n-g- -o-n- -t-r-e-e-s,- -w-h-i-l-e- -R-a-n-d-o-m- -F-o-r-e-s-t- -u-s-e-s- -b-a-g-g-i-n-g.- -S-t-a-c-k-i-n-g- -t-r-a-i-n-s- -a- -m-e-t-a---l-e-a-r-n-e-r- -(e.-g.- -L-o-g-i-s-t-i-c- -R-e-g-r-e-s-s-i-o-n)- -o-n- -t-h-e-i-r- -o-u-t---o-f---f-o-l-d- -p-r-e-d-i-c-t-i-o-n-s,- -w-h-i-c-h- -h-e-l-p-s- -s-m-o-o-t-h- -o-u-t- -i-n-d-i-v-i-d-u-a-l- -e-r-r-o-r-s- -a-n-d- -l-o-w-e-r-s- -v-a-r-i-a-n-c-e.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -T-h-i-n-k-i-n-g- -S-t-a-c-k-i-n-g- -a-l-w-a-y-s- -i-m-p-r-o-v-e-s- -p-e-r-f-o-r-m-a-n-c-e- -w-i-t-h-o-u-t- -n-o-t-i-n-g- -t-h-e- -c-o-s-t- -i-n- -c-o-m-p-l-e-x-i-t-y- -a-n-d- -t-r-a-i-n-i-n-g- -t-i-m-e.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -e-v-a-l-u-a-t-e- -S-t-a-c-k-i-n-g- -v-e-r-s-u-s- -L-i-g-h-t-G-B-M.- -S-t-a-c-k-i-n-g- -y-i-e-l-d-s- -a- +-1-.5%- -R-O-C---A-U-C- -g-a-i-n.- -l-n- -p-r-o-d-u-c-t-i-o-n,- -w-e- -e-v-a-l-u-a-t-e- -i-f- -t-h-i-s- -g-a-i-n- -o-u-t-w-e-i-g-h-s- -t-h-e- -e-x-t-r-a- -i-n-f-e-r-e-n-c-e- -l-a-t-e-n-c-y- -o-f- -r-u-n-n-i-n-g- -m-u-l-t-i-p-l-e- -b-a-s-e- -e-s-t-i-m-a-t-o-r-s.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -m-o-d-e-l- -d-i-d- -y-o-u- -u-s-e- -a-s- -t-h-e- -m-e-t-a---l-e-a-r-n-e-r- -a-n-d- -w-h-y?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -h-o-w- -m-u-l-t-i---s-t-a-g-e- -s-t-a-c-k-i-n-g- -c-a-n- -l-e-a-d- -t-o- -t-a-r-g-e-t- -l-e-a-k-a-g-e- -i-f- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -i-s- -n-o-t- -c-o-n-f-i-g-u-r-e-d- -p-r-o-p-e-r-l-y.-

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### -Q-u-e-s-t-i-o-n- -4-4:- -M-o-d-e-l- -S-e-l-e-c-t-i-o-n- -Q-u-e-s-t-i-o-n- -4-4:- -W-h-y- -d-i-d- -y-o-u- -c-h-o-o-s-e- -a- -S-t-a-c-k-i-n-g- -E-n-s-e-m-b-l-e- -o-v-e-r- -a- -s-i-n-g-l-e- -t-u-n-e-d- -L-i-g-h-t-G-B-M- -m-o-d-e-l?-

-\*- \*-R-o-u-n-d:-\*- -M-L-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -I-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -E-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g,- -s-t-a-c-k-i-n-g,- -b-i-a-s---v-a-r-i-a-n-c-e- -t-r-a-d-e-o-f-f.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -e-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g- -a-n-d- -v-a-r-i-a-n-c-e- -r-e-d-u-c-t-i-o-n.-

\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -S-t-a-c-k-i-n-g- -c-o-m-b-i-n-e-s- -m-o-d-e-l-s- -w-i-t-h- -d-i-f-f-e-r-e-n-t-  
-i-n-d-u-c-t-i-v-e- -b-i-a-s-e-s- -(e.-g.- -R-a-n-d-o-m- -F-o-r-e-s-t,- -L-i-g-h-t-G-B-M,- -E-x-t-r-a-T-r-e-e-s)-. -L-i-g-h-t-G-B-M-  
-f-o-c-u-s-e-s- -o-n- -g-r-a-d-i-e-n-t- -b-o-o-s-t-i-n-g- -o-n- -t-r-e-e-s,- -w-h-i-l-e- -R-a-n-d-o-m- -F-o-r-e-s-t- -u-s-e-s- -b-a-g-g-i-n-g.-  
-S-t-a-c-k-i-n-g- -t-r-a-i-n-s- -a- -m-e-t-a---l-e-a-r-n-e-r- -(e.-g.- -L-o-g-i-s-t-i-c- -R-e-g-r-e-s-s-i-o-n)- -o-n- -t-h-e-i-r-  
-o-u-t---o-f---f-o-l-d- -p-r-e-d-i-c-t-i-o-n-s,- -w-h-i-c-h- -h-e-l-p-s- -s-m-o-o-t-h- -o-u-t- -i-n-d-i-v-i-d-u-a-l- -e-r-r-o-r-s- -a-n-d- -l-o-w-e-r-s-  
-v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -T-h-i-n-k-i-n-g- -S-t-a-c-k-i-n-g- -a-l-w-a-y-s- -i-m-p-r-o-v-e-s- -p-e-r-f-o-r-m-a-n-c-e-  
-w-i-t-h-o-u-t- -n-o-t-i-n-g- -t-h-e- -c-o-s-t- -i-n- -c-o-m-p-l-e-x-i-t-y- -a-n-d- -t-r-a-i-n-i-n-g- -t-i-m-e.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -e-v-a-l-u-a-t-e- -S-t-a-c-k-i-n-g- -v-e-r-s-u-s-  
-L-i-g-h-t-G-B-M.- -S-t-a-c-k-i-n-g- -y-i-e-l-d-s- -a- +-1-.5-%- -R-O-C---A-U-C- -g-a-i-n.- -l-n- -p-r-o-d-u-c-t-i-o-n,- -w-e-  
-e-v-a-l-u-a-t-e- -i-f- -t-h-i-s- -g-a-i-n- -o-u-t-w-e-i-g-h-s- -t-h-e- -e-x-t-r-a- -i-n-f-e-r-e-n-c-e- -l-a-t-e-n-c-y- -o-f- -r-u-n-n-i-n-g-  
-m-u-l-t-i-p-l-e- -b-a-s-e- -e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -m-o-d-e-l- -d-i-d- -y-o-u- -u-s-e- -a-s- -t-h-e- -m-e-t-a---l-e-a-r-n-e-r- -a-n-d-  
-w-h-y?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -h-o-w- -m-u-l-t-i---s-t-a-g-e- -s-t-a-c-k-i-n-g- -c-a-n- -l-e-a-d- -t-o- -t-a-r-g-e-t-  
-l-e-a-k-a-g-e- -i-f- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -i-s- -n-o-t- -c-o-n-f-i-g-u-r-e-d- -p-r-o-p-e-r-l-y.-

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##- -Q-u-e-s-t-i-o-n- -4-5:- -M-o-d-e-l- -S-e-l-e-c-t-i-o-n- -Q-u-e-s-t-i-o-n- -4-5:- -W-h-y- -d-i-d- -y-o-u- -c-h-o-o-s-e- -a-  
-S-t-a-c-k-i-n-g- -E-n-s-e-m-b-l-e- -o-v-e-r- -a- -s-i-n-g-l-e- -t-u-n-e-d- -L-i-g-h-t-G-B-M- -m-o-d-e-l?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -E-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g,- -s-t-a-c-k-i-n-g,- -b-i-a-s---v-a-r-i-a-n-c-e- -t-r-a-d-e-o-f-f.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -e-n-s-e-m-b-l-e- -l-e-a-r-n-i-n-g- -a-n-d-  
-v-a-r-i-a-n-c-e- -r-e-d-u-c-t-i-o-n.-

\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -S-t-a-c-k-i-n-g- -c-o-m-b-i-n-e-s- -m-o-d-e-l-s- -w-i-t-h- -d-i-f-f-e-r-e-n-t-  
-i-n-d-u-c-t-i-v-e- -b-i-a-s-e-s- -(e.-g.- -R-a-n-d-o-m- -F-o-r-e-s-t,- -L-i-g-h-t-G-B-M,- -E-x-t-r-a-T-r-e-e-s)-. -L-i-g-h-t-G-B-M-  
-f-o-c-u-s-e-s- -o-n- -g-r-a-d-i-e-n-t- -b-o-o-s-t-i-n-g- -o-n- -t-r-e-e-s,- -w-h-i-l-e- -R-a-n-d-o-m- -F-o-r-e-s-t- -u-s-e-s- -b-a-g-g-i-n-g.-  
-S-t-a-c-k-i-n-g- -t-r-a-i-n-s- -a- -m-e-t-a---l-e-a-r-n-e-r- -(e.-g.- -L-o-g-i-s-t-i-c- -R-e-g-r-e-s-s-i-o-n)- -o-n- -t-h-e-i-r-  
-o-u-t---o-f---f-o-l-d- -p-r-e-d-i-c-t-i-o-n-s,- -w-h-i-c-h- -h-e-l-p-s- -s-m-o-o-t-h- -o-u-t- -i-n-d-i-v-i-d-u-a-l- -e-r-r-o-r-s- -a-n-d- -l-o-w-e-r-s-  
-v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -T-h-i-n-k-i-n-g- -S-t-a-c-k-i-n-g- -a-l-w-a-y-s- -i-m-p-r-o-v-e-s- -p-e-r-f-o-r-m-a-n-c-e-  
-w-i-t-h-o-u-t- -n-o-t-i-n-g- -t-h-e- -c-o-s-t- -i-n- -c-o-m-p-l-e-x-i-t-y- -a-n-d- -t-r-a-i-n-i-n-g- -t-i-m-e.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -e-v-a-l-u-a-t-e- -S-t-a-c-k-i-n-g- -v-e-r-s-u-s-  
-L-i-g-h-t-G-B-M.- -S-t-a-c-k-i-n-g- -y-i-e-l-d-s- -a- +-1-.5-%- -R-O-C---A-U-C- -g-a-i-n.- -l-n- -p-r-o-d-u-c-t-i-o-n,- -w-e-  
-e-v-a-l-u-a-t-e- -i-f- -t-h-i-s- -g-a-i-n- -o-u-t-w-e-i-g-h-s- -t-h-e- -e-x-t-r-a- -i-n-f-e-r-e-n-c-e- -l-a-t-e-n-c-y- -o-f- -r-u-n-n-i-n-g-  
-m-u-l-t-i-p-l-e- -b-a-s-e- -e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -m-o-d-e-l- -d-i-d- -y-o-u- -u-s-e- -a-s- -t-h-e- -m-e-t-a---l-e-a-r-n-e-r- -a-n-d-  
-w-h-y?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -h-o-w- -m-u-l-t-i---s-t-a-g-e- -s-t-a-c-k-i-n-g- -c-a-n- -l-e-a-d- -t-o- -t-a-r-g-e-t-  
-l-e-a-k-a-g-e- -i-f- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -i-s- -n-o-t- -c-o-n-f-i-g-u-r-e-d- -p-r-o-p-e-r-l-y.-

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### -Q-u-e-s-t-i-o-n- 4-6:- M-o-d-e-l- S-e-l-e-c-t-i-o-n- Q-u-e-s-t-i-o-n- 4-6:- W-h-y- d-i-d- y-o-u- c-h-o-o-s-e- a-S-t-a-c-k-i-n-g- E-n-s-e-m-b-l-e- o-v-e-r- a- s-i-n-g-l-e- t-u-n-e-d- L-i-g-h-t-G-B-M- m-o-d-e-l-?

\*- \*-R-o-u-n-d:-\*- M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- | -\*-I-m-p-o-r-t-a-n-c-e:-\*- -8/-1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- E-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g-, s-t-a-c-k-i-n-g-, b-i-a-s---v-a-r-i-a-n-c-e- t-r-a-d-e-o-f-f.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- T-e-s-t-s- u-n-d-e-r-s-t-a-n-d-i-n-g- o-f- e-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g- a-n-d- v-a-r-i-a-n-c-e- r-e-d-u-c-t-i-o-n.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- S-t-a-c-k-i-n-g- c-o-m-b-i-n-e-s- m-o-d-e-l-s- w-i-t-h- d-i-f-f-e-r-e-n-t- i-n-d-u-c-t-i-v-e- b-i-a-s-e-s- (-e.-g.- R-a-n-d-o-m- F-o-r-e-s-t-, L-i-g-h-t-G-B-M-, E-x-t-r-a-T-r-e-e-s-).- L-i-g-h-t-G-B-M- f-o-c-u-s-e-s- o-n- g-r-a-d-i-e-n-t- b-o-o-s-t-i-n-g- o-n- t-r-e-e-s-, w-h-i-l-e- R-a-n-d-o-m- F-o-r-e-s-t- u-s-e-s- b-a-g-g-i-n-g.- S-t-a-c-k-i-n-g- t-r-a-i-n-s- a- m-e-t-a---l-e-a-r-n-e-r- (-e.-g.- L-o-g-i-s-t-i-c- R-e-g-r-e-s-s-i-o-n-) o-n- t-h-e-i-r- o-u-t---o-f---f-o-l-d- p-r-e-d-i-c-t-i-o-n-s-, w-h-i-c-h- h-e-l-p-s- s-m-o-o-t-h- o-u-t- i-n-d-i-v-i-d-u-a-l- e-r-r-o-r-s- a-n-d- l-o-w-e-r-s- v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- T-h-i-n-k-i-n-g- S-t-a-c-k-i-n-g- a-l-w-a-y-s- i-m-p-r-o-v-e-s- p-e-r-f-o-r-m-a-n-c-e- w-i-t-h-o-u-t- n-o-t-i-n-g- t-h-e- c-o-s-t- i-n- c-o-m-p-l-e-x-i-t-y- a-n-d- t-r-a-i-n-i-n-g- t-i-m-e.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- e-v-a-l-u-a-t-e- S-t-a-c-k-i-n-g- v-e-r-s-u-s- L-i-g-h-t-G-B-M.- S-t-a-c-k-i-n-g- y-i-e-l-d-s- a- +1..5%- R-O-C---A-U-C- g-a-i-n.- l-n- p-r-o-d-u-c-t-i-o-n-, w-e- e-v-a-l-u-a-t-e- i-f- t-h-i-s- g-a-i-n- o-u-t-w-e-i-g-h-s- t-h-e- e-x-t-r-a- i-n-f-e-r-e-n-c-e- l-a-t-e-n-c-y- o-f- r-u-n-n-i-n-g- m-u-l-t-i-p-l-e- b-a-s-e- e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- W-h-a-t- m-o-d-e-l- d-i-d- y-o-u- u-s-e- a-s- t-h-e- m-e-t-a---l-e-a-r-n-e-r- a-n-d- w-h-y-?

\*- \*-D-e-e-p- D-i-v-e:-\*- E-x-p-l-a-i-n- h-o-w- m-u-l-t-i---s-t-a-g-e- s-t-a-c-k-i-n-g- c-a-n- l-e-a-d- t-o- t-a-r-g-e-t- l-e-a-k-a-g-e- i-f- c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- i-s- n-o-t- c-o-n-f-i-g-u-r-e-d- p-r-o-p-e-r-l-y.-

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### -Q-u-e-s-t-i-o-n- 4-7:- M-o-d-e-l- S-e-l-e-c-t-i-o-n- Q-u-e-s-t-i-o-n- 4-7:- W-h-y- d-i-d- y-o-u- c-h-o-o-s-e- a-S-t-a-c-k-i-n-g- E-n-s-e-m-b-l-e- o-v-e-r- a- s-i-n-g-l-e- t-u-n-e-d- L-i-g-h-t-G-B-M- m-o-d-e-l-?

\*- \*-R-o-u-n-d:-\*- M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- | -\*-I-m-p-o-r-t-a-n-c-e:-\*- -8/-1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- E-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g-, s-t-a-c-k-i-n-g-, b-i-a-s---v-a-r-i-a-n-c-e- t-r-a-d-e-o-f-f.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- T-e-s-t-s- u-n-d-e-r-s-t-a-n-d-i-n-g- o-f- e-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g- a-n-d- v-a-r-i-a-n-c-e- r-e-d-u-c-t-i-o-n.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- S-t-a-c-k-i-n-g- c-o-m-b-i-n-e-s- m-o-d-e-l-s- w-i-t-h- d-i-f-f-e-r-e-n-t- i-n-d-u-c-t-i-v-e- b-i-a-s-e-s- (-e.-g.- R-a-n-d-o-m- F-o-r-e-s-t-, L-i-g-h-t-G-B-M-, E-x-t-r-a-T-r-e-e-s-).- L-i-g-h-t-G-B-M- f-o-c-u-s-e-s- o-n- g-r-a-d-i-e-n-t- b-o-o-s-t-i-n-g- o-n- t-r-e-e-s-, w-h-i-l-e- R-a-n-d-o-m- F-o-r-e-s-t- u-s-e-s- b-a-g-g-i-n-g.- S-t-a-c-k-i-n-g- t-r-a-i-n-s- a- m-e-t-a---l-e-a-r-n-e-r- (-e.-g.- L-o-g-i-s-t-i-c- R-e-g-r-e-s-s-i-o-n-) o-n- t-h-e-i-r- o-u-t---o-f---f-o-l-d- p-r-e-d-i-c-t-i-o-n-s-, w-h-i-c-h- h-e-l-p-s- s-m-o-o-t-h- o-u-t- i-n-d-i-v-i-d-u-a-l- e-r-r-o-r-s- a-n-d- l-o-w-e-r-s- v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- T-h-i-n-k-i-n-g- S-t-a-c-k-i-n-g- a-l-w-a-y-s- i-m-p-r-o-v-e-s- p-e-r-f-o-r-m-a-n-c-e- w-i-t-h-o-u-t- n-o-t-i-n-g- t-h-e- c-o-s-t- i-n- c-o-m-p-l-e-x-i-t-y- a-n-d- t-r-a-i-n-i-n-g- t-i-m-e.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- e-v-a-l-u-a-t-e- S-t-a-c-k-i-n-g- v-e-r-s-u-s-  
-L-i-g-h-t-G-B-M-.- S-t-a-c-k-i-n-g- y-i-e-l-d-s- a- +-1-.5-%- R-O-C---A-U-C- g-a-i-n-.- l-n- p-r-o-d-u-c-t-i-o-n-,- w-e-  
-e-v-a-l-u-a-t-e- i-f- t-h-i-s- g-a-i-n- o-u-t-w-e-i-g-h-s- t-h-e- e-x-t-r-a- i-n-f-e-r-e-n-c-e- l-a-t-e-n-c-y- o-f- r-u-n-n-i-n-g-  
-m-u-l-t-i-p-l-e- b-a-s-e- e-s-t-i-m-a-t-o-r-s-.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- W-h-a-t- m-o-d-e-l- d-i-d- y-o-u- u-s-e- a-s- t-h-e- m-e-t-a---l-e-a-r-n-e-r- a-n-d-  
-w-h-y-?-

\*- \*-D-e-e-p- D-i-v-e:-\*- E-x-p-l-a-i-n- h-o-w- m-u-l-t-i---s-t-a-g-e- s-t-a-c-k-i-n-g- c-a-n- l-e-a-d- t-o- t-a-r-g-e-t-  
-l-e-a-k-a-g-e- i-f- c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- i-s- n-o-t- c-o-n-f-i-g-u-r-e-d- p-r-o-p-e-r-l-y-.-

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-#-#-#- Q-u-e-s-t-i-o-n- 4-8:- M-o-d-e-l- S-e-l-e-c-t-i-o-n- Q-u-e-s-t-i-o-n- 4-8:- W-h-y- d-i-d- y-o-u- c-h-o-o-s-e- a-  
-S-t-a-c-k-i-n-g- E-n-s-e-m-b-l-e- o-v-e-r- a- s-i-n-g-l-e- t-u-n-e-d- L-i-g-h-t-G-B-M- m-o-d-e-l-?-

\*- \*-R-o-u-n-d:-\*- M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- |- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8/-1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- E-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g-,- s-t-a-c-k-i-n-g-,- b-i-a-s---v-a-r-i-a-n-c-e- t-r-a-d-e-o-f-f-.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- T-e-s-t-s- u-n-d-e-r-s-t-a-n-d-i-n-g- o-f- e-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g- a-n-d-  
-v-a-r-i-a-n-c-e- r-e-d-u-c-t-i-o-n-.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- S-t-a-c-k-i-n-g- c-o-m-b-i-n-e-s- m-o-d-e-l-s- w-i-t-h- d-i-f-f-e-r-e-n-t-  
-i-n-d-u-c-t-i-v-e- b-i-a-s-e-s- -(e.-g.- R-a-n-d-o-m- F-o-r-e-s-t,- L-i-g-h-t-G-B-M,- E-x-t-r-a-T-r-e-e-s)-.- L-i-g-h-t-G-B-M-  
-f-o-c-u-s-e-s- o-n- g-r-a-d-i-e-n-t- b-o-o-s-t-i-n-g- o-n- t-r-e-e-s,- w-h-i-l-e- R-a-n-d-o-m- F-o-r-e-s-t- u-s-e-s- b-a-g-g-i-n-g-.-  
-S-t-a-c-k-i-n-g- t-r-a-i-n-s- a- m-e-t-a---l-e-a-r-n-e-r- -(e.-g.- L-o-g-i-s-t-i-c- R-e-g-r-e-s-s-i-o-n)- o-n- t-h-e-i-r-  
-o-u-t---o-f---f-o-l-d- p-r-e-d-i-c-t-i-o-n-s,- w-h-i-c-h- h-e-l-p-s- s-m-o-o-t-h- o-u-t- i-n-d-i-v-i-d-u-a-l- e-r-r-o-r-s- a-n-d- l-o-w-e-r-s-  
-v-a-r-i-a-n-c-e-.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- T-h-i-n-k-i-n-g- S-t-a-c-k-i-n-g- a-l-w-a-y-s- i-m-p-r-o-v-e-s- p-e-r-f-o-r-m-a-n-c-e-  
-w-i-t-h-o-u-t- n-o-t-i-n-g- t-h-e- c-o-s-t- i-n- c-o-m-p-l-e-x-i-t-y- a-n-d- t-r-a-i-n-i-n-g- t-i-m-e-.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- e-v-a-l-u-a-t-e- S-t-a-c-k-i-n-g- v-e-r-s-u-s-  
-L-i-g-h-t-G-B-M-.- S-t-a-c-k-i-n-g- y-i-e-l-d-s- a- +-1-.5-%- R-O-C---A-U-C- g-a-i-n-.- l-n- p-r-o-d-u-c-t-i-o-n-,- w-e-  
-e-v-a-l-u-a-t-e- i-f- t-h-i-s- g-a-i-n- o-u-t-w-e-i-g-h-s- t-h-e- e-x-t-r-a- i-n-f-e-r-e-n-c-e- l-a-t-e-n-c-y- o-f- r-u-n-n-i-n-g-  
-m-u-l-t-i-p-l-e- b-a-s-e- e-s-t-i-m-a-t-o-r-s-.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- W-h-a-t- m-o-d-e-l- d-i-d- y-o-u- u-s-e- a-s- t-h-e- m-e-t-a---l-e-a-r-n-e-r- a-n-d-  
-w-h-y-?-

\*- \*-D-e-e-p- D-i-v-e:-\*- E-x-p-l-a-i-n- h-o-w- m-u-l-t-i---s-t-a-g-e- s-t-a-c-k-i-n-g- c-a-n- l-e-a-d- t-o- t-a-r-g-e-t-  
-l-e-a-k-a-g-e- i-f- c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- i-s- n-o-t- c-o-n-f-i-g-u-r-e-d- p-r-o-p-e-r-l-y-.-

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-#-#-#- Q-u-e-s-t-i-o-n- 4-9:- M-o-d-e-l- S-e-l-e-c-t-i-o-n- Q-u-e-s-t-i-o-n- 4-9:- W-h-y- d-i-d- y-o-u- c-h-o-o-s-e- a-  
-S-t-a-c-k-i-n-g- E-n-s-e-m-b-l-e- o-v-e-r- a- s-i-n-g-l-e- t-u-n-e-d- L-i-g-h-t-G-B-M- m-o-d-e-l-?-

\*- \*-R-o-u-n-d:-\*- M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- |- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8/-1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- E-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g-,- s-t-a-c-k-i-n-g-,- b-i-a-s---v-a-r-i-a-n-c-e- t-r-a-d-e-o-f-f-.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- -T-e-s-t-s- u-n-d-e-r-s-t-a-n-d-i-n-g- o-f- e-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g- a-n-d- v-a-r-i-a-n-c-e- r-e-d-u-c-t-i-o-n-.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- -S-t-a-c-k-i-n-g- c-o-m-b-i-n-e-s- m-o-d-e-l-s- w-i-t-h- d-i-f-f-e-r-e-n-t- i-n-d-u-c-t-i-v-e- b-i-a-s-e-s- -(e-.g.- R-a-n-d-o-m- F-o-r-e-s-t,- L-i-g-h-t-G-B-M,- E-x-t-r-a-T-r-e-e-s-)-. L-i-g-h-t-G-B-M- f-o-c-u-s-e-s- o-n- g-r-a-d-i-e-n-t- b-o-o-s-t-i-n-g- o-n- t-r-e-e-s,- w-h-i-l-e- R-a-n-d-o-m- F-o-r-e-s-t- u-s-e-s- b-a-g-g-i-n-g.- S-t-a-c-k-i-n-g- t-r-a-i-n-s- a- m-e-t-a--l-e-a-r-n-e-r- -(e-.g.- L-o-g-i-s-t-i-c- R-e-g-r-e-s-s-i-o-n)- o-n- t-h-e-i-r- o-u-t--o-f--f-o-l-d- p-r-e-d-i-c-t-i-o-n-s,- w-h-i-c-h- h-e-l-p-s- s-m-o-o-t-h- o-u-t- i-n-d-i-v-i-d-u-a-l- e-r-r-o-r-s- a-n-d- l-o-w-e-r-s- v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- -T-h-i-n-k-i-n-g- S-t-a-c-k-i-n-g- a-l-w-a-y-s- i-m-p-r-o-v-e-s- p-e-r-f-o-r-m-a-n-c-e- w-i-t-h-o-u-t- n-o-t-i-n-g- t-h-e- c-o-s-t- i-n- c-o-m-p-l-e-x-i-t-y- a-n-d- t-r-a-i-n-i-n-g- t-i-m-e.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n--G-r-a-d-e- A-n-s-w-e-r:-\*- -W-e- e-v-a-l-u-a-t-e- S-t-a-c-k-i-n-g- v-e-r-s-u-s- L-i-g-h-t-G-B-M.- S-t-a-c-k-i-n-g- y-i-e-l-d-s- a- +-1-.5%- R-O-C--A-U-C- g-a-i-n.- l-n- p-r-o-d-u-c-t-i-o-n,- w-e- e-v-a-l-u-a-t-e- i-f- t-h-i-s- g-a-i-n- o-u-t-w-e-i-g-h-s- t-h-e- e-x-t-r-a- i-n-f-e-r-e-n-c-e- l-a-t-e-n-c-y- o-f- r-u-n-n-i-n-g- m-u-l-t-i-p-l-e- b-a-s-e- e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- m-o-d-e-l- d-i-d- y-o-u- u-s-e- a-s- t-h-e- m-e-t-a--l-e-a-r-n-e-r- a-n-d- w-h-y?-

\*- \*-D-e-e-p- D-i-v-e:-\*- -E-x-p-l-a-i-n- h-o-w- m-u-l-t-i---s-t-a-g-e- s-t-a-c-k-i-n-g- c-a-n- l-e-a-d- t-o- t-a-r-g-e-t- l-e-a-k-a-g-e- i-f- c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- i-s- n-o-t- c-o-n-f-i-g-u-r-e-d- p-r-o-p-e-r-l-y.-

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##-#- Q-u-e-s-t-i-o-n- 5-0:- M-o-d-e-l- S-e-l-e-c-t-i-o-n- Q-u-e-s-t-i-o-n- 5-0:- W-h-y- d-i-d- y-o-u- c-h-o-o-s-e- a- S-t-a-c-k-i-n-g- E-n-s-e-m-b-l-e- o-v-e-r- a- s-i-n-g-l-e- t-u-n-e-d- L-i-g-h-t-G-B-M- m-o-d-e-l?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- |- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- -E-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g,- s-t-a-c-k-i-n-g,- b-i-a-s--v-a-r-i-a-n-c-e- t-r-a-d-e-o-f-f.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- -T-e-s-t-s- u-n-d-e-r-s-t-a-n-d-i-n-g- o-f- e-n-s-e-m-b-l-e- l-e-a-r-n-i-n-g- a-n-d- v-a-r-i-a-n-c-e- r-e-d-u-c-t-i-o-n-.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- -S-t-a-c-k-i-n-g- c-o-m-b-i-n-e-s- m-o-d-e-l-s- w-i-t-h- d-i-f-f-e-r-e-n-t- i-n-d-u-c-t-i-v-e- b-i-a-s-e-s- -(e-.g.- R-a-n-d-o-m- F-o-r-e-s-t,- L-i-g-h-t-G-B-M,- E-x-t-r-a-T-r-e-e-s-)-. L-i-g-h-t-G-B-M- f-o-c-u-s-e-s- o-n- g-r-a-d-i-e-n-t- b-o-o-s-t-i-n-g- o-n- t-r-e-e-s,- w-h-i-l-e- R-a-n-d-o-m- F-o-r-e-s-t- u-s-e-s- b-a-g-g-i-n-g.- S-t-a-c-k-i-n-g- t-r-a-i-n-s- a- m-e-t-a--l-e-a-r-n-e-r- -(e-.g.- L-o-g-i-s-t-i-c- R-e-g-r-e-s-s-i-o-n)- o-n- t-h-e-i-r- o-u-t--o-f--f-o-l-d- p-r-e-d-i-c-t-i-o-n-s,- w-h-i-c-h- h-e-l-p-s- s-m-o-o-t-h- o-u-t- i-n-d-i-v-i-d-u-a-l- e-r-r-o-r-s- a-n-d- l-o-w-e-r-s- v-a-r-i-a-n-c-e.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- -T-h-i-n-k-i-n-g- S-t-a-c-k-i-n-g- a-l-w-a-y-s- i-m-p-r-o-v-e-s- p-e-r-f-o-r-m-a-n-c-e- w-i-t-h-o-u-t- n-o-t-i-n-g- t-h-e- c-o-s-t- i-n- c-o-m-p-l-e-x-i-t-y- a-n-d- t-r-a-i-n-i-n-g- t-i-m-e.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n--G-r-a-d-e- A-n-s-w-e-r:-\*- -W-e- e-v-a-l-u-a-t-e- S-t-a-c-k-i-n-g- v-e-r-s-u-s- L-i-g-h-t-G-B-M.- S-t-a-c-k-i-n-g- y-i-e-l-d-s- a- +-1-.5%- R-O-C--A-U-C- g-a-i-n.- l-n- p-r-o-d-u-c-t-i-o-n,- w-e- e-v-a-l-u-a-t-e- i-f- t-h-i-s- g-a-i-n- o-u-t-w-e-i-g-h-s- t-h-e- e-x-t-r-a- i-n-f-e-r-e-n-c-e- l-a-t-e-n-c-y- o-f- r-u-n-n-i-n-g- m-u-l-t-i-p-l-e- b-a-s-e- e-s-t-i-m-a-t-o-r-s.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- m-o-d-e-l- d-i-d- y-o-u- u-s-e- a-s- t-h-e- m-e-t-a--l-e-a-r-n-e-r- a-n-d- w-h-y?-

\*- \*-D-e-e-p- D-i-v-e:-\*- -E-x-p-l-a-i-n- h-o-w- m-u-l-t-i---s-t-a-g-e- s-t-a-c-k-i-n-g- c-a-n- l-e-a-d- t-o- t-a-r-g-e-t- l-e-a-k-a-g-e- i-f- c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- i-s- n-o-t- c-o-n-f-i-g-u-r-e-d- p-r-o-p-e-r-l-y.-

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##-Q-u-e-s-t-i-o-n- 5-1:- I-m-b-a-l-a-n-c-e-d- D-a-t-a- Q-u-e-s-t-i-o-n- 5-1:- H-o-w- d-o-e-s- p-r-o-b-a-b-i-l-i-t-y- c-a-l-i-b-r-a-t-i-o-n- i-m-p-a-c-t- t-h-r-e-s-h-o-l-d- o-p-t-i-m-i-z-a-t-i-o-n- o-n- i-m-b-a-l-a-n-c-e-d- d-a-t-a-s-e-t-s?-

\*- \*-R-o-u-n-d:-\*- R-e-s-e-a-r-c-h-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- |- \*-I-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- P-r-o-b-a-b-i-l-i-t-y- c-a-l-i-b-r-a-t-i-o-n- (-P-l-a-t-t/-l-s-o-t-o-n-i-c-),- B-r-i-e-r- S-c-o-r-e,- F-1- o-p-t-i-m-i-z-a-t-i-o-n.-

\*- \*-W-h-y- I-n-t-e-r-v-i-e-w-e-r- A-s-k-s- I-t:-\*- T-e-s-t-s- c-a-l-i-b-r-a-t-i-o-n- a-n-d- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- s-e-l-e-c-t-i-o-n.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- I-m-b-a-l-a-n-c-e-d- c-l-a-s-s-i-f-i-e-r-s- o-f-t-e-n- p-r-o-d-u-c-e- d-i-s-t-o-r-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s- (-e-.g.-, b-o-o-s-t-i-n-g- m-o-d-e-l-s- p-u-s-h- s-c-o-r-e-s- n-e-a-r- 0- o-r- 1).- C-a-l-i-b-r-a-t-i-o-n- m-a-p-s- t-h-e-s-e- s-c-o-r-e-s- b-a-c-k- t-o- e-m-p-i-r-i-c-a-l- p-r-o-b-a-b-i-l-i-t-i-e-s.- O-n-c-e- c-a-l-i-b-r-a-t-e-d,- w-e- c-a-n- s-e-a-r-c-h- f-o-r- a- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- (-l-i-k-e- 0.-2-0)- t-o- m-a-x-i-m-i-z-e- F-1,- k-n-o-w-i-n-g- t-h-a-t- 2-0%- m-e-a-n-s- a- 1---i-n---5- p-r-o-b-a-b-i-l-i-t-y.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- O-p-t-i-m-i-z-i-n-g- t-h-e- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- o-n- u-n-c-a-l-i-b-r-a-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s,- w-h-i-c-h- r-e-s-u-l-t-s- i-n- u-n-s-t-a-b-l-e- d-e-c-i-s-i-o-n- b-o-u-n-d-a-r-i-e-s.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- e-v-a-l-u-a-t-e- P-l-a-t-t- s-c-a-l-i-n-g- a-n-d- l-s-o-t-o-n-i-c- r-e-g-r-e-s-s-i-o-n.- S-i-n-c-e- o-u-r- d-a-t-a-s-e-t- i-s- m-e-d-i-u-m---s-i-z-e-d,- P-l-a-t-t- s-c-a-l-i-n-g- p-r-e-v-e-n-t-s- o-v-e-r-f-i-t-t-i-n-g- a-n-d- m-i-n-i-m-i-z-e-s- t-h-e- E-x-p-e-c-t-e-d- C-a-l-i-b-r-a-t-i-o-n- E-r-r-o-r- (-E-C-E)- t-o- 0.-0-2-3-3.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- W-h-y- n-o-t- u-s-e- S-M-O-T-E- t-o- r-e-s-o-l-v-e- c-l-a-s-s- i-m-b-a-l-a-n-c-e- i-n-s-t-e-a-d?-

\*- \*-D-e-e-p- D-i-v-e:-\*- P-r-o-o-f- t-h-a-t- B-r-i-e-r- S-c-o-r-e- c-a-n- b-e- d-e-c-o-m-p-o-s-e-d- i-n-t-o- r-e-l-i-a-b-i-l-i-t-y,- r-e-s-o-l-u-t-i-o-n,- a-n-d- u-n-c-e-r-t-a-i-n-t-y.-

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##-Q-u-e-s-t-i-o-n- 5-2:- I-m-b-a-l-a-n-c-e-d- D-a-t-a- Q-u-e-s-t-i-o-n- 5-2:- H-o-w- d-o-e-s- p-r-o-b-a-b-i-l-i-t-y- c-a-l-i-b-r-a-t-i-o-n- i-m-p-a-c-t- t-h-r-e-s-h-o-l-d- o-p-t-i-m-i-z-a-t-i-o-n- o-n- i-m-b-a-l-a-n-c-e-d- d-a-t-a-s-e-t-s?-

\*- \*-R-o-u-n-d:-\*- R-e-s-e-a-r-c-h-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- |- \*-I-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- P-r-o-b-a-b-i-l-i-t-y- c-a-l-i-b-r-a-t-i-o-n- (-P-l-a-t-t/-l-s-o-t-o-n-i-c-),- B-r-i-e-r- S-c-o-r-e,- F-1- o-p-t-i-m-i-z-a-t-i-o-n.-

\*- \*-W-h-y- I-n-t-e-r-v-i-e-w-e-r- A-s-k-s- I-t:-\*- T-e-s-t-s- c-a-l-i-b-r-a-t-i-o-n- a-n-d- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- s-e-l-e-c-t-i-o-n.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- I-m-b-a-l-a-n-c-e-d- c-l-a-s-s-i-f-i-e-r-s- o-f-t-e-n- p-r-o-d-u-c-e- d-i-s-t-o-r-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s- (-e-.g.-, b-o-o-s-t-i-n-g- m-o-d-e-l-s- p-u-s-h- s-c-o-r-e-s- n-e-a-r- 0- o-r- 1).- C-a-l-i-b-r-a-t-i-o-n- m-a-p-s- t-h-e-s-e- s-c-o-r-e-s- b-a-c-k- t-o- e-m-p-i-r-i-c-a-l- p-r-o-b-a-b-i-l-i-t-i-e-s.- O-n-c-e- c-a-l-i-b-r-a-t-e-d,- w-e- c-a-n- s-e-a-r-c-h- f-o-r- a- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- (-l-i-k-e- 0.-2-0)- t-o- m-a-x-i-m-i-z-e- F-1,- k-n-o-w-i-n-g- t-h-a-t- 2-0%- m-e-a-n-s- a- 1---i-n---5- p-r-o-b-a-b-i-l-i-t-y.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- -O-p-t-i-m-i-z-i-n-g- t-h-e- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- -o-n- -u-n-c-a-l-i-b-r-a-t-e-d-  
-p-r-o-b-a-b-i-l-i-t-i-e-s,- -w-h-i-c-h- -r-e-s-u-l-t-s- -i-n- -u-n-s-t-a-b-l-e- d-e-c-i-s-i-o-n- -b-o-u-n-d-a-r-i-e-s.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -e-v-a-l-u-a-t-e- -P-l-a-t-t- -s-c-a-l-i-n-g- -a-n-d-  
-l-s-o-t-o-n-i-c- -r-e-g-r-e-s-s-i-o-n,- -S-i-n-c-e- -o-u-r- -d-a-t-a-s-e-t- -i-s- -m-e-d-i-u-m---s-i-z-e-d,- -P-l-a-t-t- -s-c-a-l-i-n-g-  
-p-r-e-v-e-n-t-s- -o-v-e-r-f-i-t-t-i-n-g- -a-n-d- -m-i-n-i-m-i-z-e-s- t-h-e- -E-x-p-e-c-t-e-d- -C-a-l-i-b-r-a-t-i-o-n- -E-r-r-o-r- -(E-C-E)- -t-o-  
-0-.0-2-3-3.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-y- -n-o-t- -u-s-e- -S-M-O-T-E- -t-o- -r-e-s-o-l-v-e- -c-l-a-s-s- -i-m-b-a-l-a-n-c-e-  
-i-n-s-t-e-a-d?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -P-r-o-o-f- t-h-a-t- -B-r-i-e-r- -S-c-o-r-e- -c-a-n- -b-e- -d-e-c-o-m-p-o-s-e-d- -i-n-t-o- -r-e-l-i-a-b-i-l-i-t-y,-  
-r-e-s-o-l-u-t-i-o-n,- -a-n-d- -u-n-c-e-r-t-a-i-n-t-y.-

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###- -Q-u-e-s-t-i-o-n- -5-3:- -l-m-b-a-l-a-n-c-e-d- -D-a-t-a- -Q-u-e-s-t-i-o-n- -5-3:- -H-o-w- -d-o-e-s- -p-r-o-b-a-b-i-l-i-t-y-  
-c-a-l-i-b-r-a-t-i-o-n- -i-m-p-a-c-t- t-h-r-e-s-h-o-l-d- -o-p-t-i-m-i-z-a-t-i-o-n- -o-n- -i-m-b-a-l-a-n-c-e-d- -d-a-t-a-s-e-t-s?-

\*- \*-R-o-u-n-d:-\*- -R-e-s-e-a-r-c-h-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -P-r-o-b-a-b-i-l-i-t-y- -c-a-l-i-b-r-a-t-i-o-n- -(P-l-a-t-t-/l-s-o-t-o-n-i-c)-, -B-r-i-e-r- -S-c-o-r-e,-  
-F-1- -o-p-t-i-m-i-z-a-t-i-o-n.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -c-a-l-i-b-r-a-t-i-o-n- -a-n-d- -d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d-  
-s-e-l-e-c-t-i-o-n.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -l-m-b-a-l-a-n-c-e-d- -c-l-a-s-s-i-f-i-e-r-s- -o-f-t-e-n- -p-r-o-d-u-c-e- -d-i-s-t-o-r-t-e-d-  
-p-r-o-b-a-b-i-l-i-t-i-e-s- -(e.g.-, -b-o-o-s-t-i-n-g- -m-o-d-e-l-s- -p-u-s-h- -s-c-o-r-e-s- -n-e-a-r- -0- -o-r- -1)-, -C-a-l-i-b-r-a-t-i-o-n-  
-m-a-p-s- t-h-e-s-e- -s-c-o-r-e-s- -b-a-c-k- -t-o- -e-m-p-i-r-i-c-a-l- -p-r-o-b-a-b-i-l-i-t-i-e-s.- -O-n-c-e- -c-a-l-i-b-r-a-t-e-d,- -w-e- -c-a-n-  
-s-e-a-r-c-h- -f-o-r- -a- -d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- -(l-i-k-e- -0-.2-0)- -t-o- -m-a-x-i-m-i-z-e- -F-1,- -k-n-o-w-i-n-g- t-h-a-t-  
-2-0%- -m-e-a-n-s- -a- -1---i-n---5- -p-r-o-b-a-b-i-l-i-t-y.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- -O-p-t-i-m-i-z-i-n-g- t-h-e- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- -o-n- -u-n-c-a-l-i-b-r-a-t-e-d-  
-p-r-o-b-a-b-i-l-i-t-i-e-s,- -w-h-i-c-h- -r-e-s-u-l-t-s- -i-n- -u-n-s-t-a-b-l-e- d-e-c-i-s-i-o-n- -b-o-u-n-d-a-r-i-e-s.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -e-v-a-l-u-a-t-e- -P-l-a-t-t- -s-c-a-l-i-n-g- -a-n-d-  
-l-s-o-t-o-n-i-c- -r-e-g-r-e-s-s-i-o-n,- -S-i-n-c-e- -o-u-r- -d-a-t-a-s-e-t- -i-s- -m-e-d-i-u-m---s-i-z-e-d,- -P-l-a-t-t- -s-c-a-l-i-n-g-  
-p-r-e-v-e-n-t-s- -o-v-e-r-f-i-t-t-i-n-g- -a-n-d- -m-i-n-i-m-i-z-e-s- t-h-e- -E-x-p-e-c-t-e-d- -C-a-l-i-b-r-a-t-i-o-n- -E-r-r-o-r- -(E-C-E)- -t-o-  
-0-.0-2-3-3.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-y- -n-o-t- -u-s-e- -S-M-O-T-E- -t-o- -r-e-s-o-l-v-e- -c-l-a-s-s- -i-m-b-a-l-a-n-c-e-  
-i-n-s-t-e-a-d?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -P-r-o-o-f- t-h-a-t- -B-r-i-e-r- -S-c-o-r-e- -c-a-n- -b-e- -d-e-c-o-m-p-o-s-e-d- -i-n-t-o- -r-e-l-i-a-b-i-l-i-t-y,-  
-r-e-s-o-l-u-t-i-o-n,- -a-n-d- -u-n-c-e-r-t-a-i-n-t-y.-

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###- -Q-u-e-s-t-i-o-n- -5-4:- -l-m-b-a-l-a-n-c-e-d- -D-a-t-a- -Q-u-e-s-t-i-o-n- -5-4:- -H-o-w- -d-o-e-s- -p-r-o-b-a-b-i-l-i-t-y-  
-c-a-l-i-b-r-a-t-i-o-n- -i-m-p-a-c-t- t-h-r-e-s-h-o-l-d- -o-p-t-i-m-i-z-a-t-i-o-n- -o-n- -i-m-b-a-l-a-n-c-e-d- -d-a-t-a-s-e-t-s?-

\*- \*-R-o-u-n-d:-\*- -R-e-s-e-a-r-c-h-

Q: How do you think about the current market situation?

A: I think the current market situation is quite challenging. There are many uncertainties, and it's important to stay focused on our core business and long-term goals.

Q: Can you tell me about your experience with leading a team?

A: Yes, I have led a team of developers for the past few years. I focus on setting clear goals, providing guidance, and ensuring the team has the resources they need to succeed. I also encourage open communication and collaboration within the team.

Q: How do you handle conflict or disagreements within a team?

A: I believe in addressing conflicts directly and calmly. I listen to all sides of the story and try to find a common ground or a compromise that works for everyone. I also encourage team members to resolve conflicts on their own when possible.

Q: What is your favorite programming language and why?

A: My favorite programming language is Python. I like it because it's easy to learn and use, and it has a wide range of applications. I also enjoy the clean and readable syntax of Python.

Q: How do you stay up-to-date with the latest technology trends?

A: I stay up-to-date by reading industry news, attending conferences, and taking courses. I also like to experiment with new technologies and frameworks on my own time.

Q: Can you describe a challenging project you worked on?

A: One of the most challenging projects I worked on was developing a new feature for our product. It involved a lot of research, design, and testing, and we had to overcome many obstacles along the way.

Q: How do you think about the current market situation?

A: I think the current market situation is quite challenging. There are many uncertainties, and it's important to stay focused on our core business and long-term goals.

Q: Can you tell me about your experience with leading a team?

A: Yes, I have led a team of developers for the past few years. I focus on setting clear goals, providing guidance, and ensuring the team has the resources they need to succeed. I also encourage open communication and collaboration within the team.

Q: How do you handle conflict or disagreements within a team?

A: I believe in addressing conflicts directly and calmly. I listen to all sides of the story and try to find a common ground or a compromise that works for everyone. I also encourage team members to resolve conflicts on their own when possible.

Q: What is your favorite programming language and why?

-i-n-s-t-e-a-d-?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -P-r-o-o-f- -t-h-a-t- -B-r-i-e-r- -S-c-o-r-e- -c-a-n- -b-e- -d-e-c-o-m-p-o-s-e-d- -i-n-t-o- -r-e-l-i-a-b-i-l-i-t-y-,-  
-r-e-s-o-l-u-t-i-o-n-,- -a-n-d- -u-n-c-e-r-t-a-i-n-t-y-.-

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-##-#- -Q-u-e-s-t-i-o-n- -5-6:- -l-m-b-a-l-a-n-c-e-d- -D-a-t-a- -Q-u-e-s-t-i-o-n- -5-6:- -H-o-w- -d-o-e-s- -p-r-o-b-a-b-i-l-i-t-y-  
-c-a-l-i-b-r-a-t-i-o-n- -i-m-p-a-c-t- -t-h-r-e-s-h-o-l-d- -o-p-t-i-m-i-z-a-t-i-o-n- -o-n- -i-m-b-a-l-a-n-c-e-d- -d-a-t-a-s-e-t-s-?-

-\*- \*-R-o-u-n-d:-\*- -R-e-s-e-a-r-c-h-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -P-r-o-b-a-b-i-l-i-t-y- -c-a-l-i-b-r-a-t-i-o-n- -(P-l-a-t-t/-l-s-o-t-o-n-i-c)-,- -B-r-i-e-r- -S-c-o-r-e-,-  
-F-1- -o-p-t-i-m-i-z-a-t-i-o-n-.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -c-a-l-i-b-r-a-t-i-o-n- -a-n-d- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d-  
-s-e-l-e-c-t-i-o-n-.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -l-m-b-a-l-a-n-c-e-d- -c-l-a-s-s-i-f-i-e-r-s- -o-f-t-e-n- -p-r-o-d-u-c-e- -d-i-s-t-o-r-t-e-d-  
-p-r-o-b-a-b-i-l-i-t-i-e-s- -(e-.g.-,- -b-o-o-s-t-i-n-g- -m-o-d-e-l-s- -p-u-s-h- -s-c-o-r-e-s- -n-e-a-r- -0- -o-r- -1)-,- -C-a-l-i-b-r-a-t-i-o-n-  
-m-a-p-s- -t-h-e-s-e- -s-c-o-r-e-s- -b-a-c-k- -t-o- -e-m-p-i-r-i-c-a-l- -p-r-o-b-a-b-i-l-i-t-i-e-s-.- -O-n-c-e- -c-a-l-i-b-r-a-t-e-d,- -w-e- -c-a-n-  
-s-e-a-r-c-h- -f-o-r- -a- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d- -(l-i-k-e- -0.-2-0)- -t-o- -m-a-x-i-m-i-z-e- -F-1,- -k-n-o-w-i-n-g- -t-h-a-t-  
-2-0%- -m-e-a-n-s- -a- -1---i-n---5- -p-r-o-b-a-b-i-l-i-t-y-.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -O-p-t-i-m-i-z-i-n-g- -t-h-e- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d- -o-n- -u-n-c-a-l-i-b-r-a-t-e-d-  
-p-r-o-b-a-b-i-l-i-t-i-e-s-,- -w-h-i-c-h- -r-e-s-u-l-t-s- -i-n- -u-n-s-t-a-b-l-e- -d-e-c-i-s-i-o-n- -b-o-u-n-d-a-r-i-e-s-.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -e-v-a-l-u-a-t-e- -P-l-a-t-t- -s-c-a-l-i-n-g- -a-n-d-  
-l-s-o-t-o-n-i-c- -r-e-g-r-e-s-s-i-o-n-.- -S-i-n-c-e- -o-u-r- -d-a-t-a-s-e-t- -i-s- -m-e-d-i-u-m---s-i-z-e-d,- -P-l-a-t-t- -s-c-a-l-i-n-g-  
-p-r-e-v-e-n-t-s- -o-v-e-r-f-i-t-t-i-n-g- -a-n-d- -m-i-n-i-m-i-z-e-s- -t-h-e- -E-x-p-e-c-t-e-d- -C-a-l-i-b-r-a-t-i-o-n- -E-r-r-o-r- -(E-C-E)- -t-o-  
-0.-0-2-3-3.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-y- -n-o-t- -u-s-e- -S-M-O-T-E- -t-o- -r-e-s-o-l-v-e- -c-l-a-s-s- -i-m-b-a-l-a-n-c-e-  
-i-n-s-t-e-a-d-?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -P-r-o-o-f- -t-h-a-t- -B-r-i-e-r- -S-c-o-r-e- -c-a-n- -b-e- -d-e-c-o-m-p-o-s-e-d- -i-n-t-o- -r-e-l-i-a-b-i-l-i-t-y-,-  
-r-e-s-o-l-u-t-i-o-n-,- -a-n-d- -u-n-c-e-r-t-a-i-n-t-y-.-

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-##-#- -Q-u-e-s-t-i-o-n- -5-7:- -l-m-b-a-l-a-n-c-e-d- -D-a-t-a- -Q-u-e-s-t-i-o-n- -5-7:- -H-o-w- -d-o-e-s- -p-r-o-b-a-b-i-l-i-t-y-  
-c-a-l-i-b-r-a-t-i-o-n- -i-m-p-a-c-t- -t-h-r-e-s-h-o-l-d- -o-p-t-i-m-i-z-a-t-i-o-n- -o-n- -i-m-b-a-l-a-n-c-e-d- -d-a-t-a-s-e-t-s-?-

-\*- \*-R-o-u-n-d:-\*- -R-e-s-e-a-r-c-h-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -P-r-o-b-a-b-i-l-i-t-y- -c-a-l-i-b-r-a-t-i-o-n- -(P-l-a-t-t/-l-s-o-t-o-n-i-c)-,- -B-r-i-e-r- -S-c-o-r-e-,-  
-F-1- -o-p-t-i-m-i-z-a-t-i-o-n-.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -c-a-l-i-b-r-a-t-i-o-n- -a-n-d- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d-  
-s-e-l-e-c-t-i-o-n-.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -l-m-b-a-l-a-n-c-e-d- -c-l-a-s-s-i-f-i-e-r-s- -o-f-t-e-n- -p-r-o-d-u-c-e- -d-i-s-t-o-r-t-e-d-  
-p-r-o-b-a-b-i-l-i-t-i-e-s- -(e-.g.-,- -b-o-o-s-t-i-n-g- -m-o-d-e-l-s- -p-u-s-h- -s-c-o-r-e-s- -n-e-a-r- -0- -o-r- -1)-,- -C-a-l-i-b-r-a-t-i-o-n-

-m-a-p-s- t-h-e-s-e- -s-c-o-r-e-s- -b-a-c-k- -t-o- -e-m-p-i-r-i-c-a-l- -p-r-o-b-a-b-i-l-i-t-i-e-s-.- -O-n-c-e- -c-a-l-i-b-r-a-t-e-d-,- -w-e- -c-a-n-  
-s-e-a-r-c-h- -f-o-r- -a- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d- -(l-i-k-e- -0-.-2-0)- -t-o- -m-a-x-i-m-i-z-e- -F-1-,- -k-n-o-w-i-n-g- -t-h-a-t-  
-2-0%- -m-e-a-n-s- -a- -1---i-n---5- -p-r-o-b-a-b-i-l-i-t-y-.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-:-\*- -O-p-t-i-m-i-z-i-n-g- -t-h-e- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d- -o-n- -u-n-c-a-l-i-b-r-a-t-e-d-  
-p-r-o-b-a-b-i-l-i-t-i-e-s-,- -w-h-i-c-h- -r-e-s-u-l-t-s- -i-n- -u-n-s-t-a-b-l-e- -d-e-c-i-s-i-o-n- -b-o-u-n-d-a-r-i-e-s-.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-:-\*- -W-e- -e-v-a-l-u-a-t-e- -P-l-a-t-t- -s-c-a-l-i-n-g- -a-n-d-  
-l-s-o-t-o-n-i-c- -r-e-g-r-e-s-s-i-o-n-.- -S-i-n-c-e- -o-u-r- -d-a-t-a-s-e-t- -i-s- -m-e-d-i-u-m---s-i-z-e-d-,- -P-l-a-t-t- -s-c-a-l-i-n-g-  
-p-r-e-v-e-n-t-s- -o-v-e-r-f-i-t-t-i-n-g- -a-n-d- -m-i-n-i-m-i-z-e-s- -t-h-e- -E-x-p-e-c-t-e-d- -C-a-l-i-b-r-a-t-i-o-n- -E-r-r-o-r- -(E-C-E)- -t-o-  
-0-.-0-2-3-3-.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s-:-\*- -W-h-y- -n-o-t- -u-s-e- -S-M-O-T-E- -t-o- -r-e-s-o-l-v-e- -c-l-a-s-s- -i-m-b-a-l-a-n-c-e-  
-i-n-s-t-e-a-d-?-

-\*- \*-D-e-e-p- -D-i-v-e-:-\*- -P-r-o-o-f- -t-h-a-t- -B-r-i-e-r- -S-c-o-r-e- -c-a-n- -b-e- -d-e-c-o-m-p-o-s-e-d- -i-n-t-o- -r-e-l-i-a-b-i-l-i-t-y-,-  
-r-e-s-o-l-u-t-i-o-n-,- -a-n-d- -u-n-c-e-r-t-a-i-n-t-y-.-

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-##- -Q-u-e-s-t-i-o-n- -5-8-:- -l-m-b-a-l-a-n-c-e-d- -D-a-t-a- -Q-u-e-s-t-i-o-n- -5-8-:- -H-o-w- -d-o-e-s- -p-r-o-b-a-b-i-l-i-t-y-  
-c-a-l-i-b-r-a-t-i-o-n- -i-m-p-a-c-t- -t-h-r-e-s-h-o-l-d- -o-p-t-i-m-i-z-a-t-i-o-n- -o-n- -i-m-b-a-l-a-n-c-e-d- -d-a-t-a-s-e-t-s-?-

-\*- \*-R-o-u-n-d-:-\*- -R-e-s-e-a-r-c-h-

-\*- \*-D-i-f-f-i-c-u-l-t-y-:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e-:-\*- -9-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d-:-\*- -P-r-o-b-a-b-i-l-i-t-y- -c-a-l-i-b-r-a-t-i-o-n- -(P-l-a-t-t-/l-s-o-t-o-n-i-c)-,- -B-r-i-e-r- -S-c-o-r-e-,-  
-F-1- -o-p-t-i-m-i-z-a-t-i-o-n-.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t-:-\*- -T-e-s-t-s- -c-a-l-i-b-r-a-t-i-o-n- -a-n-d- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d-  
-s-e-l-e-c-t-i-o-n-.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-:-\*- -l-m-b-a-l-a-n-c-e-d- -c-l-a-s-s-i-f-i-e-r-s- -o-f-t-e-n- -p-r-o-d-u-c-e- -d-i-s-t-o-r-t-e-d-  
-p-r-o-b-a-b-i-l-i-t-i-e-s- -(e.-g.-,- -b-o-o-s-t-i-n-g- -m-o-d-e-l-s- -p-u-s-h- -s-c-o-r-e-s- -n-e-a-r- -0- -o-r- -1)-.- -C-a-l-i-b-r-a-t-i-o-n-  
-m-a-p-s- t-h-e-s-e- -s-c-o-r-e-s- -b-a-c-k- -t-o- -e-m-p-i-r-i-c-a-l- -p-r-o-b-a-b-i-l-i-t-i-e-s-.- -O-n-c-e- -c-a-l-i-b-r-a-t-e-d-,- -w-e- -c-a-n-  
-s-e-a-r-c-h- -f-o-r- -a- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d- -(l-i-k-e- -0-.-2-0)- -t-o- -m-a-x-i-m-i-z-e- -F-1-,- -k-n-o-w-i-n-g- -t-h-a-t-  
-2-0%- -m-e-a-n-s- -a- -1---i-n---5- -p-r-o-b-a-b-i-l-i-t-y-.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-:-\*- -O-p-t-i-m-i-z-i-n-g- -t-h-e- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d- -o-n- -u-n-c-a-l-i-b-r-a-t-e-d-  
-p-r-o-b-a-b-i-l-i-t-i-e-s-,- -w-h-i-c-h- -r-e-s-u-l-t-s- -i-n- -u-n-s-t-a-b-l-e- -d-e-c-i-s-i-o-n- -b-o-u-n-d-a-r-i-e-s-.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-:-\*- -W-e- -e-v-a-l-u-a-t-e- -P-l-a-t-t- -s-c-a-l-i-n-g- -a-n-d-  
-l-s-o-t-o-n-i-c- -r-e-g-r-e-s-s-i-o-n-.- -S-i-n-c-e- -o-u-r- -d-a-t-a-s-e-t- -i-s- -m-e-d-i-u-m---s-i-z-e-d-,- -P-l-a-t-t- -s-c-a-l-i-n-g-  
-p-r-e-v-e-n-t-s- -o-v-e-r-f-i-t-t-i-n-g- -a-n-d- -m-i-n-i-m-i-z-e-s- -t-h-e- -E-x-p-e-c-t-e-d- -C-a-l-i-b-r-a-t-i-o-n- -E-r-r-o-r- -(E-C-E)- -t-o-  
-0-.-0-2-3-3-.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s-:-\*- -W-h-y- -n-o-t- -u-s-e- -S-M-O-T-E- -t-o- -r-e-s-o-l-v-e- -c-l-a-s-s- -i-m-b-a-l-a-n-c-e-  
-i-n-s-t-e-a-d-?-

-\*- \*-D-e-e-p- -D-i-v-e-:-\*- -P-r-o-o-f- -t-h-a-t- -B-r-i-e-r- -S-c-o-r-e- -c-a-n- -b-e- -d-e-c-o-m-p-o-s-e-d- -i-n-t-o- -r-e-l-i-a-b-i-l-i-t-y-,-  
-r-e-s-o-l-u-t-i-o-n-,- -a-n-d- -u-n-c-e-r-t-a-i-n-t-y-.-

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-##- -Q-u-e-s-t-i-o-n- -5-9-:- -l-m-b-a-l-a-n-c-e-d- -D-a-t-a- -Q-u-e-s-t-i-o-n- -5-9-:- -H-o-w- -d-o-e-s- -p-r-o-b-a-b-i-l-i-t-y-

-c-a-l-i-b-r-a-t-i-o-n- -i-m-p-a-c-t- -t-h-r-e-s-h-o-l-d- -o-p-t-i-m-i-z-a-t-i-o-n- -o-n- -i-m-b-a-l-a-n-c-e-d- -d-a-t-a-s-e-t-s-?-

-\*- \*-R-o-u-n-d:-\*- -R-e-s-e-a-r-c-h-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -P-r-o-b-a-b-i-l-i-t-y- -c-a-l-i-b-r-a-t-i-o-n- -(P-l-a-t-t/-l-s-o-t-o-n-i-c)-, -B-r-i-e-r- -S-c-o-r-e-, -F-1- -o-p-t-i-m-i-z-a-t-i-o-n-.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -c-a-l-i-b-r-a-t-i-o-n- -a-n-d- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d- -s-e-l-e-c-t-i-o-n-.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -l-m-b-a-l-a-n-c-e-d- -c-l-a-s-s-i-f-i-e-r-s- -o-f-t-e-n- -p-r-o-d-u-c-e- -d-i-s-t-o-r-t-e-d- -p-r-o-b-a-b-i-l-i-t-i-e-s- -(e-.g.-, -b-o-o-s-t-i-n-g- -m-o-d-e-l-s- -p-u-s-h- -s-c-o-r-e-s- -n-e-a-r- -0- -o-r- -1)-, -C-a-l-i-b-r-a-t-i-o-n- -m-a-p-s- -t-h-e-s-e- -s-c-o-r-e-s- -b-a-c-k- -t-o- -e-m-p-i-r-i-c-a-l- -p-r-o-b-a-b-i-l-i-t-i-e-s-.- -O-n-c-e- -c-a-l-i-b-r-a-t-e-d-, -w-e- -c-a-n- -s-e-a-r-c-h- -f-o-r- -a- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d- -(l-i-k-e- -0-.-2-0)- -t-o- -m-a-x-i-m-i-z-e- -F-1-, -k-n-o-w-i-n-g- -t-h-a-t- -2-0%- -m-e-a-n-s- -a- -1---i-n---5- -p-r-o-b-a-b-i-l-i-t-y-.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -O-p-t-i-m-i-z-i-n-g- -t-h-e- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d- -o-n- -u-n-c-a-l-i-b-r-a-t-e-d- -p-r-o-b-a-b-i-l-i-t-i-e-s-, -w-h-i-c-h- -r-e-s-u-l-t-s- -i-n- -u-n-s-t-a-b-l-e- -d-e-c-i-s-i-o-n- -b-o-u-n-d-a-r-i-e-s-.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -e-v-a-l-u-a-t-e- -P-l-a-t-t- -s-c-a-l-i-n-g- -a-n-d- -l-s-o-t-o-n-i-c- -r-e-g-r-e-s-s-i-o-n-.- -S-i-n-c-e- -o-u-r- -d-a-t-a-s-e-t- -i-s- -m-e-d-i-u-m---s-i-z-e-d-, -P-l-a-t-t- -s-c-a-l-i-n-g- -p-r-e-v-e-n-t-s- -o-v-e-r-f-i-t-t-i-n-g- -a-n-d- -m-i-n-i-m-i-z-e-s- -t-h-e- -E-x-p-e-c-t-e-d- -C-a-l-i-b-r-a-t-i-o-n- -E-r-r-o-r- -(E-C-E)- -t-o- -0-.-0-2-3-.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-y- -n-o-t- -u-s-e- -S-M-O-T-E- -t-o- -r-e-s-o-l-v-e- -c-l-a-s-s- -i-m-b-a-l-a-n-c-e- -i-n-s-t-e-a-d-?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -P-r-o-o-f- -t-h-a-t- -B-r-i-e-r- -S-c-o-r-e- -c-a-n- -b-e- -d-e-c-o-m-p-o-s-e-d- -i-n-t-o- -r-e-l-i-a-b-i-l-i-t-y-, -r-e-s-o-l-u-t-i-o-n-, -a-n-d- -u-n-c-e-r-t-a-i-n-t-y-.-

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-#-#-#- -Q-u-e-s-t-i-o-n- -6-0:- -l-m-b-a-l-a-n-c-e-d- -D-a-t-a- -Q-u-e-s-t-i-o-n- -6-0:- -H-o-w- -d-o-e-s- -p-r-o-b-a-b-i-l-i-t-y- -c-a-l-i-b-r-a-t-i-o-n- -i-m-p-a-c-t- -t-h-r-e-s-h-o-l-d- -o-p-t-i-m-i-z-a-t-i-o-n- -o-n- -i-m-b-a-l-a-n-c-e-d- -d-a-t-a-s-e-t-s-?-

-\*- \*-R-o-u-n-d:-\*- -R-e-s-e-a-r-c-h-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -P-r-o-b-a-b-i-l-i-t-y- -c-a-l-i-b-r-a-t-i-o-n- -(P-l-a-t-t/-l-s-o-t-o-n-i-c)-, -B-r-i-e-r- -S-c-o-r-e-, -F-1- -o-p-t-i-m-i-z-a-t-i-o-n-.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -c-a-l-i-b-r-a-t-i-o-n- -a-n-d- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d- -s-e-l-e-c-t-i-o-n-.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -l-m-b-a-l-a-n-c-e-d- -c-l-a-s-s-i-f-i-e-r-s- -o-f-t-e-n- -p-r-o-d-u-c-e- -d-i-s-t-o-r-t-e-d- -p-r-o-b-a-b-i-l-i-t-i-e-s- -(e-.g.-, -b-o-o-s-t-i-n-g- -m-o-d-e-l-s- -p-u-s-h- -s-c-o-r-e-s- -n-e-a-r- -0- -o-r- -1)-, -C-a-l-i-b-r-a-t-i-o-n- -m-a-p-s- -t-h-e-s-e- -s-c-o-r-e-s- -b-a-c-k- -t-o- -e-m-p-i-r-i-c-a-l- -p-r-o-b-a-b-i-l-i-t-i-e-s-.- -O-n-c-e- -c-a-l-i-b-r-a-t-e-d-, -w-e- -c-a-n- -s-e-a-r-c-h- -f-o-r- -a- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d- -(l-i-k-e- -0-.-2-0)- -t-o- -m-a-x-i-m-i-z-e- -F-1-, -k-n-o-w-i-n-g- -t-h-a-t- -2-0%- -m-e-a-n-s- -a- -1---i-n---5- -p-r-o-b-a-b-i-l-i-t-y-.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -O-p-t-i-m-i-z-i-n-g- -t-h-e- -d-e-c-i-s-i-o-n- -t-h-r-e-s-h-o-l-d- -o-n- -u-n-c-a-l-i-b-r-a-t-e-d- -p-r-o-b-a-b-i-l-i-t-i-e-s-, -w-h-i-c-h- -r-e-s-u-l-t-s- -i-n- -u-n-s-t-a-b-l-e- -d-e-c-i-s-i-o-n- -b-o-u-n-d-a-r-i-e-s-.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -e-v-a-l-u-a-t-e- -P-l-a-t-t- -s-c-a-l-i-n-g- -a-n-d- -l-s-o-t-o-n-i-c- -r-e-g-r-e-s-s-i-o-n-.- -S-i-n-c-e- -o-u-r- -d-a-t-a-s-e-t- -i-s- -m-e-d-i-u-m---s-i-z-e-d-, -P-l-a-t-t- -s-c-a-l-i-n-g- -p-r-e-v-e-n-t-s- -o-v-e-r-f-i-t-t-i-n-g- -a-n-d- -m-i-n-i-m-i-z-e-s- -t-h-e- -E-x-p-e-c-t-e-d- -C-a-l-i-b-r-a-t-i-o-n- -E-r-r-o-r- -(E-C-E)- -t-o-

0.0233.

\*. \*. Follower-up Questions: \*. \*. Why not use SMOTE to resolve class imbalance in standard?

\*. \*. Deep Dive: \*. \*. Prof. that Brainer Score can be decomposed into reliability, resolution, and uncertainty.

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### Questions 61: Imbalance Data Questions 61: How does probability calibration impact threshold optimization on imbalance data datasets?

\*. \*. Round: \*. \*. Researched

\*. \*. Difficulties: \*. \*. Hard to find important cases: \*. \*. 9/10

\*. \*. Confusion Test: \*. \*. Probability calibration ( Platt/ Logistic ), Brainer Score, F1 optimization.

\*. \*. Why. Interviewer's Asks: \*. \*. Tests calibration and decision threshold selection.

\*. \*. Ideal FANG Answer: \*. \*. Imbalance classifiers often produce distorted probabilities (e.g., bootstrap models push scores near 0 or 1). Calibration maps these scores back to empirical probabilities. Once calibrated, we can research for a decision threshold (like 0.20) to maximize F1, knowing that 20% means a 1 in 5 probability.

\*. \*. Common Mistakes: \*. \*. Optimizing the decision threshold on uncalibrated probabilities, which results in unstable decision boundaries.

\*. \*. Better Production Grade Answer: \*. \*. We evaluate Platt scaling and Logistic Regression. Since our dataset is medium-sized, Platt scaling pre-vents overfitting and minimizes the expected Calibration Error (ECE) to 0.0233.

\*. \*. Follower-up Questions: \*. \*. Why not use SMOTE to resolve class imbalance in standard?

\*. \*. Deep Dive: \*. \*. Prof. that Brainer Score can be decomposed into reliability, resolution, and uncertainty.

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### Questions 62: Imbalance Data Questions 62: How does probability calibration impact threshold optimization on imbalance data datasets?

\*. \*. Round: \*. \*. Researched

\*. \*. Difficulties: \*. \*. Hard to find important cases: \*. \*. 9/10

\*. \*. Confusion Test: \*. \*. Probability calibration ( Platt/ Logistic ), Brainer Score, F1 optimization.

\*. \*. Why. Interviewer's Asks: \*. \*. Tests calibration and decision threshold selection.

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- I-m-b-a-l-a-n-c-e-d- c-l-a-s-s-i-f-i-e-r-s- o-f-t-e-n- p-r-o-d-u-c-e- d-i-s-t-o-r-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s- (-e-.g.-, b-o-o-s-t-i-n-g- m-o-d-e-l-s- p-u-s-h- s-c-o-r-e-s- n-e-a-r- 0- o-r- 1)-. C-a-l-i-b-r-a-t-i-o-n- m-a-p-s- t-h-e-s-e- s-c-o-r-e-s- b-a-c-k- t-o- e-m-p-i-r-i-c-a-l- p-r-o-b-a-b-i-l-i-t-i-e-s-. O-n-c-e- c-a-l-i-b-r-a-t-e-d-, w-e- c-a-n- s-e-a-r-c-h- f-o-r- a- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- (-l-i-k-e- 0-.2-0)- t-o- m-a-x-i-m-i-z-e- F-1-, k-n-o-w-i-n-g- t-h-a-t- 2-0%- m-e-a-n-s- a- 1---i-n---5- p-r-o-b-a-b-i-l-i-t-y.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- O-p-t-i-m-i-z-i-n-g- t-h-e- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- o-n- u-n-c-a-l-i-b-r-a-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s-, w-h-i-c-h- r-e-s-u-l-t-s- i-n- u-n-s-t-a-b-l-e- d-e-c-i-s-i-o-n- b-o-u-n-d-a-r-i-e-s.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- e-v-a-l-u-a-t-e- P-l-a-t-t- s-c-a-l-i-n-g- a-n-d- l-s-o-t-o-n-i-c- r-e-g-r-e-s-s-i-o-n-, S-i-n-c-e- o-u-r- d-a-t-a-s-e-t- i-s- m-e-d-i-u-m---s-i-z-e-d-, P-l-a-t-t- s-c-a-l-i-n-g- p-r-e-v-e-n-t-s- o-v-e-r-f-i-t-t-i-n-g- a-n-d- m-i-n-i-m-i-z-e-s- t-h-e- E-x-p-e-c-t-e-d- C-a-l-i-b-r-a-t-i-o-n- E-r-r-o-r- (-E-C-E)- t-o- 0-.0-2-3-3.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- W-h-y- n-o-t- u-s-e- S-M-O-T-E- t-o- r-e-s-o-l-v-e- c-l-a-s-s- i-m-b-a-l-a-n-c-e- i-n-s-t-e-a-d?-

\*- \*-D-e-e-p- D-i-v-e:-\*- P-r-o-o-f- t-h-a-t- B-r-i-e-r- S-c-o-r-e- c-a-n- b-e- d-e-c-o-m-p-o-s-e-d- i-n-t-o- r-e-l-i-a-b-i-l-i-t-y-, r-e-s-o-l-u-t-i-o-n-, a-n-d- u-n-c-e-r-t-a-i-n-t-y.-

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##- Q-u-e-s-t-i-o-n- 6-3:- I-m-b-a-l-a-n-c-e-d- D-a-t-a- Q-u-e-s-t-i-o-n- 6-3:- H-o-w- d-o-e-s- p-r-o-b-a-b-i-l-i-t-y- c-a-l-i-b-r-a-t-i-o-n- i-m-p-a-c-t- t-h-r-e-s-h-o-l-d- o-p-t-i-m-i-z-a-t-i-o-n- o-n- i-m-b-a-l-a-n-c-e-d- d-a-t-a-s-e-t-s?-

\*- \*-R-o-u-n-d:-\*- R-e-s-e-a-r-c-h-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- | \*-I-m-p-o-r-t-a-n-c-e:-\*- 9-/1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- P-r-o-b-a-b-i-l-i-t-y- c-a-l-i-b-r-a-t-i-o-n- (-P-l-a-t-t-/l-s-o-t-o-n-i-c-), B-r-i-e-r- S-c-o-r-e-, F-1- o-p-t-i-m-i-z-a-t-i-o-n.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- T-e-s-t-s- c-a-l-i-b-r-a-t-i-o-n- a-n-d- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- s-e-l-e-c-t-i-o-n.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- I-m-b-a-l-a-n-c-e-d- c-l-a-s-s-i-f-i-e-r-s- o-f-t-e-n- p-r-o-d-u-c-e- d-i-s-t-o-r-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s- (-e-.g.-, b-o-o-s-t-i-n-g- m-o-d-e-l-s- p-u-s-h- s-c-o-r-e-s- n-e-a-r- 0- o-r- 1)-. C-a-l-i-b-r-a-t-i-o-n- m-a-p-s- t-h-e-s-e- s-c-o-r-e-s- b-a-c-k- t-o- e-m-p-i-r-i-c-a-l- p-r-o-b-a-b-i-l-i-t-i-e-s-. O-n-c-e- c-a-l-i-b-r-a-t-e-d-, w-e- c-a-n- s-e-a-r-c-h- f-o-r- a- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- (-l-i-k-e- 0-.2-0)- t-o- m-a-x-i-m-i-z-e- F-1-, k-n-o-w-i-n-g- t-h-a-t- 2-0%- m-e-a-n-s- a- 1---i-n---5- p-r-o-b-a-b-i-l-i-t-y.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- O-p-t-i-m-i-z-i-n-g- t-h-e- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- o-n- u-n-c-a-l-i-b-r-a-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s-, w-h-i-c-h- r-e-s-u-l-t-s- i-n- u-n-s-t-a-b-l-e- d-e-c-i-s-i-o-n- b-o-u-n-d-a-r-i-e-s.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- e-v-a-l-u-a-t-e- P-l-a-t-t- s-c-a-l-i-n-g- a-n-d- l-s-o-t-o-n-i-c- r-e-g-r-e-s-s-i-o-n-, S-i-n-c-e- o-u-r- d-a-t-a-s-e-t- i-s- m-e-d-i-u-m---s-i-z-e-d-, P-l-a-t-t- s-c-a-l-i-n-g- p-r-e-v-e-n-t-s- o-v-e-r-f-i-t-t-i-n-g- a-n-d- m-i-n-i-m-i-z-e-s- t-h-e- E-x-p-e-c-t-e-d- C-a-l-i-b-r-a-t-i-o-n- E-r-r-o-r- (-E-C-E)- t-o- 0-.0-2-3-3.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- W-h-y- n-o-t- u-s-e- S-M-O-T-E- t-o- r-e-s-o-l-v-e- c-l-a-s-s- i-m-b-a-l-a-n-c-e- i-n-s-t-e-a-d?-

\*- \*-D-e-e-p- D-i-v-e:-\*- P-r-o-o-f- t-h-a-t- B-r-i-e-r- S-c-o-r-e- c-a-n- b-e- d-e-c-o-m-p-o-s-e-d- i-n-t-o- r-e-l-i-a-b-i-l-i-t-y-, r-e-s-o-l-u-t-i-o-n-, a-n-d- u-n-c-e-r-t-a-i-n-t-y.-

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### Q-u-e-s-t-i-o-n- 6-4:- I-m-b-a-l-a-n-c-e-d- D-a-t-a- Q-u-e-s-t-i-o-n- 6-4:- H-o-w- d-o-e-s- p-r-o-b-a-b-i-l-i-t-y- c-a-l-i-b-r-a-t-i-o-n- i-m-p-a-c-t- t-h-r-e-s-h-o-l-d- o-p-t-i-m-i-z-a-t-i-o-n- o-n- i-m-b-a-l-a-n-c-e-d- d-a-t-a-s-e-t-s?-

\*- \*-R-o-u-n-d:-\*- R-e-s-e-a-r-c-h-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- | \*-I-m-p-o-r-t-a-n-c-e:-\*- 9-/1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- P-r-o-b-a-b-i-l-i-t-y- c-a-l-i-b-r-a-t-i-o-n- (-P-l-a-t-t-/l-s-o-t-o-n-i-c-),- B-r-i-e-r- S-c-o-r-e,- F-1- o-p-t-i-m-i-z-a-t-i-o-n.-

\*- \*-W-h-y- I-n-t-e-r-v-i-e-w-e-r- A-s-k-s- I-t:-\*- T-e-s-t-s- c-a-l-i-b-r-a-t-i-o-n- a-n-d- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- s-e-l-e-c-t-i-o-n.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- I-m-b-a-l-a-n-c-e-d- c-l-a-s-s-i-f-i-e-r-s- o-f-t-e-n- p-r-o-d-u-c-e- d-i-s-t-o-r-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s- (-e-.g.-, b-o-o-s-t-i-n-g- m-o-d-e-l-s- p-u-s-h- s-c-o-r-e-s- n-e-a-r- 0- o-r- 1)-. C-a-l-i-b-r-a-t-i-o-n- m-a-p-s- t-h-e-s-e- s-c-o-r-e-s- b-a-c-k- t-o- e-m-p-i-r-i-c-a-l- p-r-o-b-a-b-i-l-i-t-i-e-s.- O-n-c-e- c-a-l-i-b-r-a-t-e-d,- w-e- c-a-n- s-e-a-r-c-h- f-o-r- a- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- (-l-i-k-e- 0-.2-0)- t-o- m-a-x-i-m-i-z-e- F-1,- k-n-o-w-i-n-g- t-h-a-t- 2-0%- m-e-a-n-s- a- 1---i-n---5- p-r-o-b-a-b-i-l-i-t-y.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- O-p-t-i-m-i-z-i-n-g- t-h-e- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- o-n- u-n-c-a-l-i-b-r-a-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s,- w-h-i-c-h- r-e-s-u-l-t-s- i-n- u-n-s-t-a-b-l-e- d-e-c-i-s-i-o-n- b-o-u-n-d-a-r-i-e-s.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- e-v-a-l-u-a-t-e- P-l-a-t-t- s-c-a-l-i-n-g- a-n-d- l-s-o-t-o-n-i-c- r-e-g-r-e-s-s-i-o-n.- S-i-n-c-e- o-u-r- d-a-t-a-s-e-t- i-s- m-e-d-i-u-m---s-i-z-e-d,- P-l-a-t-t- s-c-a-l-i-n-g- p-r-e-v-e-n-t-s- o-v-e-r-f-i-t-t-i-n-g- a-n-d- m-i-n-i-m-i-z-e-s- t-h-e- E-x-p-e-c-t-e-d- C-a-l-i-b-r-a-t-i-o-n- E-r-r-o-r- (-E-C-E)- t-o- 0-.0-2-3-3.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- W-h-y- n-o-t- u-s-e- S-M-O-T-E- t-o- r-e-s-o-l-v-e- c-l-a-s-s- i-m-b-a-l-a-n-c-e- i-n-s-t-e-a-d?-

\*- \*-D-e-e-p- D-i-v-e:-\*- P-r-o-o-f- t-h-a-t- B-r-i-e-r- S-c-o-r-e- c-a-n- b-e- d-e-c-o-m-p-o-s-e-d- i-n-t-o- r-e-l-i-a-b-i-l-i-t-y,- r-e-s-o-l-u-t-i-o-n,- a-n-d- u-n-c-e-r-t-a-i-n-t-y.-

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### Q-u-e-s-t-i-o-n- 6-5:- I-m-b-a-l-a-n-c-e-d- D-a-t-a- Q-u-e-s-t-i-o-n- 6-5:- H-o-w- d-o-e-s- p-r-o-b-a-b-i-l-i-t-y- c-a-l-i-b-r-a-t-i-o-n- i-m-p-a-c-t- t-h-r-e-s-h-o-l-d- o-p-t-i-m-i-z-a-t-i-o-n- o-n- i-m-b-a-l-a-n-c-e-d- d-a-t-a-s-e-t-s?-

\*- \*-R-o-u-n-d:-\*- R-e-s-e-a-r-c-h-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- | \*-I-m-p-o-r-t-a-n-c-e:-\*- 9-/1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- P-r-o-b-a-b-i-l-i-t-y- c-a-l-i-b-r-a-t-i-o-n- (-P-l-a-t-t-/l-s-o-t-o-n-i-c-),- B-r-i-e-r- S-c-o-r-e,- F-1- o-p-t-i-m-i-z-a-t-i-o-n.-

\*- \*-W-h-y- I-n-t-e-r-v-i-e-w-e-r- A-s-k-s- I-t:-\*- T-e-s-t-s- c-a-l-i-b-r-a-t-i-o-n- a-n-d- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- s-e-l-e-c-t-i-o-n.-

\*- \*-I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- I-m-b-a-l-a-n-c-e-d- c-l-a-s-s-i-f-i-e-r-s- o-f-t-e-n- p-r-o-d-u-c-e- d-i-s-t-o-r-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s- (-e-.g.-, b-o-o-s-t-i-n-g- m-o-d-e-l-s- p-u-s-h- s-c-o-r-e-s- n-e-a-r- 0- o-r- 1)-. C-a-l-i-b-r-a-t-i-o-n- m-a-p-s- t-h-e-s-e- s-c-o-r-e-s- b-a-c-k- t-o- e-m-p-i-r-i-c-a-l- p-r-o-b-a-b-i-l-i-t-i-e-s.- O-n-c-e- c-a-l-i-b-r-a-t-e-d,- w-e- c-a-n- s-e-a-r-c-h- f-o-r- a- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- (-l-i-k-e- 0-.2-0)- t-o- m-a-x-i-m-i-z-e- F-1,- k-n-o-w-i-n-g- t-h-a-t- 2-0%- m-e-a-n-s- a- 1---i-n---5- p-r-o-b-a-b-i-l-i-t-y.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- O-p-t-i-m-i-z-i-n-g- t-h-e- d-e-c-i-s-i-o-n- t-h-r-e-s-h-o-l-d- o-n- u-n-c-a-l-i-b-r-a-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s,- w-h-i-c-h- r-e-s-u-l-t-s- i-n- u-n-s-t-a-b-l-e- d-e-c-i-s-i-o-n- b-o-u-n-d-a-r-i-e-s.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- e-v-a-l-u-a-t-e- P-l-a-t-t- s-c-a-l-i-n-g- a-n-d-

-l-s-o-t-o-n-i-c- -r-e-g-r-e-s-s-i-o-n-.- -S-i-n-c-e- -o-u-r- -d-a-t-a-s-e-t- -i-s- -m-e-d-i-u-m---s-i-z-e-d,- -P-l-a-t-t- -s-c-a-l-i-n-g-  
-p-r-e-v-e-n-t-s- -o-v-e-r-f-i-t-t-i-n-g- -a-n-d- -m-i-n-i-m-i-z-e-s- -t-h-e- -E-x-p-e-c-t-e-d- -C-a-l-i-b-r-a-t-i-o-n- -E-r-r-o-r- -(E-C-E)- -t-o-  
-0-0-2-3-.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-y- -n-o-t- -u-s-e- -S-M-O-T-E- -t-o- -r-e-s-o-l-v-e- -c-l-a-s-s- -i-m-b-a-l-a-n-c-e-  
-i-n-s-t-e-a-d?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -P-r-o-o-f- -t-h-a-t- -B-r-i-e-r- -S-c-o-r-e- -c-a-n- -b-e- -d-e-c-o-m-p-o-s-e-d- -i-n-t-o- -r-e-l-i-a-b-i-l-i-t-y,-  
-r-e-s-o-l-u-t-i-o-n,- -a-n-d- -u-n-c-e-r-t-a-i-n-t-y.-

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-#-#-#- -Q-u-e-s-t-i-o-n- -6-6:- -M-o-d-e-l- -E-v-a-l-u-a-t-i-o-n- -Q-u-e-s-t-i-o-n- -6-6:- -H-o-w- -d-o- -y-o-u- -p-e-r-f-o-r-m-  
-c-h-r-o-n-o-l-o-g-i-c-a-l- -(t-i-m-e---b-a-s-e-d)- -v-a-l-i-d-a-t-i-o-n- -a-n-d- -w-h-y- -i-s- -i-t- -s-u-p-e-r-i-o-r- -t-o- -r-a-n-d-o-m-  
-s-p-l-i-t-s- -f-o-r- -R-i-s-k-P-u-l-s-e?-

-\*- \*-R-o-u-n-d:-\*- -M-L-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-e-m-p-o-r-a-l- -d-a-t-a- -s-p-l-i-t-s,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -v-a-l-i-d-a-t-i-o-n,-  
-o-u-t---o-f---t-i-m-e- -t-e-s-t-i-n-g.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -v-a-l-i-d-a-t-i-o-n- -s-t-r-a-t-e-g-y- -f-o-r- -t-i-m-e---d-e-p-e-n-d-e-n-t-  
-r-e-c-o-r-d-s.-

-\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -L-a-y-o-f-f- -t-r-e-n-d-s- -a-r-e- -s-u-b-j-e-c-t- -t-o- -m-a-c-r-o---e-c-o-n-o-m-i-c-  
-c-y-c-l-e-s,- -A- -r-a-n-d-o-m- -K---F-o-l-d- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -l-e-a-k-s- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -f-u-t-u-r-e-  
-m-a-r-k-e-t- -s-t-a-t-e-s- -i-n-t-o- -p-a-s-t- -p-r-e-d-i-c-t-i-o-n-s,- -W-e- -u-s-e- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-p-l-i-t-t-i-n-g:- -t-r-a-i-n-i-n-g-  
-o-n- -p-a-s-t- -y-e-a-r-s- -a-n-d- -v-a-l-i-d-a-t-i-n-g- -o-n- -s-u-b-s-e-q-u-e-n-t- -y-e-a-r-s,- -w-h-i-c-h- -a-c-c-u-r-a-t-e-l-y-  
-s-i-m-u-l-a-t-e-s- -h-o-w- -t-h-e- -m-o-d-e-l- -w-i-l-l- -p-e-r-f-o-r-m- -o-n- -f-u-t-u-r-e- -u-n-s-e-e-n- -d-a-t-a.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -U-s-i-n-g- -s-t-a-n-d-a-r-d- -S-t-r-a-t-i-f-i-e-d- -K---F-o-l-d- -C-V- -w-i-t-h-o-u-t-  
-c-h-e-c-k-i-n-g- -f-o-r- -t-e-m-p-o-r-a-l- -l-e-a-k-a-g-e.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -T-i-m-e-S-e-r-i-e-s-S-p-l-i-t- -f-r-o-m-  
-s-c-i-k-i-t---l-e-a-r-n- -w-i-t-h- -a- -g-a-p- -p-a-r-a-m-e-t-e-r- -t-o- -p-r-e-v-e-n-t- -b-l-e-e-d-i-n-g- -i-n-f-o- -a-c-r-o-s-s-  
-o-v-e-r-l-a-p-p-i-n-g- -e-v-e-n-t- -w-i-n-d-o-w-s.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -i-s- -t-h-e- -b-a-s-e-l-i-n-e- -p-e-r-f-o-r-m-a-n-c-e- -g-a-p- -b-e-t-w-e-e-n- -a-  
-r-a-n-d-o-m- -s-p-l-i-t- -a-n-d- -a- -t-i-m-e- -s-p-l-i-t?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -a- -c-o-m-p-a-n-y- -t-h-a-t- -c-h-a-n-g-e-s- -i-t-s- -i-n-d-u-s-t-r-y-  
-c-a-t-e-g-o-r-y- -o-v-e-r- -t-i-m-e- -d-u-r-i-n-g- -s-p-l-i-t- -s-e-t-u-p?-

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-#-#-#- -Q-u-e-s-t-i-o-n- -6-7:- -M-o-d-e-l- -E-v-a-l-u-a-t-i-o-n- -Q-u-e-s-t-i-o-n- -6-7:- -H-o-w- -d-o- -y-o-u- -p-e-r-f-o-r-m-  
-c-h-r-o-n-o-l-o-g-i-c-a-l- -(t-i-m-e---b-a-s-e-d)- -v-a-l-i-d-a-t-i-o-n- -a-n-d- -w-h-y- -i-s- -i-t- -s-u-p-e-r-i-o-r- -t-o- -r-a-n-d-o-m-  
-s-p-l-i-t-s- -f-o-r- -R-i-s-k-P-u-l-s-e?-

-\*- \*-R-o-u-n-d:-\*- -M-L-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-e-m-p-o-r-a-l- -d-a-t-a- -s-p-l-i-t-s,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -v-a-l-i-d-a-t-i-o-n,-

-o-u-t---o-f---t-i-m-e- -t-e-s-t-i-n-g.-

-\* -\*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -v-a-l-i-d-a-t-i-o-n- -s-t-r-a-t-e-g-y- -f-o-r- -t-i-m-e---d-e-p-e-n-d-e-n-t-  
-r-e-c-o-r-d-s.-

-\* -\*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -L-a-y-o-f-f- -t-r-e-n-d-s- -a-r-e- -s-u-b-j-e-c-t- -t-o- -m-a-c-r-o---e-c-o-n-o-m-i-c-  
-c-y-c-l-e-s.- -A- -r-a-n-d-o-m- -K---F-o-l-d- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -l-e-a-k-s- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -f-u-t-u-r-e-  
-m-a-r-k-e-t- -s-t-a-t-e-s- -i-n-t-o- -p-a-s-t- -p-r-e-d-i-c-t-i-o-n-s.- -W-e- -u-s-e- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-p-l-i-t-t-i-n-g:- -t-r-a-i-n-i-n-g-  
-o-n- -p-a-s-t- -y-e-a-r-s- -a-n-d- -v-a-l-i-d-a-t-i-n-g- -o-n- -s-u-b-s-e-q-u-e-n-t- -y-e-a-r-s,- -w-h-i-c-h- -a-c-c-u-r-a-t-e-l-y-  
-s-i-m-u-l-a-t-e-s- -h-o-w- -t-h-e- -m-o-d-e-l- -w-i-l-l- -p-e-r-f-o-r-m- -o-n- -f-u-t-u-r-e- -u-n-s-e-e-n- -d-a-t-a.-

-\* -\*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -U-s-i-n-g- -s-t-a-n-d-a-r-d- -S-t-r-a-t-i-f-i-e-d- -K---F-o-l-d- -C-V- -w-i-t-h-o-u-t-  
-c-h-e-c-k-i-n-g- -f-o-r- -t-e-m-p-o-r-a-l- -l-e-a-k-a-g-e.-

-\* -\*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -T-i-m-e-S-e-r-i-e-s-S-p-l-i-t- -f-r-o-m-  
-s-c-i-k-i-t---l-e-a-r-n- -w-i-t-h- -a- -g-a-p- -p-a-r-a-m-e-t-e-r- -t-o- -p-r-e-v-e-n-t- -b-l-e-e-d-i-n-g- -i-n-f-o- -a-c-r-o-s-s-  
-o-v-e-r-l-a-p-p-i-n-g- -e-v-e-n-t- -w-i-n-d-o-w-s.-

-\* -\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -i-s- -t-h-e- -b-a-s-e-l-i-n-e- -p-e-r-f-o-r-m-a-n-c-e- -g-a-p- -b-e-t-w-e-e-n- -a-  
-r-a-n-d-o-m- -s-p-l-i-t- -a-n-d- -a- -t-i-m-e- -s-p-l-i-t?-

-\* -\*-D-e-e-p- -D-i-v-e:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -a- -c-o-m-p-a-n-y- -t-h-a-t- -c-h-a-n-g-e-s- -i-t-s- -i-n-d-u-s-t-r-y-  
-c-a-t-e-g-o-r-y- -o-v-e-r- -t-i-m-e- -d-u-r-i-n-g- -s-p-l-i-t- -s-e-t-u-p?-

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-#-#-#- -Q-u-e-s-t-i-o-n- -6-8:- -M-o-d-e-l- -E-v-a-l-u-a-t-i-o-n- -Q-u-e-s-t-i-o-n- -6-8:- -H-o-w- -d-o- -y-o-u- -p-e-r-f-o-r-m-  
-c-h-r-o-n-o-l-o-g-i-c-a-l- -(t-i-m-e---b-a-s-e-d)- -v-a-l-i-d-a-t-i-o-n- -a-n-d- -w-h-y- -i-s- -i-t- -s-u-p-e-r-i-o-r- -t-o- -r-a-n-d-o-m-  
-s-p-l-i-t-s- -f-o-r- -R-i-s-k-P-u-l-s-e?-

-\* -\*-R-o-u-n-d:-\*- -M-L-

-\* -\*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-

-\* -\*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-e-m-p-o-r-a-l- -d-a-t-a- -s-p-l-i-t-s,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -v-a-l-i-d-a-t-i-o-n,-  
-o-u-t---o-f---t-i-m-e- -t-e-s-t-i-n-g.-

-\* -\*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -v-a-l-i-d-a-t-i-o-n- -s-t-r-a-t-e-g-y- -f-o-r- -t-i-m-e---d-e-p-e-n-d-e-n-t-  
-r-e-c-o-r-d-s.-

-\* -\*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -L-a-y-o-f-f- -t-r-e-n-d-s- -a-r-e- -s-u-b-j-e-c-t- -t-o- -m-a-c-r-o---e-c-o-n-o-m-i-c-  
-c-y-c-l-e-s.- -A- -r-a-n-d-o-m- -K---F-o-l-d- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -l-e-a-k-s- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -f-u-t-u-r-e-  
-m-a-r-k-e-t- -s-t-a-t-e-s- -i-n-t-o- -p-a-s-t- -p-r-e-d-i-c-t-i-o-n-s.- -W-e- -u-s-e- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-p-l-i-t-t-i-n-g:- -t-r-a-i-n-i-n-g-  
-o-n- -p-a-s-t- -y-e-a-r-s- -a-n-d- -v-a-l-i-d-a-t-i-n-g- -o-n- -s-u-b-s-e-q-u-e-n-t- -y-e-a-r-s,- -w-h-i-c-h- -a-c-c-u-r-a-t-e-l-y-  
-s-i-m-u-l-a-t-e-s- -h-o-w- -t-h-e- -m-o-d-e-l- -w-i-l-l- -p-e-r-f-o-r-m- -o-n- -f-u-t-u-r-e- -u-n-s-e-e-n- -d-a-t-a.-

-\* -\*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -U-s-i-n-g- -s-t-a-n-d-a-r-d- -S-t-r-a-t-i-f-i-e-d- -K---F-o-l-d- -C-V- -w-i-t-h-o-u-t-  
-c-h-e-c-k-i-n-g- -f-o-r- -t-e-m-p-o-r-a-l- -l-e-a-k-a-g-e.-

-\* -\*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -T-i-m-e-S-e-r-i-e-s-S-p-l-i-t- -f-r-o-m-  
-s-c-i-k-i-t---l-e-a-r-n- -w-i-t-h- -a- -g-a-p- -p-a-r-a-m-e-t-e-r- -t-o- -p-r-e-v-e-n-t- -b-l-e-e-d-i-n-g- -i-n-f-o- -a-c-r-o-s-s-  
-o-v-e-r-l-a-p-p-i-n-g- -e-v-e-n-t- -w-i-n-d-o-w-s.-

-\* -\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -i-s- -t-h-e- -b-a-s-e-l-i-n-e- -p-e-r-f-o-r-m-a-n-c-e- -g-a-p- -b-e-t-w-e-e-n- -a-  
-r-a-n-d-o-m- -s-p-l-i-t- -a-n-d- -a- -t-i-m-e- -s-p-l-i-t?-

-\* -\*-D-e-e-p- -D-i-v-e:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -a- -c-o-m-p-a-n-y- -t-h-a-t- -c-h-a-n-g-e-s- -i-t-s- -i-n-d-u-s-t-r-y-  
-c-a-t-e-g-o-r-y- -o-v-e-r- -t-i-m-e- -d-u-r-i-n-g- -s-p-l-i-t- -s-e-t-u-p?-

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### -Q-u-e-s-t-i-o-n- 6-9:- -M-o-d-e-l- -E-v-a-l-u-a-t-i-o-n- -Q-u-e-s-t-i-o-n- 6-9:- -H-o-w- -d-o- -y-o-u- -p-e-r-f-o-r-m-  
-c-h-r-o-n-o-l-o-g-i-c-a-l- -(t-i-m-e---b-a-s-e-d)- -v-a-l-i-d-a-t-i-o-n- -a-n-d- -w-h-y- -i-s- -i-t- -s-u-p-e-r-i-o-r- -t-o- -r-a-n-d-o-m-  
-s-p-l-i-t-s- -f-o-r- -R-i-s-k-P-u-l-s-e?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -| -\*-I-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-e-m-p-o-r-a-l- -d-a-t-a- -s-p-l-i-t-s,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -v-a-l-i-d-a-t-i-o-n,-  
-o-u-t---o-f---t-i-m-e- -t-e-s-t-i-n-g.-

\*- \*-W-h-y- -I-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -v-a-l-i-d-a-t-i-o-n- -s-t-r-a-t-e-g-y- -f-o-r- -t-i-m-e---d-e-p-e-n-d-e-n-t-  
-r-e-c-o-r-d-s.-

\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -L-a-y-o-f-f- -t-r-e-n-d-s- -a-r-e- -s-u-b-j-e-c-t- -t-o- -m-a-c-r-o---e-c-o-n-o-m-i-c-  
-c-y-c-l-e-s.- -A- -r-a-n-d-o-m- -K---F-o-l-d- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -l-e-a-k-s- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -f-u-t-u-r-e-  
-m-a-r-k-e-t- -s-t-a-t-e-s- -i-n-t-o- -p-a-s-t- -p-r-e-d-i-c-t-i-o-n-s.- -W-e- -u-s-e- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-p-l-i-t-t-i-n-g:- -t-r-a-i-n-i-n-g-  
-o-n- -p-a-s-t- -y-e-a-r-s- -a-n-d- -v-a-l-i-d-a-t-i-n-g- -o-n- -s-u-b-s-e-q-u-e-n-t- -y-e-a-r-s,- -w-h-i-c-h- -a-c-c-u-r-a-t-e-l-y-  
-s-i-m-u-l-a-t-e-s- -h-o-w- -t-h-e- -m-o-d-e-l- -w-i-l-l- -p-e-r-f-o-r-m- -o-n- -f-u-t-u-r-e- -u-n-s-e-e-n- -d-a-t-a.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -U-s-i-n-g- -s-t-a-n-d-a-r-d- -S-t-r-a-t-i-f-i-e-d- -K---F-o-l-d- -C-V- -w-i-t-h-o-u-t-  
-c-h-e-c-k-i-n-g- -f-o-r- -t-e-m-p-o-r-a-l- -l-e-a-k-a-g-e.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -T-i-m-e-S-e-r-i-e-s-S-p-l-i-t- -f-r-o-m-  
-s-c-i-k-i-t---l-e-a-r-n- -w-i-t-h- -a- -g-a-p- -p-a-r-a-m-e-t-e-r- -t-o- -p-r-e-v-e-n-t- -b-l-e-e-d-i-n-g- -i-n-f-o- -a-c-r-o-s-s-  
-o-v-e-r-l-a-p-p-i-n-g- -e-v-e-n-t- -w-i-n-d-o-w-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -i-s- -t-h-e- -b-a-s-e-l-i-n-e- -p-e-r-f-o-r-m-a-n-c-e- -g-a-p- -b-e-t-w-e-e-n- -a-  
-r-a-n-d-o-m- -s-p-l-i-t- -a-n-d- -a- -t-i-m-e- -s-p-l-i-t?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -a- -c-o-m-p-a-n-y- -t-h-a-t- -c-h-a-n-g-e-s- -i-t-s- -i-n-d-u-s-t-r-y-  
-c-a-t-e-g-o-r-y- -o-v-e-r- -t-i-m-e- -d-u-r-i-n-g- -s-p-l-i-t- -s-e-t-u-p?-

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### -Q-u-e-s-t-i-o-n- 7-0:- -M-o-d-e-l- -E-v-a-l-u-a-t-i-o-n- -Q-u-e-s-t-i-o-n- 7-0:- -H-o-w- -d-o- -y-o-u- -p-e-r-f-o-r-m-  
-c-h-r-o-n-o-l-o-g-i-c-a-l- -(t-i-m-e---b-a-s-e-d)- -v-a-l-i-d-a-t-i-o-n- -a-n-d- -w-h-y- -i-s- -i-t- -s-u-p-e-r-i-o-r- -t-o- -r-a-n-d-o-m-  
-s-p-l-i-t-s- -f-o-r- -R-i-s-k-P-u-l-s-e?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -| -\*-I-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-e-m-p-o-r-a-l- -d-a-t-a- -s-p-l-i-t-s,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -v-a-l-i-d-a-t-i-o-n,-  
-o-u-t---o-f---t-i-m-e- -t-e-s-t-i-n-g.-

\*- \*-W-h-y- -I-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -v-a-l-i-d-a-t-i-o-n- -s-t-r-a-t-e-g-y- -f-o-r- -t-i-m-e---d-e-p-e-n-d-e-n-t-  
-r-e-c-o-r-d-s.-

\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -L-a-y-o-f-f- -t-r-e-n-d-s- -a-r-e- -s-u-b-j-e-c-t- -t-o- -m-a-c-r-o---e-c-o-n-o-m-i-c-  
-c-y-c-l-e-s.- -A- -r-a-n-d-o-m- -K---F-o-l-d- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -l-e-a-k-s- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -f-u-t-u-r-e-  
-m-a-r-k-e-t- -s-t-a-t-e-s- -i-n-t-o- -p-a-s-t- -p-r-e-d-i-c-t-i-o-n-s.- -W-e- -u-s-e- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-p-l-i-t-t-i-n-g:- -t-r-a-i-n-i-n-g-  
-o-n- -p-a-s-t- -y-e-a-r-s- -a-n-d- -v-a-l-i-d-a-t-i-n-g- -o-n- -s-u-b-s-e-q-u-e-n-t- -y-e-a-r-s,- -w-h-i-c-h- -a-c-c-u-r-a-t-e-l-y-  
-s-i-m-u-l-a-t-e-s- -h-o-w- -t-h-e- -m-o-d-e-l- -w-i-l-l- -p-e-r-f-o-r-m- -o-n- -f-u-t-u-r-e- -u-n-s-e-e-n- -d-a-t-a.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -U-s-i-n-g- -s-t-a-n-d-a-r-d- -S-t-r-a-t-i-f-i-e-d- -K---F-o-l-d- -C-V- -w-i-t-h-o-u-t-  
-c-h-e-c-k-i-n-g- -f-o-r- -t-e-m-p-o-r-a-l- -l-e-a-k-a-g-e.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -T-i-m-e-S-e-r-i-e-s-S-p-l-i-t- -f-r-o-m-  
-s-c-i-k-i-t---l-e-a-r-n- -w-i-t-h- -a- -g-a-p- -p-a-r-a-m-e-t-e-r- -t-o- -p-r-e-v-e-n-t- -b-l-e-e-d-i-n-g- -i-n-f-o- -a-c-r-o-s-s-  
-o-v-e-r-l-a-p-p-i-n-g- -e-v-e-n-t- -w-i-n-d-o-w-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -i-s- -t-h-e- -b-a-s-e-l-i-n-e- -p-e-r-f-o-r-m-a-n-c-e- -g-a-p- -b-e-t-w-e-e-n- -a-  
-r-a-n-d-o-m- -s-p-l-i-t- -a-n-d- -a- -t-i-m-e- -s-p-l-i-t?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -a- -c-o-m-p-a-n-y- -t-h-a-t- -c-h-a-n-g-e-s- -i-t-s- -i-n-d-u-s-t-r-y-  
-c-a-t-e-g-o-r-y- -o-v-e-r- -t-i-m-e- -d-u-r-i-n-g- -s-p-l-i-t- -s-e-t-u-p?-

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-#-#-#- -Q-u-e-s-t-i-o-n- -7-1:- -M-o-d-e-l- -E-v-a-l-u-a-t-i-o-n- -Q-u-e-s-t-i-o-n- -7-1:- -H-o-w- -d-o- -y-o-u- -p-e-r-f-o-r-m-  
-c-h-r-o-n-o-l-o-g-i-c-a-l- -(t-i-m-e---b-a-s-e-d)- -v-a-l-i-d-a-t-i-o-n- -a-n-d- -w-h-y- -i-s- -i-t- -s-u-p-e-r-i-o-r- -t-o- -r-a-n-d-o-m-  
-s-p-l-i-t-s- -f-o-r- -R-i-s-k-P-u-l-s-e?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-e-m-p-o-r-a-l- -d-a-t-a- -s-p-l-i-t-s,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -v-a-l-i-d-a-t-i-o-n,-  
-o-u-t---o-f---t-i-m-e- -t-e-s-t-i-n-g.-

\*- \*-W-h-y- -I-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -v-a-l-i-d-a-t-i-o-n- -s-t-r-a-t-e-g-y- -f-o-r- -t-i-m-e---d-e-p-e-n-d-e-n-t-  
-r-e-c-o-r-d-s.-

\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -L-a-y-o-f-f- -t-r-e-n-d-s- -a-r-e- -s-u-b-j-e-c-t- -t-o- -m-a-c-r-o---e-c-o-n-o-m-i-c-  
-c-y-c-l-e-s.- -A- -r-a-n-d-o-m- -K---F-o-l-d- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -l-e-a-k-s- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -f-u-t-u-r-e-  
-m-a-r-k-e-t- -s-t-a-t-e-s- -i-n-t-o- -p-a-s-t- -p-r-e-d-i-c-t-i-o-n-s.- -W-e- -u-s-e- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-p-l-i-t-t-i-n-g:- -t-r-a-i-n-i-n-g-  
-o-n- -p-a-s-t- -y-e-a-r-s- -a-n-d- -v-a-l-i-d-a-t-i-n-g- -o-n- -s-u-b-s-e-q-u-e-n-t- -y-e-a-r-s,- -w-h-i-c-h- -a-c-c-u-r-a-t-e-l-y-  
-s-i-m-u-l-a-t-e-s- -h-o-w- -t-h-e- -m-o-d-e-l- -w-i-l-l- -p-e-r-f-o-r-m- -o-n- -f-u-t-u-r-e- -u-n-s-e-e-n- -d-a-t-a.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -U-s-i-n-g- -s-t-a-n-d-a-r-d- -S-t-r-a-t-i-f-i-e-d- -K---F-o-l-d- -C-V- -w-i-t-h-o-u-t-  
-c-h-e-c-k-i-n-g- -f-o-r- -t-e-m-p-o-r-a-l- -l-e-a-k-a-g-e.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -T-i-m-e-S-e-r-i-e-s-S-p-l-i-t- -f-r-o-m-  
-s-c-i-k-i-t---l-e-a-r-n- -w-i-t-h- -a- -g-a-p- -p-a-r-a-m-e-t-e-r- -t-o- -p-r-e-v-e-n-t- -b-l-e-e-d-i-n-g- -i-n-f-o- -a-c-r-o-s-s-  
-o-v-e-r-l-a-p-p-i-n-g- -e-v-e-n-t- -w-i-n-d-o-w-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -i-s- -t-h-e- -b-a-s-e-l-i-n-e- -p-e-r-f-o-r-m-a-n-c-e- -g-a-p- -b-e-t-w-e-e-n- -a-  
-r-a-n-d-o-m- -s-p-l-i-t- -a-n-d- -a- -t-i-m-e- -s-p-l-i-t?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -a- -c-o-m-p-a-n-y- -t-h-a-t- -c-h-a-n-g-e-s- -i-t-s- -i-n-d-u-s-t-r-y-  
-c-a-t-e-g-o-r-y- -o-v-e-r- -t-i-m-e- -d-u-r-i-n-g- -s-p-l-i-t- -s-e-t-u-p?-

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-#-#-#- -Q-u-e-s-t-i-o-n- -7-2:- -M-o-d-e-l- -E-v-a-l-u-a-t-i-o-n- -Q-u-e-s-t-i-o-n- -7-2:- -H-o-w- -d-o- -y-o-u- -p-e-r-f-o-r-m-  
-c-h-r-o-n-o-l-o-g-i-c-a-l- -(t-i-m-e---b-a-s-e-d)- -v-a-l-i-d-a-t-i-o-n- -a-n-d- -w-h-y- -i-s- -i-t- -s-u-p-e-r-i-o-r- -t-o- -r-a-n-d-o-m-  
-s-p-l-i-t-s- -f-o-r- -R-i-s-k-P-u-l-s-e?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-e-m-p-o-r-a-l- -d-a-t-a- -s-p-l-i-t-s,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -v-a-l-i-d-a-t-i-o-n,-  
-o-u-t---o-f---t-i-m-e- -t-e-s-t-i-n-g.-

\*- \*-W-h-y- -I-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -v-a-l-i-d-a-t-i-o-n- -s-t-r-a-t-e-g-y- -f-o-r- -t-i-m-e---d-e-p-e-n-d-e-n-t-  
-r-e-c-o-r-d-s.-

\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -L-a-y-o-f-f- -t-r-e-n-d-s- -a-r-e- -s-u-b-j-e-c-t- -t-o- -m-a-c-r-o---e-c-o-n-o-m-i-c-  
-c-y-c-l-e-s,- -A- -r-a-n-d-o-m- -K---F-o-l-d- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -l-e-a-k-s- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -f-u-t-u-r-e-  
-m-a-r-k-e-t- -s-t-a-t-e-s- -i-n-t-o- -p-a-s-t- -p-r-e-d-i-c-t-i-o-n-s,- -W-e- -u-s-e- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-p-l-i-t-t-i-n-g:- -t-r-a-i-n-i-n-g-  
-o-n- -p-a-s-t- -y-e-a-r-s- -a-n-d- -v-a-l-i-d-a-t-i-n-g- -o-n- -s-u-b-s-e-q-u-e-n-t- -y-e-a-r-s,- -w-h-i-c-h- -a-c-c-u-r-a-t-e-l-y-  
-s-i-m-u-l-a-t-e-s- -h-o-w- -t-h-e- -m-o-d-e-l- -w-i-l-l- -p-e-r-f-o-r-m- -o-n- -f-u-t-u-r-e- -u-n-s-e-e-n- -d-a-t-a.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -U-s-i-n-g- -s-t-a-n-d-a-r-d- -S-t-r-a-t-i-f-i-e-d- -K---F-o-l-d- -C-V- -w-i-t-h-o-u-t-  
-c-h-e-c-k-i-n-g- -f-o-r- -t-e-m-p-o-r-a-l- -l-e-a-k-a-g-e.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -T-i-m-e-S-e-r-i-e-s-S-p-l-i-t- -f-r-o-m-  
-s-c-i-k-i-t---l-e-a-r-n- -w-i-t-h- -a- -g-a-p- -p-a-r-a-m-e-t-e-r- -t-o- -p-r-e-v-e-n-t- -b-l-e-e-d-i-n-g- -i-n-f-o- -a-c-r-o-s-s-  
-o-v-e-r-l-a-p-p-i-n-g- -e-v-e-n-t- -w-i-n-d-o-w-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -i-s- -t-h-e- -b-a-s-e-l-i-n-e- -p-e-r-f-o-r-m-a-n-c-e- -g-a-p- -b-e-t-w-e-e-n- -a-  
-r-a-n-d-o-m- -s-p-l-i-t- -a-n-d- -a- -t-i-m-e- -s-p-l-i-t?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -a- -c-o-m-p-a-n-y- -t-h-a-t- -c-h-a-n-g-e-s- -i-t-s- -i-n-d-u-s-t-r-y-  
-c-a-t-e-g-o-r-y- -o-v-e-r- -t-i-m-e- -d-u-r-i-n-g- -s-p-l-i-t- -s-e-t-u-p?-

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### -Q-u-e-s-t-i-o-n- -7-3:- -M-o-d-e-l- -E-v-a-l-u-a-t-i-o-n- -Q-u-e-s-t-i-o-n- -7-3:- -H-o-w- -d-o- -y-o-u- -p-e-r-f-o-r-m-  
-c-h-r-o-n-o-l-o-g-i-c-a-l- -(t-i-m-e---b-a-s-e-d)- -v-a-l-i-d-a-t-i-o-n- -a-n-d- -w-h-y- -i-s- -i-t- -s-u-p-e-r-i-o-r- -t-o- -r-a-n-d-o-m-  
-s-p-l-i-t-s- -f-o-r- -R-i-s-k-P-u-l-s-e?-

\*- \*-R-o-u-n-d:-\*- -M-L-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-e-m-p-o-r-a-l- -d-a-t-a- -s-p-l-i-t-s,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -v-a-l-i-d-a-t-i-o-n,-  
-o-u-t---o-f---t-i-m-e- -t-e-s-t-i-n-g.-

\*- \*-W-h-y- -I-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -v-a-l-i-d-a-t-i-o-n- -s-t-r-a-t-e-g-y- -f-o-r- -t-i-m-e---d-e-p-e-n-d-e-n-t-  
-r-e-c-o-r-d-s.-

\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -L-a-y-o-f-f- -t-r-e-n-d-s- -a-r-e- -s-u-b-j-e-c-t- -t-o- -m-a-c-r-o---e-c-o-n-o-m-i-c-  
-c-y-c-l-e-s,- -A- -r-a-n-d-o-m- -K---F-o-l-d- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -l-e-a-k-s- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -f-u-t-u-r-e-  
-m-a-r-k-e-t- -s-t-a-t-e-s- -i-n-t-o- -p-a-s-t- -p-r-e-d-i-c-t-i-o-n-s,- -W-e- -u-s-e- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-p-l-i-t-t-i-n-g:- -t-r-a-i-n-i-n-g-  
-o-n- -p-a-s-t- -y-e-a-r-s- -a-n-d- -v-a-l-i-d-a-t-i-n-g- -o-n- -s-u-b-s-e-q-u-e-n-t- -y-e-a-r-s,- -w-h-i-c-h- -a-c-c-u-r-a-t-e-l-y-  
-s-i-m-u-l-a-t-e-s- -h-o-w- -t-h-e- -m-o-d-e-l- -w-i-l-l- -p-e-r-f-o-r-m- -o-n- -f-u-t-u-r-e- -u-n-s-e-e-n- -d-a-t-a.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -U-s-i-n-g- -s-t-a-n-d-a-r-d- -S-t-r-a-t-i-f-i-e-d- -K---F-o-l-d- -C-V- -w-i-t-h-o-u-t-  
-c-h-e-c-k-i-n-g- -f-o-r- -t-e-m-p-o-r-a-l- -l-e-a-k-a-g-e.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -T-i-m-e-S-e-r-i-e-s-S-p-l-i-t- -f-r-o-m-  
-s-c-i-k-i-t---l-e-a-r-n- -w-i-t-h- -a- -g-a-p- -p-a-r-a-m-e-t-e-r- -t-o- -p-r-e-v-e-n-t- -b-l-e-e-d-i-n-g- -i-n-f-o- -a-c-r-o-s-s-  
-o-v-e-r-l-a-p-p-i-n-g- -e-v-e-n-t- -w-i-n-d-o-w-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -i-s- -t-h-e- -b-a-s-e-l-i-n-e- -p-e-r-f-o-r-m-a-n-c-e- -g-a-p- -b-e-t-w-e-e-n- -a-  
-r-a-n-d-o-m- -s-p-l-i-t- -a-n-d- -a- -t-i-m-e- -s-p-l-i-t?-

-\*-D-e-e-p- -D-i-v-e:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -a- -c-o-m-p-a-n-y- -t-h-a-t- -c-h-a-n-g-e-s- -i-t-s- -i-n-d-u-s-t-r-y-  
-c-a-t-e-g-o-r-y- -o-v-e-r- -t-i-m-e- -d-u-r-i-n-g- -s-p-l-i-t- -s-e-t-u-p-?-  
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-##- -Q-u-e-s-t-i-o-n- -7-4:- -M-o-d-e-l- -E-v-a-l-u-a-t-i-o-n- -Q-u-e-s-t-i-o-n- -7-4:- -H-o-w- -d-o- -y-o-u- -p-e-r-f-o-r-m-  
-c-h-r-o-n-o-l-o-g-i-c-a-l- -(t-i-m-e---b-a-s-e-d)- -v-a-l-i-d-a-t-i-o-n- -a-n-d- -w-h-y- -i-s- -i-t- -s-u-p-e-r-i-o-r- -t-o- -r-a-n-d-o-m-  
-s-p-l-i-t-s- -f-o-r- -R-i-s-k-P-u-l-s-e-?-  
-\*- -R-o-u-n-d:-\*- -M-L-

-\*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-

-\*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-e-m-p-o-r-a-l- -d-a-t-a- -s-p-l-i-t-s,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -v-a-l-i-d-a-t-i-o-n,-  
-o-u-t---o-f---t-i-m-e- -t-e-s-t-i-n-g.-

-\*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -v-a-l-i-d-a-t-i-o-n- -s-t-r-a-t-e-g-y- -f-o-r- -t-i-m-e---d-e-p-e-n-d-e-n-t-  
-r-e-c-o-r-d-s.-

-\*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -L-a-y-o-f-f- -t-r-e-n-d-s- -a-r-e- -s-u-b-j-e-c-t- -t-o- -m-a-c-r-o---e-c-o-n-o-m-i-c-  
-c-y-c-l-e-s.- -A- -r-a-n-d-o-m- -K---F-o-l-d- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -l-e-a-k-s- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -f-u-t-u-r-e-  
-m-a-r-k-e-t- -s-t-a-t-e-s- -i-n-t-o- -p-a-s-t- -p-r-e-d-i-c-t-i-o-n-s.- -W-e- -u-s-e- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-p-l-i-t-t-i-n-g:- -t-r-a-i-n-i-n-g-  
-o-n- -p-a-s-t- -y-e-a-r-s- -a-n-d- -v-a-l-i-d-a-t-i-n-g- -o-n- -s-u-b-s-e-q-u-e-n-t- -y-e-a-r-s,- -w-h-i-c-h- -a-c-c-u-r-a-t-e-l-y-  
-s-i-m-u-l-a-t-e-s- -h-o-w- -t-h-e- -m-o-d-e-l- -w-i-l-l- -p-e-r-f-o-r-m- -o-n- -f-u-t-u-r-e- -u-n-s-e-e-n- -d-a-t-a.-

-\*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -U-s-i-n-g- -s-t-a-n-d-a-r-d- -S-t-r-a-t-i-f-i-e-d- -K---F-o-l-d- -C-V- -w-i-t-h-o-u-t-  
-c-h-e-c-k-i-n-g- -f-o-r- -t-e-m-p-o-r-a-l- -l-e-a-k-a-g-e.-

-\*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -u-s-e- -T-i-m-e-S-e-r-i-e-s-S-p-l-i-t- -f-r-o-m-  
-s-c-i-k-i-t---l-e-a-r-n- -w-i-t-h- -a- -g-a-p- -p-a-r-a-m-e-t-e-r- -t-o- -p-r-e-v-e-n-t- -b-l-e-e-d-i-n-g- -i-n-f-o- -a-c-r-o-s-s-  
-o-v-e-r-l-a-p-p-i-n-g- -e-v-e-n-t- -w-i-n-d-o-w-s.-

-\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -i-s- -t-h-e- -b-a-s-e-l-i-n-e- -p-e-r-f-o-r-m-a-n-c-e- -g-a-p- -b-e-t-w-e-e-n- -a-  
-r-a-n-d-o-m- -s-p-l-i-t- -a-n-d- -a- -t-i-m-e- -s-p-l-i-t-?-  
-\*- -D-e-e-p- -D-i-v-e:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -a- -c-o-m-p-a-n-y- -t-h-a-t- -c-h-a-n-g-e-s- -i-t-s- -i-n-d-u-s-t-r-y-  
-c-a-t-e-g-o-r-y- -o-v-e-r- -t-i-m-e- -d-u-r-i-n-g- -s-p-l-i-t- -s-e-t-u-p-?-  
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-##- -Q-u-e-s-t-i-o-n- -7-5:- -M-o-d-e-l- -E-v-a-l-u-a-t-i-o-n- -Q-u-e-s-t-i-o-n- -7-5:- -H-o-w- -d-o- -y-o-u- -p-e-r-f-o-r-m-  
-c-h-r-o-n-o-l-o-g-i-c-a-l- -(t-i-m-e---b-a-s-e-d)- -v-a-l-i-d-a-t-i-o-n- -a-n-d- -w-h-y- -i-s- -i-t- -s-u-p-e-r-i-o-r- -t-o- -r-a-n-d-o-m-  
-s-p-l-i-t-s- -f-o-r- -R-i-s-k-P-u-l-s-e-?-  
-\*- -R-o-u-n-d:-\*- -M-L-

-\*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -9-/1-0-

-\*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -T-e-m-p-o-r-a-l- -d-a-t-a- -s-p-l-i-t-s,- -c-h-r-o-n-o-l-o-g-i-c-a-l- -v-a-l-i-d-a-t-i-o-n,-  
-o-u-t---o-f---t-i-m-e- -t-e-s-t-i-n-g.-

-\*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -v-a-l-i-d-a-t-i-o-n- -s-t-r-a-t-e-g-y- -f-o-r- -t-i-m-e---d-e-p-e-n-d-e-n-t-  
-r-e-c-o-r-d-s.-

-\*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -L-a-y-o-f-f- -t-r-e-n-d-s- -a-r-e- -s-u-b-j-e-c-t- -t-o- -m-a-c-r-o---e-c-o-n-o-m-i-c-  
-c-y-c-l-e-s.- -A- -r-a-n-d-o-m- -K---F-o-l-d- -c-r-o-s-s---v-a-l-i-d-a-t-i-o-n- -l-e-a-k-s- -i-n-f-o-r-m-a-t-i-o-n- -f-r-o-m- -f-u-t-u-r-e-

-m-a-r-k-e-t -s-t-a-t-e-s- i-n-t-o- -p-a-s-t- -p-r-e-d-i-c-t-i-o-n-s-.- -W-e- -u-s-e- -c-h-r-o-n-o-l-o-g-i-c-a-l- -s-p-l-i-t-t-i-n-g-:- -t-r-a-i-n-i-n-g-  
-o-n- -p-a-s-t- -y-e-a-r-s- -a-n-d- -v-a-l-i-d-a-t-i-n-g- -o-n- -s-u-b-s-e-q-u-e-n-t- -y-e-a-r-s-,- -w-h-i-c-h- -a-c-c-u-r-a-t-e-l-y-  
-s-i-m-u-l-a-t-e-s- -h-o-w- -t-h-e- -m-o-d-e-l- -w-i-l-l- -p-e-r-f-o-r-m- -o-n- -f-u-t-u-r-e- -u-n-s-e-e-n- -d-a-t-a-.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-:-\*- -U-s-i-n-g- -s-t-a-n-d-a-r-d- -S-t-r-a-t-i-f-i-e-d- -K---F-o-l-d- -C-V- -w-i-t-h-o-u-t-  
-c-h-e-c-k-i-n-g- -f-o-r- -t-e-m-p-o-r-a-l- -l-e-a-k-a-g-e-.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-:-\*- -W-e- -u-s-e- -T-i-m-e-S-e-r-i-e-s-S-p-l-i-t- -f-r-o-m-  
-s-c-i-k-i-t---l-e-a-r-n- -w-i-t-h- -a- -g-a-p- -p-a-r-a-m-e-t-e-r- -t-o- -p-r-e-v-e-n-t- -b-l-e-e-d-i-n-g- -i-n-f-o- -a-c-r-o-s-s-  
-o-v-e-r-l-a-p-p-i-n-g- -e-v-e-n-t- -w-i-n-d-o-w-s-.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s-:-\*- -W-h-a-t- -i-s- -t-h-e- -b-a-s-e-l-i-n-e- -p-e-r-f-o-r-m-a-n-c-e- -g-a-p- -b-e-t-w-e-e-n- -a-  
-r-a-n-d-o-m- -s-p-l-i-t- -a-n-d- -a- -t-i-m-e- -s-p-l-i-t-?-

-\*- \*-D-e-e-p- -D-i-v-e-:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -a- -c-o-m-p-a-n-y- -t-h-a-t- -c-h-a-n-g-e-s- -i-t-s- -i-n-d-u-s-t-r-y-  
-c-a-t-e-g-o-r-y- -o-v-e-r- -t-i-m-e- -d-u-r-i-n-g- -s-p-l-i-t- -s-e-t-u-p-?-  
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-##-#- -Q-u-e-s-t-i-o-n- -7-6-:- -E-x-p-l-a-i-n-a-b-l-e- -A-l- -Q-u-e-s-t-i-o-n- -7-6-:- -H-o-w- -d-o- -y-o-u- -c-o-m-p-u-t-e- -S-H-A-P-  
-v-a-l-u-e-s- -f-o-r- -a-n- -e-n-s-e-m-b-l-e- -S-t-a-c-k-i-n-g- -c-l-a-s-s-i-f-i-e-r-?-  
-\*- \*-R-o-u-n-d-:-\*- -R-e-s-e-a-r-c-h-  
-\*- \*-D-i-f-f-i-c-u-l-t-y-:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e-:-\*- -9-/1-0-  
-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d-:-\*- -K-e-r-n-e-l-S-H-A-P-,- -T-r-e-e-S-H-A-P-,- -c-o-a-l-i-t-i-o-n- -g-a-m-e- -t-h-e-o-r-y-.-  
-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t-:-\*- -T-e-s-t-s- -d-e-e-p- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -S-H-A-P-  
-c-o-m-p-u-t-a-t-i-o-n-s-.-  
-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-:-\*- -W-h-i-l-e- -T-r-e-e-S-H-A-P- -i-s- -f-a-s-t- -f-o-r- -t-r-e-e---b-a-s-e-d- -m-o-d-e-l-s-,- -i-t-  
-c-a-n-n-o-t- -b-e- -r-u-n- -o-n- -a- -S-t-a-c-k-i-n-g- -c-l-a-s-s-i-f-i-e-r- -w-i-t-h- -a- -n-o-n---t-r-e-e- -m-e-t-a---l-e-a-r-n-e-r-,- -W-e-  
-e-i-t-h-e-r- -c-a-l-c-u-l-a-t-e- -T-r-e-e-S-H-A-P- -o-n- -t-h-e- -p-r-i-m-a-r-y- -b-a-s-e- -m-o-d-e-l-s- -(l-i-k-e- -X-G-B-o-o-s-t)- -a-n-d-  
-a-g-g-r-e-g-a-t-e- -t-h-e-i-r- -e-x-p-l-a-n-a-t-i-o-n-s-,- -o-r- -w-e- -r-u-n- -K-e-r-n-e-l-S-H-A-P- -o-n- -t-h-e- -e-n-t-i-r-e- -S-t-a-c-k-i-n-g-  
-p-i-p-e-l-i-n-e-,- -w-h-i-c-h- -i-s- -m-o-r-e- -e-x-p-e-n-s-i-v-e-.-  
-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-:-\*- -T-r-y-i-n-g- -t-o- -r-u-n- -T-r-e-e-S-H-A-P- -d-i-r-e-c-t-l-y- -o-n- -t-h-e-  
-S-t-a-c-k-i-n-g-C-l-a-s-s-i-f-i-e-r- -w-r-a-p-p-e-r-.-  
-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-:-\*- -T-o- -m-a-i-n-t-a-i-n- -r-e-a-l---t-i-m-e- -p-e-r-f-o-r-m-a-n-c-e- -i-n-  
-p-r-o-d-u-c-t-i-o-n-,- -w-e- -e-x-t-r-a-c-t- -t-h-e- -b-a-s-e- -X-G-B-o-o-s-t- -m-o-d-e-l- -a-n-d- -p-r-e---g-e-n-e-r-a-t-e- -i-t-s- -S-H-A-P-  
-e-x-p-l-a-i-n-e-r- -t-o- -c-o-m-p-u-t-e- -l-o-c-a-l- -w-a-t-e-r-f-a-l-l-s-,- -k-e-e-p-i-n-g- -p-r-e-d-i-c-t-i-o-n- -l-a-t-e-n-c-y- -u-n-d-e-r-  
-1-0-0-m-s-.-  
-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s-:-\*- -W-h-a-t- -i-s- -t-h-e- -m-a-t-h-e-m-a-t-i-c-a-l- -d-e-f-i-n-i-t-i-o-n- -o-f- -a- -S-h-a-p-l-e-y-  
-v-a-l-u-e-?-  
-\*- \*-D-e-e-p- -D-i-v-e-:-\*- -H-o-w- -d-o-e-s- -T-r-e-e-S-H-A-P- -a-c-h-i-e-v-e- -p-o-l-y-n-o-m-i-a-l- -t-i-m-e- -c-o-m-p-l-e-x-i-t-y-  
-O-(-T- -L- -D-^2)- -c-o-m-p-a-r-e-d- -t-o- -K-e-r-n-e-l-S-H-A-P-'s- -e-x-p-o-n-e-n-t-i-a-l- -t-i-m-e-?-  
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-##-#- -Q-u-e-s-t-i-o-n- -7-7-:- -E-x-p-l-a-i-n-a-b-l-e- -A-l- -Q-u-e-s-t-i-o-n- -7-7-:- -H-o-w- -d-o- -y-o-u- -c-o-m-p-u-t-e- -S-H-A-P-  
-v-a-l-u-e-s- -f-o-r- -a-n- -e-n-s-e-m-b-l-e- -S-t-a-c-k-i-n-g- -c-l-a-s-s-i-f-i-e-r-?-  
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\*\*-\*\*-R-o-u-n-d:-\*\*- R-e-s-e-a-r-c-h-

\*\*-\*\*-D-i-f-f-i-c-u-l-t-y:-\*\*- H-a-r-d-|-\*\*-l-m-p-o-r-t-a-n-c-e:-\*\*- -9/-1-0-

\*\*-\*\*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*\*- K-e-r-n-e-l-S-H-A-P,- T-r-e-e-S-H-A-P,- c-o-a-l-i-t-i-o-n- g-a-m-e- t-h-e-o-r-y.-

\*\*-\*\*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*\*- T-e-s-t-s- d-e-e-p- u-n-d-e-r-s-t-a-n-d-i-n-g- o-f- S-H-A-P- c-o-m-p-u-t-a-t-i-o-n-s.-

\*\*-\*\*-l-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*\*- W-h-i-l-e- T-r-e-e-S-H-A-P- i-s- f-a-s-t- f-o-r- t-r-e-e---b-a-s-e-d- m-o-d-e-l-s,- i-t- c-a-n-n-o-t- b-e- r-u-n- o-n- a- S-t-a-c-k-i-n-g- c-l-a-s-s-i-f-i-e-r- w-i-t-h- a- n-o-n---t-r-e-e- m-e-t-a---l-e-a-r-n-e-r.- W-e- e-i-t-h-e-r- c-a-l-c-u-l-a-t-e- T-r-e-e-S-H-A-P- o-n- t-h-e- p-r-i-m-a-r-y- b-a-s-e- m-o-d-e-l-s- (-l-i-k-e- X-G-B-o-o-s-t)- a-n-d- a-g-g-r-e-g-a-t-e- t-h-e-i-r- e-x-p-l-a-n-a-t-i-o-n-s,- o-r- w-e- r-u-n- K-e-r-n-e-l-S-H-A-P- o-n- t-h-e- e-n-t-i-r-e- S-t-a-c-k-i-n-g- p-i-p-e-l-i-n-e,- w-h-i-c-h- i-s- m-o-r-e- e-x-p-e-n-s-i-v-e.-

\*\*-\*\*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*\*- T-r-y-i-n-g- t-o- r-u-n- T-r-e-e-S-H-A-P- d-i-r-e-c-t-l-y- o-n- t-h-e- S-t-a-c-k-i-n-g-C-l-a-s-s-i-f-i-e-r- w-r-a-p-p-e-r.-

\*\*-\*\*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*\*- T-o- m-a-i-n-t-a-i-n- r-e-a-l---t-i-m-e- p-e-r-f-o-r-m-a-n-c-e- i-n- p-r-o-d-u-c-t-i-o-n,- w-e- e-x-t-r-a-c-t- t-h-e- b-a-s-e- X-G-B-o-o-s-t- m-o-d-e-l- a-n-d- p-r-e---g-e-n-e-r-a-t-e- i-t-s- S-H-A-P- e-x-p-l-a-i-n-e-r- t-o- c-o-m-p-u-t-e- l-o-c-a-l- w-a-t-e-r-f-a-l-l-s,- k-e-e-p-i-n-g- p-r-e-d-i-c-t-i-o-n- l-a-t-e-n-c-y- u-n-d-e-r- 1-0-0-m-s.-

\*\*-\*\*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*\*- W-h-a-t- i-s- t-h-e- m-a-t-h-e-m-a-t-i-c-a-l- d-e-f-i-n-i-t-i-o-n- o-f- a- S-h-a-p-l-e-y- v-a-l-u-e?-

\*\*-\*\*-D-e-e-p- D-i-v-e:-\*\*- H-o-w- d-o-e-s- T-r-e-e-S-H-A-P- a-c-h-i-e-v-e- p-o-l-y-n-o-m-i-a-l- t-i-m-e- c-o-m-p-l-e-x-i-t-y- O-(-T- L- D^2)- c-o-m-p-a-r-e-d- t-o- K-e-r-n-e-l-S-H-A-P-'s- e-x-p-o-n-e-n-t-i-a-l- t-i-m-e?-  
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##-#- Q-u-e-s-t-i-o-n- 7-8:- E-x-p-l-a-i-n-a-b-l-e- A-l- Q-u-e-s-t-i-o-n- 7-8:- H-o-w- d-o- y-o-u- c-o-m-p-u-t-e- S-H-A-P- v-a-l-u-e-s- f-o-r- a-n- e-n-s-e-m-b-l-e- S-t-a-c-k-i-n-g- c-l-a-s-s-i-f-i-e-r?-  
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\*\*-\*\*-R-o-u-n-d:-\*\*- R-e-s-e-a-r-c-h-

\*\*-\*\*-D-i-f-f-i-c-u-l-t-y:-\*\*- H-a-r-d-|-\*\*-l-m-p-o-r-t-a-n-c-e:-\*\*- -9/-1-0-

\*\*-\*\*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*\*- K-e-r-n-e-l-S-H-A-P,- T-r-e-e-S-H-A-P,- c-o-a-l-i-t-i-o-n- g-a-m-e- t-h-e-o-r-y.-

\*\*-\*\*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*\*- T-e-s-t-s- d-e-e-p- u-n-d-e-r-s-t-a-n-d-i-n-g- o-f- S-H-A-P- c-o-m-p-u-t-a-t-i-o-n-s.-

\*\*-\*\*-l-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*\*- W-h-i-l-e- T-r-e-e-S-H-A-P- i-s- f-a-s-t- f-o-r- t-r-e-e---b-a-s-e-d- m-o-d-e-l-s,- i-t- c-a-n-n-o-t- b-e- r-u-n- o-n- a- S-t-a-c-k-i-n-g- c-l-a-s-s-i-f-i-e-r- w-i-t-h- a- n-o-n---t-r-e-e- m-e-t-a---l-e-a-r-n-e-r.- W-e- e-i-t-h-e-r- c-a-l-c-u-l-a-t-e- T-r-e-e-S-H-A-P- o-n- t-h-e- p-r-i-m-a-r-y- b-a-s-e- m-o-d-e-l-s- (-l-i-k-e- X-G-B-o-o-s-t)- a-n-d- a-g-g-r-e-g-a-t-e- t-h-e-i-r- e-x-p-l-a-n-a-t-i-o-n-s,- o-r- w-e- r-u-n- K-e-r-n-e-l-S-H-A-P- o-n- t-h-e- e-n-t-i-r-e- S-t-a-c-k-i-n-g- p-i-p-e-l-i-n-e,- w-h-i-c-h- i-s- m-o-r-e- e-x-p-e-n-s-i-v-e.-

\*\*-\*\*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*\*- T-r-y-i-n-g- t-o- r-u-n- T-r-e-e-S-H-A-P- d-i-r-e-c-t-l-y- o-n- t-h-e- S-t-a-c-k-i-n-g-C-l-a-s-s-i-f-i-e-r- w-r-a-p-p-e-r.-

\*\*-\*\*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*\*- T-o- m-a-i-n-t-a-i-n- r-e-a-l---t-i-m-e- p-e-r-f-o-r-m-a-n-c-e- i-n- p-r-o-d-u-c-t-i-o-n,- w-e- e-x-t-r-a-c-t- t-h-e- b-a-s-e- X-G-B-o-o-s-t- m-o-d-e-l- a-n-d- p-r-e---g-e-n-e-r-a-t-e- i-t-s- S-H-A-P- e-x-p-l-a-i-n-e-r- t-o- c-o-m-p-u-t-e- l-o-c-a-l- w-a-t-e-r-f-a-l-l-s,- k-e-e-p-i-n-g- p-r-e-d-i-c-t-i-o-n- l-a-t-e-n-c-y- u-n-d-e-r- 1-0-0-m-s.-

\*\*-\*\*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*\*- W-h-a-t- i-s- t-h-e- m-a-t-h-e-m-a-t-i-c-a-l- d-e-f-i-n-i-t-i-o-n- o-f- a- S-h-a-p-l-e-y- v-a-l-u-e?-

\*- \*-D-e-e-p- -D-i-v-e-:-\*- -H-o-w- -d-o-e-s- -T-r-e-e-S-H-A-P- -a-c-h-i-e-v-e- -p-o-l-y-n-o-m-i-a-l- -t-i-m-e- -c-o-m-p-l-e-x-i-t-y-  
-O-(-T- -L- -D^2-) -c-o-m-p-a-r-e-d- -t-o- -K-e-r-n-e-l-S-H-A-P-'s- -e-x-p-o-n-e-n-t-i-a-l- -t-i-m-e-?-  
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###- -Q-u-e-s-t-i-o-n- -7-9:- -E-x-p-l-a-i-n-a-b-l-e- -A-l- -Q-u-e-s-t-i-o-n- -7-9:- -H-o-w- -d-o- -y-o-u- -c-o-m-p-u-t-e- -S-H-A-P-  
-v-a-l-u-e-s- -f-o-r- -a-n- -e-n-s-e-m-b-l-e- -S-t-a-c-k-i-n-g- -c-l-a-s-s-i-f-i-e-r-?-  
\*- \*-R-o-u-n-d:-\*- -R-e-s-e-a-r-c-h-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -K-e-r-n-e-l-S-H-A-P-, -T-r-e-e-S-H-A-P-, -c-o-a-l-i-t-i-o-n- -g-a-m-e- -t-h-e-o-r-y.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -d-e-e-p- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -S-H-A-P-  
-c-o-m-p-u-t-a-t-i-o-n-s.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -W-h-i-l-e- -T-r-e-e-S-H-A-P- -i-s- -f-a-s-t- -f-o-r- -t-r-e-e---b-a-s-e-d- -m-o-d-e-l-s-, -i-t-  
-c-a-n-n-o-t- -b-e- -r-u-n- -o-n- -a- -S-t-a-c-k-i-n-g- -c-l-a-s-s-i-f-i-e-r- -w-i-t-h- -a- -n-o-n---t-r-e-e- -m-e-t-a---l-e-a-r-n-e-r-, -W-e-  
-e-i-t-h-e-r- -c-a-l-c-u-l-a-t-e- -T-r-e-e-S-H-A-P- -o-n- -t-h-e- -p-r-i-m-a-r-y- -b-a-s-e- -m-o-d-e-l-s- -(l-i-k-e- -X-G-B-o-o-s-t)- -a-n-d-  
-a-g-g-r-e-g-a-t-e- -t-h-e-i-r- -e-x-p-l-a-n-a-t-i-o-n-s-, -o-r- -w-e- -r-u-n- -K-e-r-n-e-l-S-H-A-P- -o-n- -t-h-e- -e-n-t-i-r-e- -S-t-a-c-k-i-n-g-  
-p-i-p-e-l-i-n-e-, -w-h-i-c-h- -i-s- -m-o-r-e- -e-x-p-e-n-s-i-v-e.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -T-r-y-i-n-g- -t-o- -r-u-n- -T-r-e-e-S-H-A-P- -d-i-r-e-c-t-l-y- -o-n- -t-h-e-  
-S-t-a-c-k-i-n-g-C-l-a-s-s-i-f-i-e-r- -w-r-a-p-p-e-r.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -T-o- -m-a-i-n-t-a-i-n- -r-e-a-l---t-i-m-e- -p-e-r-f-o-r-m-a-n-c-e- -i-n-  
-p-r-o-d-u-c-t-i-o-n-, -w-e- -e-x-t-r-a-c-t- -t-h-e- -b-a-s-e- -X-G-B-o-o-s-t- -m-o-d-e-l- -a-n-d- -p-r-e---g-e-n-e-r-a-t-e- -i-t-s- -S-H-A-P-  
-e-x-p-l-a-i-n-e-r- -t-o- -c-o-m-p-u-t-e- -l-o-c-a-l- -w-a-t-e-r-f-a-l-l-s-, -k-e-e-p-i-n-g- -p-r-e-d-i-c-t-i-o-n- -l-a-t-e-n-c-y- -u-n-d-e-r-  
-1-0-0-m-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -i-s- -t-h-e- -m-a-t-h-e-m-a-t-i-c-a-l- -d-e-f-i-n-i-t-i-o-n- -o-f- -a- -S-h-a-p-l-e-y-  
-v-a-l-u-e-?-  
\*- \*-D-e-e-p- -D-i-v-e-:-\*- -H-o-w- -d-o-e-s- -T-r-e-e-S-H-A-P- -a-c-h-i-e-v-e- -p-o-l-y-n-o-m-i-a-l- -t-i-m-e- -c-o-m-p-l-e-x-i-t-y-  
-O-(-T- -L- -D^2-) -c-o-m-p-a-r-e-d- -t-o- -K-e-r-n-e-l-S-H-A-P-'s- -e-x-p-o-n-e-n-t-i-a-l- -t-i-m-e-?-  
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###- -Q-u-e-s-t-i-o-n- -8-0:- -E-x-p-l-a-i-n-a-b-l-e- -A-l- -Q-u-e-s-t-i-o-n- -8-0:- -H-o-w- -d-o- -y-o-u- -c-o-m-p-u-t-e- -S-H-A-P-  
-v-a-l-u-e-s- -f-o-r- -a-n- -e-n-s-e-m-b-l-e- -S-t-a-c-k-i-n-g- -c-l-a-s-s-i-f-i-e-r-?-  
\*- \*-R-o-u-n-d:-\*- -R-e-s-e-a-r-c-h-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -9/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -K-e-r-n-e-l-S-H-A-P-, -T-r-e-e-S-H-A-P-, -c-o-a-l-i-t-i-o-n- -g-a-m-e- -t-h-e-o-r-y.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -d-e-e-p- -u-n-d-e-r-s-t-a-n-d-i-n-g- -o-f- -S-H-A-P-  
-c-o-m-p-u-t-a-t-i-o-n-s.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -W-h-i-l-e- -T-r-e-e-S-H-A-P- -i-s- -f-a-s-t- -f-o-r- -t-r-e-e---b-a-s-e-d- -m-o-d-e-l-s-, -i-t-  
-c-a-n-n-o-t- -b-e- -r-u-n- -o-n- -a- -S-t-a-c-k-i-n-g- -c-l-a-s-s-i-f-i-e-r- -w-i-t-h- -a- -n-o-n---t-r-e-e- -m-e-t-a---l-e-a-r-n-e-r-, -W-e-  
-e-i-t-h-e-r- -c-a-l-c-u-l-a-t-e- -T-r-e-e-S-H-A-P- -o-n- -t-h-e- -p-r-i-m-a-r-y- -b-a-s-e- -m-o-d-e-l-s- -(l-i-k-e- -X-G-B-o-o-s-t)- -a-n-d-  
-a-g-g-r-e-g-a-t-e- -t-h-e-i-r- -e-x-p-l-a-n-a-t-i-o-n-s-, -o-r- -w-e- -r-u-n- -K-e-r-n-e-l-S-H-A-P- -o-n- -t-h-e- -e-n-t-i-r-e- -S-t-a-c-k-i-n-g-  
-p-i-p-e-l-i-n-e-, -w-h-i-c-h- -i-s- -m-o-r-e- -e-x-p-e-n-s-i-v-e.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -T-r-y-i-n-g- -t-o- -r-u-n- -T-r-e-e-S-H-A-P- -d-i-r-e-c-t-l-y- -o-n- -t-h-e-  
-S-t-a-c-k-i-n-g-C-l-a-s-s-i-f-i-e-r- -w-r-a-p-p-e-r.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -T-o- -m-a-i-n-t-a-i-n- -r-e-a-l---t-i-m-e- -p-e-r-f-o-r-m-a-n-c-e- -i-n-  
-p-r-o-d-u-c-t-i-o-n,- -w-e- -e-x-t-r-a-c-t- -t-h-e- -b-a-s-e- -X-G-B-o-o-s-t- -m-o-d-e-l- -a-n-d- -p-r-e---g-e-n-e-r-a-t-e- -i-t-s- -S-H-A-P-  
-e-x-p-l-a-i-n-e-r- -t-o- -c-o-m-p-u-t-e- -l-o-c-a-l- -w-a-t-e-r-f-a-l-l-s,- -k-e-e-p-i-n-g- -p-r-e-d-i-c-t-i-o-n- -l-a-t-e-n-c-y- -u-n-d-e-r-  
-1-0-0-m-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -W-h-a-t- -i-s- -t-h-e- -m-a-t-h-e-m-a-t-i-c-a-l- -d-e-f-i-n-i-t-i-o-n- -o-f- -a- -S-h-a-p-l-e-y-  
-v-a-l-u-e?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -H-o-w- -d-o-e-s- -T-r-e-e-S-H-A-P- -a-c-h-i-e-v-e- -p-o-l-y-n-o-m-i-a-l- -t-i-m-e- -c-o-m-p-l-e-x-i-t-y-  
-O-(-T- -L- -D^2)- -c-o-m-p-a-r-e-d- -t-o- -K-e-r-n-e-l-S-H-A-P-'s- -e-x-p-o-n-e-n-t-i-a-l- -t-i-m-e?-

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###- -Q-u-e-s-t-i-o-n- -8-1:- -N-e-w-s- -S-e-n-t-i-m-e-n-t- -Q-u-e-s-t-i-o-n- -8-1:- -H-o-w- -d-o- -y-o-u- -h-a-n-d-l-e- -s-e-n-t-i-m-e-n-t-  
-s-c-o-r-i-n-g- -d-r-i-f-t- -w-h-e-n- -m-a-c-r-o---e-c-o-n-o-m-i-c- -j-a-r-g-o-n- -s-h-i-f-t-s?-

\*- \*-R-o-u-n-d:-\*- -R-e-s-e-a-r-c-h-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -S-e-n-t-i-m-e-n-t- -d-r-i-f-t,- -l-e-x-i-c-o-n- -u-p-d-a-t-e-s,- -L-L-M- -f-i-n-e---t-u-n-i-n-g.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -N-L-P- -p-i-p-e-l-i-n-e- -r-o-b-u-s-t-n-e-s-s.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -O-u-r- -s-e-n-t-i-m-e-n-t- -e-n-g-i-n-e- -c-u-r-r-e-n-t-l-y- -u-s-e-s- -a- -k-e-y-w-o-r-d-  
-h-e-u-r-i-s-t-i-c- -w-e-i-g-h-t-i-n-g- -l-a-y-o-f-f- -a-n-d- -g-r-o-w-t-h- -w-o-r-d-s.- -O-v-e-r- -t-i-m-e,- -t-e-r-m-i-n-o-l-o-g-y- -c-h-a-n-g-e-s-  
-(e.-g.- -'r-i-g-h-t---s-i-z-i-n-g',- -'r-e---a-l-l-o-c-a-t-i-n-g- -r-e-s-o-u-r-c-e-s').- -H-e-u-r-i-s-t-i-c-s- -m-u-s-t- -b-e-  
-p-e-r-i-o-d-i-c-a-l-l-y- -u-p-d-a-t-e-d,- -o-r- -r-e-p-l-a-c-e-d- -b-y- -a- -p-r-e-t-r-a-i-n-e-d- -T-r-a-n-s-f-o-r-m-e-r- -(l-i-k-e-  
-F-i-n-B-E-R-T)- -t-h-a-t- -g-e-n-e-r-a-l-i-z-e-s- -b-e-t-t-e-r- -t-o- -s-e-m-a-n-t-i-c- -n-u-a-n-c-e-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -R-e-l-y-i-n-g- -o-n- -a- -s-t-a-t-i-c- -v-o-c-a-b-u-l-a-r-y- -w-i-t-h-o-u-t- -c-h-e-c-k-i-n-g- -f-o-r-  
-u-n-s-e-e-n- -s-e-n-t-i-m-e-n-t- -v-o-c-a-b-u-l-a-r-y.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -r-u-n- -a- -w-e-e-k-l-y- -a-u-d-i-t- -s-c-r-i-p-t- -t-h-a-t-  
-f-l-a-g-s- -a-r-t-i-c-l-e-s- -w-i-t-h- -n-e-u-t-r-a-l- -s-c-o-r-e-s- -b-u-t- -h-i-g-h- -r-i-s-k- -i-n-d-i-c-a-t-o-r-s,- -a-n-d- -f-e-e-d- -t-h-e-m-  
-i-n-t-o- -a- -v-a-l-i-d-a-t-i-o-n- -s-e-t- -t-o- -c-h-e-c-k- -i-f- -o-u-r- -v-o-c-a-b-u-l-a-r-y- -n-e-e-d-s- -u-p-d-a-t-e-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -d-o- -y-o-u- -a-v-o-i-d- -b-i-a-s- -w-h-e-n- -a- -c-o-m-p-a-n-y- -g-e-t-s-  
-n-e-g-a-t-i-v-e- -c-o-v-e-r-a-g-e- -u-n-r-e-l-a-t-e-d- -t-o- -w-o-r-k-f-o-r-c-e- -r-e-d-u-c-t-i-o-n-s?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -h-o-w- -y-o-u- -w-o-u-l-d- -d-e-s-i-g-n- -a- -F-i-n-B-E-R-T- -f-i-n-e---t-u-n-i-n-g-  
-p-i-p-e-l-i-n-e- -u-s-i-n-g- -H-u-g-g-i-n-g- -F-a-c-e.-

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###- -Q-u-e-s-t-i-o-n- -8-2:- -N-e-w-s- -S-e-n-t-i-m-e-n-t- -Q-u-e-s-t-i-o-n- -8-2:- -H-o-w- -d-o- -y-o-u- -h-a-n-d-l-e- -s-e-n-t-i-m-e-n-t-  
-s-c-o-r-i-n-g- -d-r-i-f-t- -w-h-e-n- -m-a-c-r-o---e-c-o-n-o-m-i-c- -j-a-r-g-o-n- -s-h-i-f-t-s?-

\*- \*-R-o-u-n-d:-\*- -R-e-s-e-a-r-c-h-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -S-e-n-t-i-m-e-n-t- -d-r-i-f-t,- -l-e-x-i-c-o-n- -u-p-d-a-t-e-s,- -L-L-M- -f-i-n-e---t-u-n-i-n-g.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -N-L-P- -p-i-p-e-l-i-n-e- -r-o-b-u-s-t-n-e-s-s.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -O-u-r- -s-e-n-t-i-m-e-n-t- -e-n-g-i-n-e- -c-u-r-r-e-n-t-l-y- -u-s-e-s- -a- -k-e-y-w-o-r-d- -h-e-u-r-i-s-t-i-c- -w-e-i-g-h-t-i-n-g- -l-a-y-o-f-f- -a-n-d- -g-r-o-w-t-h- -w-o-r-d-s.- -O-v-e-r- -t-i-m-e,- -t-e-r-m-i-n-o-l-o-g-y- -c-h-a-n-g-e-s- -(-e-.g.- -'r-i-g-h-t---s-i-z-i-n-g',- -'r-e---a-l-l-o-c-a-t-i-n-g- -r-e-s-o-u-r-c-e-s').- -H-e-u-r-i-s-t-i-c-s- -m-u-s-t- -b-e- -p-e-r-i-o-d-i-c-a-l-l-y- -u-p-d-a-t-e-d,- -o-r- -r-e-p-l-a-c-e-d- -b-y- -a- -p-r-e-t-r-a-i-n-e-d- -T-r-a-n-s-f-o-r-m-e-r- -(l-i-k-e- -F-i-n-B-E-R-T)- -t-h-a-t- -g-e-n-e-r-a-l-i-z-e-s- -b-e-t-t-e-r- -t-o- -s-e-m-a-n-t-i-c- -n-u-a-n-c-e-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -R-e-l-y-i-n-g- -o-n- -a- -s-t-a-t-i-c- -v-o-c-a-b-u-l-a-r-y- -w-i-t-h-o-u-t- -c-h-e-c-k-i-n-g- -f-o-r- -u-n-s-e-e-n- -s-e-n-t-i-m-e-n-t- -v-o-c-a-b-u-l-a-r-y.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -r-u-n- -a- -w-e-e-k-l-y- -a-u-d-i-t- -s-c-r-i-p-t- -t-h-a-t- -f-l-a-g-s- -a-r-t-i-c-l-e-s- -w-i-t-h- -n-e-u-t-r-a-l- -s-c-o-r-e-s- -b-u-t- -h-i-g-h- -r-i-s-k- -i-n-d-i-c-a-t-o-r-s,- -a-n-d- -f-e-e-d- -t-h-e-m- -i-n-t-o- -a- -v-a-l-i-d-a-t-i-o-n- -s-e-t- -t-o- -c-h-e-c-k- -i-f- -o-u-r- -v-o-c-a-b-u-l-a-r-y- -n-e-e-d-s- -u-p-d-a-t-e-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -d-o- -y-o-u- -a-v-o-i-d- -b-i-a-s- -w-h-e-n- -a- -c-o-m-p-a-n-y- -g-e-t-s- -n-e-g-a-t-i-v-e- -c-o-v-e-r-a-g-e- -u-n-r-e-l-a-t-e-d- -t-o- -w-o-r-k-f-o-r-c-e- -r-e-d-u-c-t-i-o-n-s?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -h-o-w- -y-o-u- -w-o-u-l-d- -d-e-s-i-g-n- -a- -F-i-n-B-E-R-T- -f-i-n-e---t-u-n-i-n-g- -p-i-p-e-l-i-n-e- -u-s-i-n-g- -H-u-g-g-i-n-g- -F-a-c-e.-

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###- -Q-u-e-s-t-i-o-n- -8-3:- -N-e-w-s- -S-e-n-t-i-m-e-n-t- -Q-u-e-s-t-i-o-n- -8-3:- -H-o-w- -d-o- -y-o-u- -h-a-n-d-l-e- -s-e-n-t-i-m-e-n-t- -s-c-o-r-i-n-g- -d-r-i-f-t- -w-h-e-n- -m-a-c-r-o---e-c-o-n-o-m-i-c- -j-a-r-g-o-n- -s-h-i-f-t-s?-

\*- \*-R-o-u-n-d:-\*- -R-e-s-e-a-r-c-h-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -M-e-d-i-u-m- -|- \*-l-m-p-o-r-t-a-n-c-e:-\*- -8-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -S-e-n-t-i-m-e-n-t- -d-r-i-f-t,- -l-e-x-i-c-o-n- -u-p-d-a-t-e-s,- -L-L-M- -f-i-n-e---t-u-n-i-n-g.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -N-L-P- -p-i-p-e-l-i-n-e- -r-o-b-u-s-t-n-e-s-s.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -O-u-r- -s-e-n-t-i-m-e-n-t- -e-n-g-i-n-e- -c-u-r-r-e-n-t-l-y- -u-s-e-s- -a- -k-e-y-w-o-r-d- -h-e-u-r-i-s-t-i-c- -w-e-i-g-h-t-i-n-g- -l-a-y-o-f-f- -a-n-d- -g-r-o-w-t-h- -w-o-r-d-s.- -O-v-e-r- -t-i-m-e,- -t-e-r-m-i-n-o-l-o-g-y- -c-h-a-n-g-e-s- -(-e-.g.- -'r-i-g-h-t---s-i-z-i-n-g',- -'r-e---a-l-l-o-c-a-t-i-n-g- -r-e-s-o-u-r-c-e-s').- -H-e-u-r-i-s-t-i-c-s- -m-u-s-t- -b-e- -p-e-r-i-o-d-i-c-a-l-l-y- -u-p-d-a-t-e-d,- -o-r- -r-e-p-l-a-c-e-d- -b-y- -a- -p-r-e-t-r-a-i-n-e-d- -T-r-a-n-s-f-o-r-m-e-r- -(l-i-k-e- -F-i-n-B-E-R-T)- -t-h-a-t- -g-e-n-e-r-a-l-i-z-e-s- -b-e-t-t-e-r- -t-o- -s-e-m-a-n-t-i-c- -n-u-a-n-c-e-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -R-e-l-y-i-n-g- -o-n- -a- -s-t-a-t-i-c- -v-o-c-a-b-u-l-a-r-y- -w-i-t-h-o-u-t- -c-h-e-c-k-i-n-g- -f-o-r- -u-n-s-e-e-n- -s-e-n-t-i-m-e-n-t- -v-o-c-a-b-u-l-a-r-y.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -r-u-n- -a- -w-e-e-k-l-y- -a-u-d-i-t- -s-c-r-i-p-t- -t-h-a-t- -f-l-a-g-s- -a-r-t-i-c-l-e-s- -w-i-t-h- -n-e-u-t-r-a-l- -s-c-o-r-e-s- -b-u-t- -h-i-g-h- -r-i-s-k- -i-n-d-i-c-a-t-o-r-s,- -a-n-d- -f-e-e-d- -t-h-e-m- -i-n-t-o- -a- -v-a-l-i-d-a-t-i-o-n- -s-e-t- -t-o- -c-h-e-c-k- -i-f- -o-u-r- -v-o-c-a-b-u-l-a-r-y- -n-e-e-d-s- -u-p-d-a-t-e-s.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -d-o- -y-o-u- -a-v-o-i-d- -b-i-a-s- -w-h-e-n- -a- -c-o-m-p-a-n-y- -g-e-t-s- -n-e-g-a-t-i-v-e- -c-o-v-e-r-a-g-e- -u-n-r-e-l-a-t-e-d- -t-o- -w-o-r-k-f-o-r-c-e- -r-e-d-u-c-t-i-o-n-s?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -E-x-p-l-a-i-n- -h-o-w- -y-o-u- -w-o-u-l-d- -d-e-s-i-g-n- -a- -F-i-n-B-E-R-T- -f-i-n-e---t-u-n-i-n-g- -p-i-p-e-l-i-n-e- -u-s-i-n-g- -H-u-g-g-i-n-g- -F-a-c-e.-

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###- -Q-u-e-s-t-i-o-n- -8-4:- -N-e-w-s- -S-e-n-t-i-m-e-n-t- -Q-u-e-s-t-i-o-n- -8-4:- -H-o-w- -d-o- -y-o-u- -h-a-n-d-l-e- -s-e-n-t-i-m-e-n-t-

-s-c-o-r-i-n-g- -d-r-i-f-t- -w-h-e-n- -m-a-c-r-o---e-c-o-n-o-m-i-c- -j-a-r-g-o-n- -s-h-i-f-t-s-?-

-\*- \*-R-o-u-n-d-:-\*- -R-e-s-e-a-r-c-h-

-\*- \*-D-i-f-f-i-c-u-l-t-y-:-\*- -M-e-d-i-u-m- -|- \*- \*-l-m-p-o-r-t-a-n-c-e-:-\*- -8-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d-:-\*- -S-e-n-t-i-m-e-n-t- -d-r-i-f-t-, -l-e-x-i-c-o-n- -u-p-d-a-t-e-s-, -L-L-M- -f-i-n-e---t-u-n-i-n-g.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t-:-\*- -T-e-s-t-s- -N-L-P- -p-i-p-e-l-i-n-e- -r-o-b-u-s-t-n-e-s-s.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-:-\*- -O-u-r- -s-e-n-t-i-m-e-n-t- -e-n-g-i-n-e- -c-u-r-r-e-n-t-l-y- -u-s-e-s- -a- -k-e-y-w-o-r-d-  
-h-e-u-r-i-s-t-i-c- -w-e-i-g-h-t-i-n-g- -l-a-y-o-f-f- -a-n-d- -g-r-o-w-t-h- -w-o-r-d-s.- -O-v-e-r- -t-i-m-e-, -t-e-r-m-i-n-o-l-o-g-y- -c-h-a-n-g-e-s-  
-(e-.g.- -'r-i-g-h-t---s-i-z-i-n-g',- -'r-e---a-l-l-o-c-a-t-i-n-g- -r-e-s-o-u-r-c-e-s')-.- -H-e-u-r-i-s-t-i-c-s- -m-u-s-t- -b-e-  
-p-e-r-i-o-d-i-c-a-l-l-y- -u-p-d-a-t-e-d-, -o-r- -r-e-p-l-a-c-e-d- -b-y- -a- -p-r-e-t-r-a-i-n-e-d- -T-r-a-n-s-f-o-r-m-e-r- -(l-i-k-e-  
-F-i-n-B-E-R-T)- -t-h-a-t- -g-e-n-e-r-a-l-i-z-e-s- -b-e-t-t-e-r- -t-o- -s-e-m-a-n-t-i-c- -n-u-a-n-c-e-s.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-:-\*- -R-e-l-y-i-n-g- -o-n- -a- -s-t-a-t-i-c- -v-o-c-a-b-u-l-a-r-y- -w-i-t-h-o-u-t- -c-h-e-c-k-i-n-g- -f-o-r-  
-u-n-s-e-e-n- -s-e-n-t-i-m-e-n-t- -v-o-c-a-b-u-l-a-r-y.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-:-\*- -W-e- -r-u-n- -a- -w-e-e-k-l-y- -a-u-d-i-t- -s-c-r-i-p-t- -t-h-a-t-  
-f-l-a-g-s- -a-r-t-i-c-l-e-s- -w-i-t-h- -n-e-u-t-r-a-l- -s-c-o-r-e-s- -b-u-t- -h-i-g-h- -r-i-s-k- -i-n-d-i-c-a-t-o-r-s-, -a-n-d- -f-e-e-d- -t-h-e-m-  
-i-n-t-o- -a- -v-a-l-i-d-a-t-i-o-n- -s-e-t- -t-o- -c-h-e-c-k- -i-f- -o-u-r- -v-o-c-a-b-u-l-a-r-y- -n-e-e-d-s- -u-p-d-a-t-e-s.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s-:-\*- -H-o-w- -d-o- -y-o-u- -a-v-o-i-d- -b-i-a-s- -w-h-e-n- -a- -c-o-m-p-a-n-y- -g-e-t-s-  
-n-e-g-a-t-i-v-e- -c-o-v-e-r-a-g-e- -u-n-r-e-l-a-t-e-d- -t-o- -w-o-r-k-f-o-r-c-e- -r-e-d-u-c-t-i-o-n-s-?-

-\*- \*-D-e-e-p- -D-i-v-e-:-\*- -E-x-p-l-a-i-n- -h-o-w- -y-o-u- -w-o-u-l-d- -d-e-s-i-g-n- -a- -F-i-n-B-E-R-T- -f-i-n-e---t-u-n-i-n-g-  
-p-i-p-e-l-i-n-e- -u-s-i-n-g- -H-u-g-g-i-n-g- -F-a-c-e.-

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-##- -Q-u-e-s-t-i-o-n- -8-5-:- -N-e-w-s- -S-e-n-t-i-m-e-n-t- -Q-u-e-s-t-i-o-n- -8-5-:- -H-o-w- -d-o- -y-o-u- -h-a-n-d-l-e- -s-e-n-t-i-m-e-n-t-  
-s-c-o-r-i-n-g- -d-r-i-f-t- -w-h-e-n- -m-a-c-r-o---e-c-o-n-o-m-i-c- -j-a-r-g-o-n- -s-h-i-f-t-s-?-

-\*- \*-R-o-u-n-d-:-\*- -R-e-s-e-a-r-c-h-

-\*- \*-D-i-f-f-i-c-u-l-t-y-:-\*- -M-e-d-i-u-m- -|- \*- \*-l-m-p-o-r-t-a-n-c-e-:-\*- -8-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d-:-\*- -S-e-n-t-i-m-e-n-t- -d-r-i-f-t-, -l-e-x-i-c-o-n- -u-p-d-a-t-e-s-, -L-L-M- -f-i-n-e---t-u-n-i-n-g.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t-:-\*- -T-e-s-t-s- -N-L-P- -p-i-p-e-l-i-n-e- -r-o-b-u-s-t-n-e-s-s.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-:-\*- -O-u-r- -s-e-n-t-i-m-e-n-t- -e-n-g-i-n-e- -c-u-r-r-e-n-t-l-y- -u-s-e-s- -a- -k-e-y-w-o-r-d-  
-h-e-u-r-i-s-t-i-c- -w-e-i-g-h-t-i-n-g- -l-a-y-o-f-f- -a-n-d- -g-r-o-w-t-h- -w-o-r-d-s.- -O-v-e-r- -t-i-m-e-, -t-e-r-m-i-n-o-l-o-g-y- -c-h-a-n-g-e-s-  
-(e-.g.- -'r-i-g-h-t---s-i-z-i-n-g',- -'r-e---a-l-l-o-c-a-t-i-n-g- -r-e-s-o-u-r-c-e-s')-.- -H-e-u-r-i-s-t-i-c-s- -m-u-s-t- -b-e-  
-p-e-r-i-o-d-i-c-a-l-l-y- -u-p-d-a-t-e-d-, -o-r- -r-e-p-l-a-c-e-d- -b-y- -a- -p-r-e-t-r-a-i-n-e-d- -T-r-a-n-s-f-o-r-m-e-r- -(l-i-k-e-  
-F-i-n-B-E-R-T)- -t-h-a-t- -g-e-n-e-r-a-l-i-z-e-s- -b-e-t-t-e-r- -t-o- -s-e-m-a-n-t-i-c- -n-u-a-n-c-e-s.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-:-\*- -R-e-l-y-i-n-g- -o-n- -a- -s-t-a-t-i-c- -v-o-c-a-b-u-l-a-r-y- -w-i-t-h-o-u-t- -c-h-e-c-k-i-n-g- -f-o-r-  
-u-n-s-e-e-n- -s-e-n-t-i-m-e-n-t- -v-o-c-a-b-u-l-a-r-y.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-:-\*- -W-e- -r-u-n- -a- -w-e-e-k-l-y- -a-u-d-i-t- -s-c-r-i-p-t- -t-h-a-t-  
-f-l-a-g-s- -a-r-t-i-c-l-e-s- -w-i-t-h- -n-e-u-t-r-a-l- -s-c-o-r-e-s- -b-u-t- -h-i-g-h- -r-i-s-k- -i-n-d-i-c-a-t-o-r-s-, -a-n-d- -f-e-e-d- -t-h-e-m-  
-i-n-t-o- -a- -v-a-l-i-d-a-t-i-o-n- -s-e-t- -t-o- -c-h-e-c-k- -i-f- -o-u-r- -v-o-c-a-b-u-l-a-r-y- -n-e-e-d-s- -u-p-d-a-t-e-s.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s-:-\*- -H-o-w- -d-o- -y-o-u- -a-v-o-i-d- -b-i-a-s- -w-h-e-n- -a- -c-o-m-p-a-n-y- -g-e-t-s-  
-n-e-g-a-t-i-v-e- -c-o-v-e-r-a-g-e- -u-n-r-e-l-a-t-e-d- -t-o- -w-o-r-k-f-o-r-c-e- -r-e-d-u-c-t-i-o-n-s-?-

-\*- \*-D-e-e-p- -D-i-v-e-:-\*- -E-x-p-l-a-i-n- -h-o-w- -y-o-u- -w-o-u-l-d- -d-e-s-i-g-n- -a- -F-i-n-B-E-R-T- -f-i-n-e---t-u-n-i-n-g-  
-p-i-p-e-l-i-n-e- -u-s-i-n-g- -H-u-g-g-i-n-g- -F-a-c-e.-

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##- Q-u-e-s-t-i-o-n- 8-6:- S-y-s-t-e-m- D-e-s-i-g-n- Q-u-e-s-t-i-o-n- 8-6:- H-o-w- w-o-u-l-d- y-o-u- s-c-a-l-e- t-h-e- p-r-e-d-i-c-t-i-o-n- a-r-c-h-i-t-e-c-t-u-r-e- t-o- s-u-p-p-o-r-t- 1-0- m-i-l-l-i-o-n- c-o-m-p-a-n-i-e-s-?

\*- \*-R-o-u-n-d:-\*- S-y-s-t-e-m- D-e-s-i-g-n-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- | \*-\*-l-m-p-o-r-t-a-n-c-e:-\*- 1-0-/1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- C-a-c-h-i-n-g,- w-o-r-k-e-r- q-u-e-u-e-s,- l-o-a-d- b-a-l-a-n-c-i-n-g,- h-o-r-i-z-o-n-t-a-l- s-c-a-l-i-n-g,-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- T-e-s-t-s- h-o-r-i-z-o-n-t-a-l- s-c-a-l-i-n-g- a-n-d- h-i-g-h---t-h-r-o-u-g-h-p-u-t- d-e-s-i-g-n.-

\*- \*-l-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- S-c-a-l-i-n-g- t-o- 1-0-M- c-o-m-p-a-n-i-e-s- r-e-q-u-i-r-e-s- s-e-p-a-r-a-t-i-n-g- o-n-l-i-n-e- r-e-q-u-e-s-t- h-a-n-d-l-i-n-g- f-r-o-m- e-x-p-e-n-s-i-v-e- m-o-d-e-l- c-o-m-p-u-t-a-t-i-o-n.- W-e- r-u-n- b-a-t-c-h- p-r-e-d-i-c-t-i-o-n-s- o-f-f-l-i-n-e- n-i-g-h-t-l-y- f-o-r- a-l-l- 1-0-M- c-o-m-p-a-n-i-e-s- u-s-i-n-g- A-p-a-c-h-e- S-p-a-r-k- o-r- C-e-l-e-r-y- w-o-r-k-e-r-s,- a-n-d- s-t-o-r-e- t-h-e- c-o-m-p-u-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s,- S-H-A-P- v-a-l-u-e-s,- a-n-d- l-o-c-a-l- p-l-o-t-s- i-n- a- f-a-s-t- c-a-c-h-i-n-g- l-a-y-e-r- (R-e-d-i-s)- o-r- d-a-t-a-b-a-s-e.- T-h-e- o-n-l-i-n-e- S-t-r-e-a-m-l-i-t- U-l- s-i-m-p-l-y- f-e-t-c-h-e-s- p-r-e---c-o-m-p-u-t-e-d- r-e-c-o-r-d-s.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- P-r-o-p-o-s-i-n-g- t-o- c-a-l-c-u-l-a-t-e- S-H-A-P- v-a-l-u-e-s- i-n- r-e-a-l---t-i-m-e- o-n- e-v-e-r-y- p-a-g-e- l-o-a-d- f-o-r- m-i-l-l-i-o-n-s- o-f- a-c-t-i-v-e- r-e-q-u-e-s-t-s.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- i-m-p-l-e-m-e-n-t- a- H-y-b-r-i-d- l-n-f-e-r-e-n-c-e- p-a-t-h:- p-r-e-c-o-m-p-u-t-e- p-r-e-d-i-c-t-i-o-n-s- f-o-r- a-c-t-i-v-e- c-o-m-p-a-n-i-e-s,- a-n-d- r-u-n- o-n---d-e-m-a-n-d- i-n-f-e-r-e-n-c-e- f-o-r- l-o-n-g---t-a-i-l- c-o-m-p-a-n-i-e-s- u-s-i-n-g- a- q-u-e-u-e- s-y-s-t-e-m- b-a-c-k-e-d- b-y- R-a-b-b-i-t-M-Q.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- H-o-w- w-o-u-l-d- y-o-u- h-a-n-d-l-e- i-n-v-a-l-i-d-a-t-i-o-n- o-f- t-h-e- c-a-c-h-e- w-h-e-n- a- n-e-w- l-a-y-o-f-f- i-s- l-o-g-g-e-d-?

\*- \*-D-e-e-p- D-i-v-e:-\*- D-e-s-i-g-n- a- g-e-o---d-i-s-t-r-i-b-u-t-e-d- d-a-t-a-b-a-s-e- r-e-p-l-i-c-a-t-i-o-n- s-t-r-a-t-e-g-y- t-o- s-u-p-p-o-r-t- l-o-w- l-a-t-e-n-c-y- r-e-a-d-s- w-o-r-l-d-w-i-d-e.-

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##- Q-u-e-s-t-i-o-n- 8-7:- S-y-s-t-e-m- D-e-s-i-g-n- Q-u-e-s-t-i-o-n- 8-7:- H-o-w- w-o-u-l-d- y-o-u- s-c-a-l-e- t-h-e- p-r-e-d-i-c-t-i-o-n- a-r-c-h-i-t-e-c-t-u-r-e- t-o- s-u-p-p-o-r-t- 1-0- m-i-l-l-i-o-n- c-o-m-p-a-n-i-e-s-?

\*- \*-R-o-u-n-d:-\*- S-y-s-t-e-m- D-e-s-i-g-n-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- | \*-\*-l-m-p-o-r-t-a-n-c-e:-\*- 1-0-/1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- C-a-c-h-i-n-g,- w-o-r-k-e-r- q-u-e-u-e-s,- l-o-a-d- b-a-l-a-n-c-i-n-g,- h-o-r-i-z-o-n-t-a-l- s-c-a-l-i-n-g,-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- T-e-s-t-s- h-o-r-i-z-o-n-t-a-l- s-c-a-l-i-n-g- a-n-d- h-i-g-h---t-h-r-o-u-g-h-p-u-t- d-e-s-i-g-n.-

\*- \*-l-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- S-c-a-l-i-n-g- t-o- 1-0-M- c-o-m-p-a-n-i-e-s- r-e-q-u-i-r-e-s- s-e-p-a-r-a-t-i-n-g- o-n-l-i-n-e- r-e-q-u-e-s-t- h-a-n-d-l-i-n-g- f-r-o-m- e-x-p-e-n-s-i-v-e- m-o-d-e-l- c-o-m-p-u-t-a-t-i-o-n.- W-e- r-u-n- b-a-t-c-h- p-r-e-d-i-c-t-i-o-n-s- o-f-f-l-i-n-e- n-i-g-h-t-l-y- f-o-r- a-l-l- 1-0-M- c-o-m-p-a-n-i-e-s- u-s-i-n-g- A-p-a-c-h-e- S-p-a-r-k- o-r- C-e-l-e-r-y- w-o-r-k-e-r-s,- a-n-d- s-t-o-r-e- t-h-e- c-o-m-p-u-t-e-d- p-r-o-b-a-b-i-l-i-t-i-e-s,- S-H-A-P- v-a-l-u-e-s,- a-n-d- l-o-c-a-l- p-l-o-t-s- i-n- a- f-a-s-t- c-a-c-h-i-n-g- l-a-y-e-r- (R-e-d-i-s)- o-r- d-a-t-a-b-a-s-e.- T-h-e- o-n-l-i-n-e- S-t-r-e-a-m-l-i-t-

-U-I- -s-i-m-p-l-y- -f-e-t-c-h-e-s- -p-r-e---c-o-m-p-u-t-e-d- -r-e-c-o-r-d-s.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -P-r-o-p-o-s-i-n-g- -t-o- -c-a-l-c-u-l-a-t-e- -S-H-A-P- -v-a-l-u-e-s- -i-n- -r-e-a-l---t-i-m-e- -o-n-  
-e-v-e-r-y- -p-a-g-e- -l-o-a-d- -f-o-r- -m-i-l-l-i-o-n-s- -o-f- -a-c-t-i-v-e- -r-e-q-u-e-s-t-s.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -i-m-p-l-e-m-e-n-t- -a- -H-y-b-r-i-d- -l-n-f-e-r-e-n-c-e-  
-p-a-t-h:- -p-r-e-c-o-m-p-u-t-e- -p-r-e-d-i-c-t-i-o-n-s- -f-o-r- -a-c-t-i-v-e- -c-o-m-p-a-n-i-e-s,- -a-n-d- -r-u-n- -o-n---d-e-m-a-n-d-  
-i-n-f-e-r-e-n-c-e- -f-o-r- -l-o-n-g---t-a-i-l- -c-o-m-p-a-n-i-e-s- -u-s-i-n-g- -a- -q-u-e-u-e- -s-y-s-t-e-m- -b-a-c-k-e-d- -b-y-  
-R-a-b-b-i-t-M-Q.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -i-n-v-a-l-i-d-a-t-i-o-n- -o-f- -t-h-e- -c-a-c-h-e-  
-w-h-e-n- -a- -n-e-w- -l-a-y-o-f-f- -i-s- -l-o-g-g-e-d?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -D-e-s-i-g-n- -a- -g-e-o---d-i-s-t-r-i-b-u-t-e-d- -d-a-t-a-b-a-s-e- -r-e-p-l-i-c-a-t-i-o-n- -s-t-r-a-t-e-g-y- -t-o-  
-s-u-p-p-o-r-t- -l-o-w- -l-a-t-e-n-c-y- -r-e-a-d-s- -w-o-r-l-d-w-i-d-e.-

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-##- -Q-u-e-s-t-i-o-n- -8-8:- -S-y-s-t-e-m- -D-e-s-i-g-n- -Q-u-e-s-t-i-o-n- -8-8:- -H-o-w- -w-o-u-l-d- -y-o-u- -s-c-a-l-e- -t-h-e-  
-p-r-e-d-i-c-t-i-o-n- -a-r-c-h-i-t-e-c-t-u-r-e- -t-o- -s-u-p-p-o-r-t- -1-0- -m-i-l-l-i-o-n- -c-o-m-p-a-n-i-e-s?-

-\*- \*-R-o-u-n-d:-\*- -S-y-s-t-e-m- -D-e-s-i-g-n-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -1-0-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -C-a-c-h-i-n-g,- -w-o-r-k-e-r- -q-u-e-u-e-s,- -l-o-a-d- -b-a-l-a-n-c-i-n-g,- -h-o-r-i-z-o-n-t-a-l-  
-s-c-a-l-i-n-g.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -h-o-r-i-z-o-n-t-a-l- -s-c-a-l-i-n-g- -a-n-d- -h-i-g-h---t-h-r-o-u-g-h-p-u-t-  
-d-e-s-i-g-n.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -S-c-a-l-i-n-g- -t-o- -1-0-M- -c-o-m-p-a-n-i-e-s- -r-e-q-u-i-r-e-s- -s-e-p-a-r-a-t-i-n-g-  
-o-n-l-i-n-e- -r-e-q-u-e-s-t- -h-a-n-d-l-i-n-g- -f-r-o-m- -e-x-p-e-n-s-i-v-e- -m-o-d-e-l- -c-o-m-p-u-t-a-t-i-o-n.- -W-e- -r-u-n- -b-a-t-c-h-  
-p-r-e-d-i-c-t-i-o-n-s- -o-f-f-l-i-n-e- -n-i-g-h-t-l-y- -f-o-r- -a-l-l- -1-0-M- -c-o-m-p-a-n-i-e-s- -u-s-i-n-g- -A-p-a-c-h-e- -S-p-a-r-k- -o-r-  
-C-e-l-e-r-y- -w-o-r-k-e-r-s,- -a-n-d- -s-t-o-r-e- -t-h-e- -c-o-m-p-u-t-e-d- -p-r-o-b-a-b-i-l-i-t-i-e-s,- -S-H-A-P- -v-a-l-u-e-s,- -a-n-d-  
-l-o-c-a-l- -p-l-o-t-s- -i-n- -a- -f-a-s-t- -c-a-c-h-i-n-g- -l-a-y-e-r- -(R-e-d-i-s)- -o-r- -d-a-t-a-b-a-s-e.- -T-h-e- -o-n-l-i-n-e- -S-t-r-e-a-m-l-i-t-  
-U-I- -s-i-m-p-l-y- -f-e-t-c-h-e-s- -p-r-e---c-o-m-p-u-t-e-d- -r-e-c-o-r-d-s.-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -P-r-o-p-o-s-i-n-g- -t-o- -c-a-l-c-u-l-a-t-e- -S-H-A-P- -v-a-l-u-e-s- -i-n- -r-e-a-l---t-i-m-e- -o-n-  
-e-v-e-r-y- -p-a-g-e- -l-o-a-d- -f-o-r- -m-i-l-l-i-o-n-s- -o-f- -a-c-t-i-v-e- -r-e-q-u-e-s-t-s.-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -i-m-p-l-e-m-e-n-t- -a- -H-y-b-r-i-d- -l-n-f-e-r-e-n-c-e-  
-p-a-t-h:- -p-r-e-c-o-m-p-u-t-e- -p-r-e-d-i-c-t-i-o-n-s- -f-o-r- -a-c-t-i-v-e- -c-o-m-p-a-n-i-e-s,- -a-n-d- -r-u-n- -o-n---d-e-m-a-n-d-  
-i-n-f-e-r-e-n-c-e- -f-o-r- -l-o-n-g---t-a-i-l- -c-o-m-p-a-n-i-e-s- -u-s-i-n-g- -a- -q-u-e-u-e- -s-y-s-t-e-m- -b-a-c-k-e-d- -b-y-  
-R-a-b-b-i-t-M-Q.-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -i-n-v-a-l-i-d-a-t-i-o-n- -o-f- -t-h-e- -c-a-c-h-e-  
-w-h-e-n- -a- -n-e-w- -l-a-y-o-f-f- -i-s- -l-o-g-g-e-d?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -D-e-s-i-g-n- -a- -g-e-o---d-i-s-t-r-i-b-u-t-e-d- -d-a-t-a-b-a-s-e- -r-e-p-l-i-c-a-t-i-o-n- -s-t-r-a-t-e-g-y- -t-o-  
-s-u-p-p-o-r-t- -l-o-w- -l-a-t-e-n-c-y- -r-e-a-d-s- -w-o-r-l-d-w-i-d-e.-

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-##- -Q-u-e-s-t-i-o-n- -8-9:- -S-y-s-t-e-m- -D-e-s-i-g-n- -Q-u-e-s-t-i-o-n- -8-9:- -H-o-w- -w-o-u-l-d- -y-o-u- -s-c-a-l-e- -t-h-e-  
-p-r-e-d-i-c-t-i-o-n- -a-r-c-h-i-t-e-c-t-u-r-e- -t-o- -s-u-p-p-o-r-t- -1-0- -m-i-l-l-i-o-n- -c-o-m-p-a-n-i-e-s?-

\*- \*-R-o-u-n-d:-\*- -S-y-s-t-e-m- -D-e-s-i-g-n-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -1-0/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -C-a-c-h-i-n-g,- -w-o-r-k-e-r- -q-u-e-u-e-s,- -l-o-a-d- -b-a-l-a-n-c-i-n-g,- -h-o-r-i-z-o-n-t-a-l- -s-c-a-l-i-n-g.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -h-o-r-i-z-o-n-t-a-l- -s-c-a-l-i-n-g- -a-n-d- -h-i-g-h---t-h-r-o-u-g-h-p-u-t- -d-e-s-i-g-n.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -S-c-a-l-i-n-g- -t-o- -1-0-M- -c-o-m-p-a-n-i-e-s- -r-e-q-u-i-r-e-s- -s-e-p-a-r-a-t-i-n-g- -o-n-l-i-n-e- -r-e-q-u-e-s-t- -h-a-n-d-l-i-n-g- -f-r-o-m- -e-x-p-e-n-s-i-v-e- -m-o-d-e-l- -c-o-m-p-u-t-a-t-i-o-n.- -W-e- -r-u-n- -b-a-t-c-h- -p-r-e-d-i-c-t-i-o-n-s- -o-f-f-l-i-n-e- -n-i-g-h-t-l-y- -f-o-r- -a-l-l- -1-0-M- -c-o-m-p-a-n-i-e-s- -u-s-i-n-g- -A-p-a-c-h-e- -S-p-a-r-k- -o-r- -C-e-l-e-r-y- -w-o-r-k-e-r-s,- -a-n-d- -s-t-o-r-e- -t-h-e- -c-o-m-p-u-t-e-d- -p-r-o-b-a-b-i-l-i-t-i-e-s,- -S-H-A-P- -v-a-l-u-e-s,- -a-n-d- -l-o-c-a-l- -p-l-o-t-s- -i-n- -a- -f-a-s-t- -c-a-c-h-i-n-g- -l-a-y-e-r- -(R-e-d-i-s)- -o-r- -d-a-t-a-b-a-s-e.- -T-h-e- -o-n-l-i-n-e- -S-t-r-e-a-m-l-i-t- -U-l- -s-i-m-p-l-y- -f-e-t-c-h-e-s- -p-r-e---c-o-m-p-u-t-e-d- -r-e-c-o-r-d-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -P-r-o-p-o-s-i-n-g- -t-o- -c-a-l-c-u-l-a-t-e- -S-H-A-P- -v-a-l-u-e-s- -i-n- -r-e-a-l---t-i-m-e- -o-n- -e-v-e-r-y- -p-a-g-e- -l-o-a-d- -f-o-r- -m-i-l-l-i-o-n-s- -o-f- -a-c-t-i-v-e- -r-e-q-u-e-s-t-s.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -i-m-p-l-e-m-e-n-t- -a- -H-y-b-r-i-d- -l-n-f-e-r-e-n-c-e- -p-a-t-h:- -p-r-e-c-o-m-p-u-t-e- -p-r-e-d-i-c-t-i-o-n-s- -f-o-r- -a-c-t-i-v-e- -c-o-m-p-a-n-i-e-s,- -a-n-d- -r-u-n- -o-n---d-e-m-a-n-d- -i-n-f-e-r-e-n-c-e- -f-o-r- -l-o-n-g---t-a-i-l- -c-o-m-p-a-n-i-e-s- -u-s-i-n-g- -a- -q-u-e-u-e- -s-y-s-t-e-m- -b-a-c-k-e-d- -b-y- -R-a-b-b-i-t-M-Q.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -i-n-v-a-l-i-d-a-t-i-o-n- -o-f- -t-h-e- -c-a-c-h-e- -w-h-e-n- -a- -n-e-w- -l-a-y-o-f-f- -i-s- -l-o-g-g-e-d?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -D-e-s-i-g-n- -a- -g-e-o---d-i-s-t-r-i-b-u-t-e-d- -d-a-t-a-b-a-s-e- -r-e-p-l-i-c-a-t-i-o-n- -s-t-r-a-t-e-g-y- -t-o- -s-u-p-p-o-r-t- -l-o-w- -l-a-t-e-n-c-y- -r-e-a-d-s- -w-o-r-l-d-w-i-d-e.-

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### -Q-u-e-s-t-i-o-n- -9-0:- -S-y-s-t-e-m- -D-e-s-i-g-n- -Q-u-e-s-t-i-o-n- -9-0:- -H-o-w- -w-o-u-l-d- -y-o-u- -s-c-a-l-e- -t-h-e- -p-r-e-d-i-c-t-i-o-n- -a-r-c-h-i-t-e-c-t-u-r-e- -t-o- -s-u-p-p-o-r-t- -1-0- -m-i-l-l-i-o-n- -c-o-m-p-a-n-i-e-s?-

\*- \*-R-o-u-n-d:-\*- -S-y-s-t-e-m- -D-e-s-i-g-n-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -1-0/-1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -C-a-c-h-i-n-g,- -w-o-r-k-e-r- -q-u-e-u-e-s,- -l-o-a-d- -b-a-l-a-n-c-i-n-g,- -h-o-r-i-z-o-n-t-a-l- -s-c-a-l-i-n-g.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -h-o-r-i-z-o-n-t-a-l- -s-c-a-l-i-n-g- -a-n-d- -h-i-g-h---t-h-r-o-u-g-h-p-u-t- -d-e-s-i-g-n.-

\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -S-c-a-l-i-n-g- -t-o- -1-0-M- -c-o-m-p-a-n-i-e-s- -r-e-q-u-i-r-e-s- -s-e-p-a-r-a-t-i-n-g- -o-n-l-i-n-e- -r-e-q-u-e-s-t- -h-a-n-d-l-i-n-g- -f-r-o-m- -e-x-p-e-n-s-i-v-e- -m-o-d-e-l- -c-o-m-p-u-t-a-t-i-o-n.- -W-e- -r-u-n- -b-a-t-c-h- -p-r-e-d-i-c-t-i-o-n-s- -o-f-f-l-i-n-e- -n-i-g-h-t-l-y- -f-o-r- -a-l-l- -1-0-M- -c-o-m-p-a-n-i-e-s- -u-s-i-n-g- -A-p-a-c-h-e- -S-p-a-r-k- -o-r- -C-e-l-e-r-y- -w-o-r-k-e-r-s,- -a-n-d- -s-t-o-r-e- -t-h-e- -c-o-m-p-u-t-e-d- -p-r-o-b-a-b-i-l-i-t-i-e-s,- -S-H-A-P- -v-a-l-u-e-s,- -a-n-d- -l-o-c-a-l- -p-l-o-t-s- -i-n- -a- -f-a-s-t- -c-a-c-h-i-n-g- -l-a-y-e-r- -(R-e-d-i-s)- -o-r- -d-a-t-a-b-a-s-e.- -T-h-e- -o-n-l-i-n-e- -S-t-r-e-a-m-l-i-t- -U-l- -s-i-m-p-l-y- -f-e-t-c-h-e-s- -p-r-e---c-o-m-p-u-t-e-d- -r-e-c-o-r-d-s.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -P-r-o-p-o-s-i-n-g- -t-o- -c-a-l-c-u-l-a-t-e- -S-H-A-P- -v-a-l-u-e-s- -i-n- -r-e-a-l---t-i-m-e- -o-n- -e-v-e-r-y- -p-a-g-e- -l-o-a-d- -f-o-r- -m-i-l-l-i-o-n-s- -o-f- -a-c-t-i-v-e- -r-e-q-u-e-s-t-s.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -i-m-p-l-e-m-e-n-t- -a- -H-y-b-r-i-d- -l-n-f-e-r-e-n-c-e- -p-a-t-h:- -p-r-e-c-o-m-p-u-t-e- -p-r-e-d-i-c-t-i-o-n-s- -f-o-r- -a-c-t-i-v-e- -c-o-m-p-a-n-i-e-s,- -a-n-d- -r-u-n- -o-n---d-e-m-a-n-d-

-i-n-f-e-r-e-n-c-e- f-o-r- l-o-n-g---t-a-i-l- -c-o-m-p-a-n-i-e-s- -u-s-i-n-g- -a- -q-u-e-u-e- -s-y-s-t-e-m- -b-a-c-k-e-d- -b-y-  
-R-a-b-b-i-t-M-Q--

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -w-o-u-l-d- -y-o-u- -h-a-n-d-l-e- -i-n-v-a-l-i-d-a-t-i-o-n- -o-f- -t-h-e- -c-a-c-h-e-  
-w-h-e-n- -a- -n-e-w- -l-a-y-o-f-f- -i-s- -l-o-g-g-e-d?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -D-e-s-i-g-n- -a- -g-e-o---d-i-s-t-r-i-b-u-t-e-d- -d-a-t-a-b-a-s-e- -r-e-p-l-i-c-a-t-i-o-n- -s-t-r-a-t-e-g-y- -t-o-  
-s-u-p-p-o-r-t- -l-o-w- -l-a-t-e-n-c-y- -r-e-a-d-s- -w-o-r-l-d-w-i-d-e.-

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-##-#- -Q-u-e-s-t-i-o-n- -9-1:- -S-e-n-i-o-r-/S-t-a-f-f- -O-u-t-l-i-e-r- -Q-u-e-s-t-i-o-n- -9-1:- -W-h-a-t- -h-a-p-p-e-n-s- -i-f- -a-  
-c-o-m-p-a-n-y- -d-y-n-a-m-i-c-a-l-l-y- -a-l-t-e-r-s- -i-t-s- -p-u-b-l-i-c- -P-R- -m-e-s-s-a-g-i-n-g- -t-o- -t-r-i-c-k- -y-o-u-r- -s-e-n-t-i-m-e-n-t-  
-m-o-d-e-l?-

\*- \*-R-o-u-n-d:-\*- -H-i-r-i-n-g- -C-o-m-m-i-t-t-e-e-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -1-0-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -A-d-v-e-r-s-a-r-i-a-l- -a-t-t-a-c-k-s,- -d-a-t-a- -m-a-n-i-p-u-l-a-t-i-o-n,- -v-a-l-i-d-a-t-i-o-n-  
-c-o-n-s-t-r-a-i-n-t-s.-

\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -a-d-v-e-r-s-a-r-i-a-l- -r-o-b-u-s-t-n-e-s-s- -a-n-d- -s-y-s-t-e-m-  
-b-o-u-n-d-a-r-i-e-s.-

\*- \*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -A-d-v-e-r-s-a-r-i-a-l- -P-R- -(e-.g.- -p-a-c-k-i-n-g- -r-e-l-e-a-s-e-s- -w-i-t-h-  
-p-o-s-i-t-i-v-e- -b-u-z-z-w-o-r-d-s- -l-i-k-e- -'s-y-n-e-r-g-y'- -a-n-d- -'h-y-p-e-r---g-r-o-w-t-h'- -t-o- -m-a-s-k- -l-a-y-o-f-f-s)- -w-o-u-l-d-  
-i-n-f-l-a-t-e- -t-h-e- -s-e-n-t-i-m-e-n-t- -s-c-o-r-e.- -T-o- -c-o-u-n-t-e-r- -t-h-i-s,- -w-e- -a-s-s-i-g-n- -d-i-f-f-e-r-e-n-t- -w-e-i-g-h-t-s:-  
-d-i-r-e-c-t- -D-B- -f-i-l-i-n-g-s- -(W-A-R-N- -n-o-t-i-c-e-s)- -a-n-d- -a-c-t-u-a-l- -l-a-y-o-f-f- -r-e-c-o-r-d-s- -h-a-v-e- -1-0-x- -t-h-e-  
-w-e-i-g-h-t- -o-f- -n-e-w-s- -s-e-n-t-i-m-e-n-t.- -N-e-w-s- -s-e-n-t-i-m-e-n-t- -a-c-t-s- -a-s- -a- -m-o-d-i-f-i-e-r,- -n-o-t- -t-h-e-  
-p-r-i-m-a-r-y- -s-i-g-n-a-l.-

\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -B-e-l-i-e-v-i-n-g- -n-e-w-s- -s-e-n-t-i-m-e-n-t- -i-s- -a-n- -u-n---g-a-m-e-a-b-l-e- -m-e-t-r-i-c.-

\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -i-m-p-l-e-m-e-n-t- -a- -d-i-s-c-r-e-p-a-n-c-y- -a-n-o-m-a-l-y-  
-d-e-t-e-c-t-o-r:- -i-f- -a- -c-o-m-p-a-n-y-'s- -p-r-e-d-i-c-t-e-d- -r-i-s-k- -f-r-o-m- -t-a-b-u-l-a-r- -d-a-t-a- -i-s- -h-i-g-h,- -b-u-t- -n-e-w-s-  
-s-e-n-t-i-m-e-n-t- -i-s- -h-i-g-h-l-y- -p-o-s-i-t-i-v-e,- -w-e- -f-l-a-g- -t-h-e- -r-e-c-o-r-d- -f-o-r- -r-e-v-i-e-w- -a-n-d- -p-u-l-l-  
-t-h-i-r-d---p-a-r-t-y- -e-m-p-l-o-y-e-e- -r-e-v-i-e-w-s- -(e-.g.-, -G-l-a-s-s-d-o-o-r)- -t-o- -v-e-r-i-f-y.-

\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -d-o- -y-o-u- -m-a-t-h-e-m-a-t-i-c-a-l-l-y- -f-o-r-m-a-l-i-z-e- -t-h-e- -w-e-i-g-h-t-  
-p-e-n-a-l-t-y- -f-o-r- -n-e-w-s- -d-i-s-c-r-e-p-a-n-c-y?-

\*- \*-D-e-e-p- -D-i-v-e:-\*- -D-e-s-i-g-n- -a-n- -o-n-l-i-n-e- -a-d-v-e-r-s-a-r-i-a-l- -t-r-a-i-n-i-n-g- -m-e-t-h-o-d- -t-o- -m-a-k-e- -t-h-e-  
-t-e-x-t- -c-l-a-s-s-i-f-i-e-r- -r-o-b-u-s-t- -a-g-a-i-n-s-t- -w-o-r-d- -s-u-b-s-t-i-t-u-t-i-o-n-s.-

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-##-#- -Q-u-e-s-t-i-o-n- -9-2:- -S-e-n-i-o-r-/S-t-a-f-f- -O-u-t-l-i-e-r- -Q-u-e-s-t-i-o-n- -9-2:- -W-h-a-t- -h-a-p-p-e-n-s- -i-f- -a-  
-c-o-m-p-a-n-y- -d-y-n-a-m-i-c-a-l-l-y- -a-l-t-e-r-s- -i-t-s- -p-u-b-l-i-c- -P-R- -m-e-s-s-a-g-i-n-g- -t-o- -t-r-i-c-k- -y-o-u-r- -s-e-n-t-i-m-e-n-t-  
-m-o-d-e-l?-

\*- \*-R-o-u-n-d:-\*- -H-i-r-i-n-g- -C-o-m-m-i-t-t-e-e-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*-I-m-p-o-r-t-a-n-c-e:-\*- -1-0-/1-0-

\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -A-d-v-e-r-s-a-r-i-a-l- -a-t-t-a-c-k-s,- -d-a-t-a- -m-a-n-i-p-u-l-a-t-i-o-n,- -v-a-l-i-d-a-t-i-o-n-

-c-o-n-s-t-r-a-i-n-t-s.-

-\*-\*W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*-\* -T-e-s-t-s- -a-d-v-e-r-s-a-r-i-a-l- -r-o-b-u-s-t-n-e-s-s- -a-n-d- -s-y-s-t-e-m-  
-b-o-u-n-d-a-r-i-e-s.-

-\*-\*I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*-\* -A-d-v-e-r-s-a-r-i-a-l- -P-R- -(e-.g.- -p-a-c-k-i-n-g- -r-e-l-e-a-s-e-s- -w-i-t-h-  
-p-o-s-i-t-i-v-e- -b-u-z-z-w-o-r-d-s- -l-i-k-e- -'s-y-n-e-r-g-y-' -a-n-d- -'h-y-p-e-r---g-r-o-w-t-h-' -t-o- -m-a-s-k- -l-a-y-o-f-f-s)- -w-o-u-l-d-  
-i-n-f-l-a-t-e- -t-h-e- -s-e-n-t-i-m-e-n-t- -s-c-o-r-e.- -T-o- -c-o-u-n-t-e-r- -t-h-i-s,- -w-e- -a-s-s-i-g-n- -d-i-f-f-e-r-e-n-t- -w-e-i-g-h-t-s:-  
-d-i-r-e-c-t- -D-B- -f-i-l-i-n-g-s- -(W-A-R-N- -n-o-t-i-c-e-s)- -a-n-d- -a-c-t-u-a-l- -l-a-y-o-f-f- -r-e-c-o-r-d-s- -h-a-v-e- -1-0-x- -t-h-e-  
-w-e-i-g-h-t- -o-f- -n-e-w-s- -s-e-n-t-i-m-e-n-t.- -N-e-w-s- -s-e-n-t-i-m-e-n-t- -a-c-t-s- -a-s- -a- -m-o-d-i-f-i-e-r,- -n-o-t- -t-h-e-  
-p-r-i-m-a-r-y- -s-i-g-n-a-l.-

-\*-\*C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*-\* -B-e-l-i-e-v-i-n-g- -n-e-w-s- -s-e-n-t-i-m-e-n-t- -i-s- -a-n- -u-n---g-a-m-e-a-b-l-e- -m-e-t-r-i-c.-

-\*-\*B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*-\* -W-e- -i-m-p-l-e-m-e-n-t- -a- -d-i-s-c-r-e-p-a-n-c-y- -a-n-o-m-a-l-y-  
-d-e-t-e-c-t-o-r:- -i-f- -a- -c-o-m-p-a-n-y-'s- -p-r-e-d-i-c-t-e-d- -r-i-s-k- -f-r-o-m- -t-a-b-u-l-a-r- -d-a-t-a- -i-s- -h-i-g-h,- -b-u-t- -n-e-w-s-  
-s-e-n-t-i-m-e-n-t- -i-s- -h-i-g-h-l-y- -p-o-s-i-t-i-v-e,- -w-e- -f-l-a-g- -t-h-e- -r-e-c-o-r-d- -f-o-r- -r-e-v-i-e-w- -a-n-d- -p-u-l-l-  
-t-h-i-r-d---p-a-r-t-y- -e-m-p-l-o-y-e-e- -r-e-v-i-e-w-s- -(e-.g.-,- -G-l-a-s-s-d-o-o-r)- -t-o- -v-e-r-i-f-y.-

-\*-\*F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*-\* -H-o-w- -d-o- -y-o-u- -m-a-t-h-e-m-a-t-i-c-a-l-l-y- -f-o-r-m-a-l-i-z-e- -t-h-e- -w-e-i-g-h-t-  
-p-e-n-a-l-t-y- -f-o-r- -n-e-w-s- -d-i-s-c-r-e-p-a-n-c-y-?

-\*-\*D-e-e-p- -D-i-v-e:-\*-\* -D-e-s-i-g-n- -a-n- -o-n-l-i-n-e- -a-d-v-e-r-s-a-r-i-a-l- -t-r-a-i-n-i-n-g- -m-e-t-h-o-d- -t-o- -m-a-k-e- -t-h-e-  
-t-e-x-t- -c-l-a-s-s-i-f-i-e-r- -r-o-b-u-s-t- -a-g-a-i-n-s-t- -w-o-r-d- -s-u-b-s-t-i-t-u-t-i-o-n-s.-

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### -Q-u-e-s-t-i-o-n- -9-3:- -S-e-n-i-o-r/-S-t-a-f-f- -O-u-t-l-i-e-r- -Q-u-e-s-t-i-o-n- -9-3:- -W-h-a-t- -h-a-p-p-e-n-s- -i-f- -a-  
-c-o-m-p-a-n-y- -d-y-n-a-m-i-c-a-l-l-y- -a-l-t-e-r-s- -i-t-s- -p-u-b-l-i-c- -P-R- -m-e-s-s-a-g-i-n-g- -t-o- -t-r-i-c-k- -y-o-u-r- -s-e-n-t-i-m-e-n-t-  
-m-o-d-e-l?-

-\*-\*R-o-u-n-d:-\*-\* -H-i-r-i-n-g- -C-o-m-m-i-t-t-e-e-

-\*-\*D-i-f-f-i-c-u-l-t-y:-\*-\* -H-a-r-d- -|- -\*-\*I-m-p-o-r-t-a-n-c-e:-\*-\* -1-0/-1-0-

-\*-\*C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*-\* -A-d-v-e-r-s-a-r-i-a-l- -a-t-t-a-c-k-s,- -d-a-t-a- -m-a-n-i-p-u-l-a-t-i-o-n,- -v-a-l-i-d-a-t-i-o-n-  
-c-o-n-s-t-r-a-i-n-t-s.-

-\*-\*W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*-\* -T-e-s-t-s- -a-d-v-e-r-s-a-r-i-a-l- -r-o-b-u-s-t-n-e-s-s- -a-n-d- -s-y-s-t-e-m-  
-b-o-u-n-d-a-r-i-e-s.-

-\*-\*I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*-\* -A-d-v-e-r-s-a-r-i-a-l- -P-R- -(e-.g.- -p-a-c-k-i-n-g- -r-e-l-e-a-s-e-s- -w-i-t-h-  
-p-o-s-i-t-i-v-e- -b-u-z-z-w-o-r-d-s- -l-i-k-e- -'s-y-n-e-r-g-y-' -a-n-d- -'h-y-p-e-r---g-r-o-w-t-h-' -t-o- -m-a-s-k- -l-a-y-o-f-f-s)- -w-o-u-l-d-  
-i-n-f-l-a-t-e- -t-h-e- -s-e-n-t-i-m-e-n-t- -s-c-o-r-e.- -T-o- -c-o-u-n-t-e-r- -t-h-i-s,- -w-e- -a-s-s-i-g-n- -d-i-f-f-e-r-e-n-t- -w-e-i-g-h-t-s:-  
-d-i-r-e-c-t- -D-B- -f-i-l-i-n-g-s- -(W-A-R-N- -n-o-t-i-c-e-s)- -a-n-d- -a-c-t-u-a-l- -l-a-y-o-f-f- -r-e-c-o-r-d-s- -h-a-v-e- -1-0-x- -t-h-e-  
-w-e-i-g-h-t- -o-f- -n-e-w-s- -s-e-n-t-i-m-e-n-t.- -N-e-w-s- -s-e-n-t-i-m-e-n-t- -a-c-t-s- -a-s- -a- -m-o-d-i-f-i-e-r,- -n-o-t- -t-h-e-  
-p-r-i-m-a-r-y- -s-i-g-n-a-l.-

-\*-\*C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*-\* -B-e-l-i-e-v-i-n-g- -n-e-w-s- -s-e-n-t-i-m-e-n-t- -i-s- -a-n- -u-n---g-a-m-e-a-b-l-e- -m-e-t-r-i-c.-

-\*-\*B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*-\* -W-e- -i-m-p-l-e-m-e-n-t- -a- -d-i-s-c-r-e-p-a-n-c-y- -a-n-o-m-a-l-y-  
-d-e-t-e-c-t-o-r:- -i-f- -a- -c-o-m-p-a-n-y-'s- -p-r-e-d-i-c-t-e-d- -r-i-s-k- -f-r-o-m- -t-a-b-u-l-a-r- -d-a-t-a- -i-s- -h-i-g-h,- -b-u-t- -n-e-w-s-  
-s-e-n-t-i-m-e-n-t- -i-s- -h-i-g-h-l-y- -p-o-s-i-t-i-v-e,- -w-e- -f-l-a-g- -t-h-e- -r-e-c-o-r-d- -f-o-r- -r-e-v-i-e-w- -a-n-d- -p-u-l-l-  
-t-h-i-r-d---p-a-r-t-y- -e-m-p-l-o-y-e-e- -r-e-v-i-e-w-s- -(e-.g.-,- -G-l-a-s-s-d-o-o-r)- -t-o- -v-e-r-i-f-y.-

-\*-\*F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*-\* -H-o-w- -d-o- -y-o-u- -m-a-t-h-e-m-a-t-i-c-a-l-l-y- -f-o-r-m-a-l-i-z-e- -t-h-e- -w-e-i-g-h-t-  
-p-e-n-a-l-t-y- -f-o-r- -n-e-w-s- -d-i-s-c-r-e-p-a-n-c-y-?

-\*-D-e-e-p- -D-i-v-e-:-\*- -D-e-s-i-g-n- -a-n- -o-n-l-i-n-e- -a-d-v-e-r-s-a-r-i-a-l- -t-r-a-i-n-i-n-g- -m-e-t-h-o-d- -t-o- -m-a-k-e- -t-h-e-  
-t-e-x-t- -c-l-a-s-s-i-f-i-e-r- -r-o-b-u-s-t- -a-g-a-i-n-s-t- -w-o-r-d- -s-u-b-s-t-i-t-u-t-i-o-n-s-.

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-#-#-#- -Q-u-e-s-t-i-o-n- -9-4-:- -S-e-n-i-o-r-/S-t-a-f-f- -O-u-t-l-i-e-r- -Q-u-e-s-t-i-o-n- -9-4-:- -W-h-a-t- -h-a-p-p-e-n-s- -i-f- -a-  
-c-o-m-p-a-n-y- -d-y-n-a-m-i-c-a-l-l-y- -a-l-t-e-r-s- -i-t-s- -p-u-b-l-i-c- -P-R- -m-e-s-s-a-g-i-n-g- -t-o- -t-r-i-c-k- -y-o-u-r- -s-e-n-t-i-m-e-n-t-  
-m-o-d-e-l-?

-\*-R-o-u-n-d-:-\*- -H-i-r-i-n-g- -C-o-m-m-i-t-t-e-e-

-\*-D-i-f-f-i-c-u-l-t-y-:-\*- -H-a-r-d- -|- \*-l-m-p-o-r-t-a-n-c-e-:-\*- -1-0-/1-0-

-\*-C-o-n-c-e-p-t- -T-e-s-t-e-d-:-\*- -A-d-v-e-r-s-a-r-i-a-l- -a-t-t-a-c-k-s,- -d-a-t-a- -m-a-n-i-p-u-l-a-t-i-o-n,- -v-a-l-i-d-a-t-i-o-n-  
-c-o-n-s-t-r-a-i-n-t-s-.

-\*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t-:-\*- -T-e-s-t-s- -a-d-v-e-r-s-a-r-i-a-l- -r-o-b-u-s-t-n-e-s-s- -a-n-d- -s-y-s-t-e-m-  
-b-o-u-n-d-a-r-i-e-s-.

-\*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-:-\*- -A-d-v-e-r-s-a-r-i-a-l- -P-R- -(e-.g.- -p-a-c-k-i-n-g- -r-e-l-e-a-s-e-s- -w-i-t-h-  
-p-o-s-i-t-i-v-e- -b-u-z-z-w-o-r-d-s- -l-i-k-e- -'s-y-n-e-r-g-y'- -a-n-d- -'h-y-p-e-r---g-r-o-w-t-h'- -t-o- -m-a-s-k- -l-a-y-o-f-f-s)- -w-o-u-l-d-  
-i-n-f-l-a-t-e- -t-h-e- -s-e-n-t-i-m-e-n-t- -s-c-o-r-e- -T-o- -c-o-u-n-t-e-r- -t-h-i-s,- -w-e- -a-s-s-i-g-n- -d-i-f-f-e-r-e-n-t- -w-e-i-g-h-t-s:-  
-d-i-r-e-c-t- -D-B- -f-i-l-i-n-g-s- -(W-A-R-N- -n-o-t-i-c-e-s)- -a-n-d- -a-c-t-u-a-l- -l-a-y-o-f-f- -r-e-c-o-r-d-s- -h-a-v-e- -1-0-x- -t-h-e-  
-w-e-i-g-h-t- -o-f- -n-e-w-s- -s-e-n-t-i-m-e-n-t- -N-e-w-s- -s-e-n-t-i-m-e-n-t- -a-c-t-s- -a-s- -a- -m-o-d-i-f-i-e-r,- -n-o-t- -t-h-e-  
-p-r-i-m-a-r-y- -s-i-g-n-a-l-.

-\*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s-:-\*- -B-e-l-i-e-v-i-n-g- -n-e-w-s- -s-e-n-t-i-m-e-n-t- -i-s- -a-n- -u-n---g-a-m-e-a-b-l-e- -m-e-t-r-i-c-.

-\*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r-:-\*- -W-e- -i-m-p-l-e-m-e-n-t- -a- -d-i-s-c-r-e-p-a-n-c-y- -a-n-o-m-a-l-y-  
-d-e-t-e-c-t-o-r:- -i-f- -a- -c-o-m-p-a-n-y-'s- -p-r-e-d-i-c-t-e-d- -r-i-s-k- -f-r-o-m- -t-a-b-u-l-a-r- -d-a-t-a- -i-s- -h-i-g-h,- -b-u-t- -n-e-w-s-  
-s-e-n-t-i-m-e-n-t- -i-s- -h-i-g-h-l-y- -p-o-s-i-t-i-v-e,- -w-e- -f-l-a-g- -t-h-e- -r-e-c-o-r-d- -f-o-r- -r-e-v-i-e-w- -a-n-d- -p-u-l-l-  
-t-h-i-r-d---p-a-r-t-y- -e-m-p-l-o-y-e-e- -r-e-v-i-e-w-s- -(e-.g.-, -G-l-a-s-s-d-o-o-r)- -t-o- -v-e-r-i-f-y-.

-\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s-:-\*- -H-o-w- -d-o- -y-o-u- -m-a-t-h-e-m-a-t-i-c-a-l-l-y- -f-o-r-m-a-l-i-z-e- -t-h-e- -w-e-i-g-h-t-  
-p-e-n-a-l-t-y- -f-o-r- -n-e-w-s- -d-i-s-c-r-e-p-a-n-c-y-?

-\*-D-e-e-p- -D-i-v-e-:-\*- -D-e-s-i-g-n- -a-n- -o-n-l-i-n-e- -a-d-v-e-r-s-a-r-i-a-l- -t-r-a-i-n-i-n-g- -m-e-t-h-o-d- -t-o- -m-a-k-e- -t-h-e-  
-t-e-x-t- -c-l-a-s-s-i-f-i-e-r- -r-o-b-u-s-t- -a-g-a-i-n-s-t- -w-o-r-d- -s-u-b-s-t-i-t-u-t-i-o-n-s-.

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-#-#-#- -Q-u-e-s-t-i-o-n- -9-5-:- -S-e-n-i-o-r-/S-t-a-f-f- -O-u-t-l-i-e-r- -Q-u-e-s-t-i-o-n- -9-5-:- -W-h-a-t- -h-a-p-p-e-n-s- -i-f- -a-  
-c-o-m-p-a-n-y- -d-y-n-a-m-i-c-a-l-l-y- -a-l-t-e-r-s- -i-t-s- -p-u-b-l-i-c- -P-R- -m-e-s-s-a-g-i-n-g- -t-o- -t-r-i-c-k- -y-o-u-r- -s-e-n-t-i-m-e-n-t-  
-m-o-d-e-l-?

-\*-R-o-u-n-d-:-\*- -H-i-r-i-n-g- -C-o-m-m-i-t-t-e-e-

-\*-D-i-f-f-i-c-u-l-t-y-:-\*- -H-a-r-d- -|- \*-l-m-p-o-r-t-a-n-c-e-:-\*- -1-0-/1-0-

-\*-C-o-n-c-e-p-t- -T-e-s-t-e-d-:-\*- -A-d-v-e-r-s-a-r-i-a-l- -a-t-t-a-c-k-s,- -d-a-t-a- -m-a-n-i-p-u-l-a-t-i-o-n,- -v-a-l-i-d-a-t-i-o-n-  
-c-o-n-s-t-r-a-i-n-t-s-.

-\*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t-:-\*- -T-e-s-t-s- -a-d-v-e-r-s-a-r-i-a-l- -r-o-b-u-s-t-n-e-s-s- -a-n-d- -s-y-s-t-e-m-  
-b-o-u-n-d-a-r-i-e-s-.

-\*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r-:-\*- -A-d-v-e-r-s-a-r-i-a-l- -P-R- -(e-.g.- -p-a-c-k-i-n-g- -r-e-l-e-a-s-e-s- -w-i-t-h-

-p-o-s-i-t-i-v-e- -b-u-z-z-w-o-r-d-s- -l-i-k-e- -'s-y-n-e-r-g-y'- -a-n-d- -'h-y-p-e-r---g-r-o-w-t-h'- -t-o- -m-a-s-k- -l-a-y-o-f-f-s-) -w-o-u-l-d-  
-i-n-f-l-a-t-e- -t-h-e- -s-e-n-t-i-m-e-n-t- -s-c-o-r-e-,- -T-o- -c-o-u-n-t-e-r- -t-h-i-s,- -w-e- -a-s-s-i-g-n- -d-i-f-f-e-r-e-n-t- -w-e-i-g-h-t-s:-  
-d-i-r-e-c-t- -D-B- -f-i-l-i-n-g-s- -(W-A-R-N- -n-o-t-i-c-e-s)- -a-n-d- -a-c-t-u-a-l- -l-a-y-o-f-f- -r-e-c-o-r-d-s- -h-a-v-e- -1-0-x- -t-h-e-  
-w-e-i-g-h-t- -o-f- -n-e-w-s- -s-e-n-t-i-m-e-n-t-,- -N-e-w-s- -s-e-n-t-i-m-e-n-t- -a-c-t-s- -a-s- -a- -m-o-d-i-f-i-e-r,- -n-o-t- -t-h-e-  
-p-r-i-m-a-r-y- -s-i-g-n-a-l-,-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -B-e-l-i-e-v-i-n-g- -n-e-w-s- -s-e-n-t-i-m-e-n-t- -i-s- -a-n- -u-n---g-a-m-e-a-b-l-e- -m-e-t-r-i-c-,-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -i-m-p-l-e-m-e-n-t- -a- -d-i-s-c-r-e-p-a-n-c-y- -a-n-o-m-a-l-y-  
-d-e-t-e-c-t-o-r:- -i-f- -a- -c-o-m-p-a-n-y-'s- -p-r-e-d-i-c-t-e-d- -r-i-s-k- -f-r-o-m- -t-a-b-u-l-a-r- -d-a-t-a- -i-s- -h-i-g-h,- -b-u-t- -n-e-w-s-  
-s-e-n-t-i-m-e-n-t- -i-s- -h-i-g-h-l-y- -p-o-s-i-t-i-v-e,- -w-e- -f-l-a-g- -t-h-e- -r-e-c-o-r-d- -f-o-r- -r-e-v-i-e-w- -a-n-d- -p-u-l-l-  
-t-h-i-r-d---p-a-r-t-y- -e-m-p-l-o-y-e-e- -r-e-v-i-e-w-s- -(e.-g.-,- -G-l-a-s-s-d-o-o-r)- -t-o- -v-e-r-i-f-y-,-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -d-o- -y-o-u- -m-a-t-h-e-m-a-t-i-c-a-l-l-y- -f-o-r-m-a-l-i-z-e- -t-h-e- -w-e-i-g-h-t-  
-p-e-n-a-l-t-y- -f-o-r- -n-e-w-s- -d-i-s-c-r-e-p-a-n-c-y-?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -D-e-s-i-g-n- -a-n- -o-n-l-i-n-e- -a-d-v-e-r-s-a-r-i-a-l- -t-r-a-i-n-i-n-g- -m-e-t-h-o-d- -t-o- -m-a-k-e- -t-h-e-  
-t-e-x-t- -c-l-a-s-s-i-f-i-e-r- -r-o-b-u-s-t- -a-g-a-i-n-s-t- -w-o-r-d- -s-u-b-s-t-i-t-u-t-i-o-n-s.-

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-#-#-#- -Q-u-e-s-t-i-o-n- -9-6:- -S-e-n-i-o-r-/S-t-a-f-f- -O-u-t-l-i-e-r- -Q-u-e-s-t-i-o-n- -9-6:- -W-h-a-t- -h-a-p-p-e-n-s- -i-f- -a-  
-c-o-m-p-a-n-y- -d-y-n-a-m-i-c-a-l-l-y- -a-l-t-e-r-s- -i-t-s- -p-u-b-l-i-c- -P-R- -m-e-s-s-a-g-i-n-g- -t-o- -t-r-i-c-k- -y-o-u-r- -s-e-n-t-i-m-e-n-t-  
-m-o-d-e-l-?-

-\*- \*-R-o-u-n-d:-\*- -H-i-r-i-n-g- -C-o-m-m-i-t-t-e-e-

-\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|- \*- -l-m-p-o-r-t-a-n-c-e:-\*- -1-0-/1-0-

-\*- \*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -A-d-v-e-r-s-a-r-i-a-l- -a-t-t-a-c-k-s,- -d-a-t-a- -m-a-n-i-p-u-l-a-t-i-o-n,- -v-a-l-i-d-a-t-i-o-n-  
-c-o-n-s-t-r-a-i-n-t-s.-

-\*- \*-W-h-y- -l-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -a-d-v-e-r-s-a-r-i-a-l- -r-o-b-u-s-t-n-e-s-s- -a-n-d- -s-y-s-t-e-m-  
-b-o-u-n-d-a-r-i-e-s.-

-\*- \*-l-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -A-d-v-e-r-s-a-r-i-a-l- -P-R- -(e.-g.- -p-a-c-k-i-n-g- -r-e-l-e-a-s-e-s- -w-i-t-h-  
-p-o-s-i-t-i-v-e- -b-u-z-z-w-o-r-d-s- -l-i-k-e- -'s-y-n-e-r-g-y'- -a-n-d- -'h-y-p-e-r---g-r-o-w-t-h'- -t-o- -m-a-s-k- -l-a-y-o-f-f-s-) -w-o-u-l-d-  
-i-n-f-l-a-t-e- -t-h-e- -s-e-n-t-i-m-e-n-t- -s-c-o-r-e-,- -T-o- -c-o-u-n-t-e-r- -t-h-i-s,- -w-e- -a-s-s-i-g-n- -d-i-f-f-e-r-e-n-t- -w-e-i-g-h-t-s:-  
-d-i-r-e-c-t- -D-B- -f-i-l-i-n-g-s- -(W-A-R-N- -n-o-t-i-c-e-s)- -a-n-d- -a-c-t-u-a-l- -l-a-y-o-f-f- -r-e-c-o-r-d-s- -h-a-v-e- -1-0-x- -t-h-e-  
-w-e-i-g-h-t- -o-f- -n-e-w-s- -s-e-n-t-i-m-e-n-t-,- -N-e-w-s- -s-e-n-t-i-m-e-n-t- -a-c-t-s- -a-s- -a- -m-o-d-i-f-i-e-r,- -n-o-t- -t-h-e-  
-p-r-i-m-a-r-y- -s-i-g-n-a-l-,-

-\*- \*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -B-e-l-i-e-v-i-n-g- -n-e-w-s- -s-e-n-t-i-m-e-n-t- -i-s- -a-n- -u-n---g-a-m-e-a-b-l-e- -m-e-t-r-i-c-,-

-\*- \*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -i-m-p-l-e-m-e-n-t- -a- -d-i-s-c-r-e-p-a-n-c-y- -a-n-o-m-a-l-y-  
-d-e-t-e-c-t-o-r:- -i-f- -a- -c-o-m-p-a-n-y-'s- -p-r-e-d-i-c-t-e-d- -r-i-s-k- -f-r-o-m- -t-a-b-u-l-a-r- -d-a-t-a- -i-s- -h-i-g-h,- -b-u-t- -n-e-w-s-  
-s-e-n-t-i-m-e-n-t- -i-s- -h-i-g-h-l-y- -p-o-s-i-t-i-v-e,- -w-e- -f-l-a-g- -t-h-e- -r-e-c-o-r-d- -f-o-r- -r-e-v-i-e-w- -a-n-d- -p-u-l-l-  
-t-h-i-r-d---p-a-r-t-y- -e-m-p-l-o-y-e-e- -r-e-v-i-e-w-s- -(e.-g.-,- -G-l-a-s-s-d-o-o-r)- -t-o- -v-e-r-i-f-y-,-

-\*- \*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -d-o- -y-o-u- -m-a-t-h-e-m-a-t-i-c-a-l-l-y- -f-o-r-m-a-l-i-z-e- -t-h-e- -w-e-i-g-h-t-  
-p-e-n-a-l-t-y- -f-o-r- -n-e-w-s- -d-i-s-c-r-e-p-a-n-c-y-?-

-\*- \*-D-e-e-p- -D-i-v-e:-\*- -D-e-s-i-g-n- -a-n- -o-n-l-i-n-e- -a-d-v-e-r-s-a-r-i-a-l- -t-r-a-i-n-i-n-g- -m-e-t-h-o-d- -t-o- -m-a-k-e- -t-h-e-  
-t-e-x-t- -c-l-a-s-s-i-f-i-e-r- -r-o-b-u-s-t- -a-g-a-i-n-s-t- -w-o-r-d- -s-u-b-s-t-i-t-u-t-i-o-n-s.-

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### -Q-u-e-s-t-i-o-n- 9-7:- S-e-n-i-o-r-/S-t-a-f-f- O-u-t-l-i-e-r- Q-u-e-s-t-i-o-n- 9-7:- W-h-a-t- h-a-p-p-e-n-s- i-f- a-c-o-m-p-a-n-y- d-y-n-a-m-i-c-a-l-l-y- a-l-t-e-r-s- i-t-s- p-u-b-l-i-c- P-R- m-e-s-s-a-g-i-n-g- t-o- t-r-i-c-k- y-o-u-r- s-e-n-t-i-m-e-n-t- m-o-d-e-l?-

\*- \*-R-o-u-n-d:-\*- H-i-r-i-n-g- C-o-m-m-i-t-t-e-e-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- |- \*-l-m-p-o-r-t-a-n-c-e:-\*- 1-0-/1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- A-d-v-e-r-s-a-r-i-a-l- a-t-t-a-c-k-s,- d-a-t-a- m-a-n-i-p-u-l-a-t-i-o-n,- v-a-l-i-d-a-t-i-o-n- c-o-n-s-t-r-a-i-n-t-s.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- T-e-s-t-s- a-d-v-e-r-s-a-r-i-a-l- r-o-b-u-s-t-n-e-s-s- a-n-d- s-y-s-t-e-m- b-o-u-n-d-a-r-i-e-s.-

\*- \*-l-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- A-d-v-e-r-s-a-r-i-a-l- P-R- -(e.-g.- p-a-c-k-i-n-g- r-e-l-e-a-s-e-s- w-i-t-h- p-o-s-i-t-i-v-e- b-u-z-z-w-o-r-d-s- l-i-k-e- 's-y-n-e-r-g-y'- a-n-d- 'h-y-p-e-r---g-r-o-w-t-h'- t-o- m-a-s-k- l-a-y-o-f-f-s)- w-o-u-l-d- i-n-f-l-a-t-e- t-h-e- s-e-n-t-i-m-e-n-t- s-c-o-r-e.- T-o- c-o-u-n-t-e-r- t-h-i-s,- w-e- a-s-s-i-g-n- d-i-f-f-e-r-e-n-t- w-e-i-g-h-t-s:- d-i-r-e-c-t- D-B- f-i-l-i-n-g-s- (-W-A-R-N- n-o-t-i-c-e-s)- a-n-d- a-c-t-u-a-l- l-a-y-o-f-f- r-e-c-o-r-d-s- h-a-v-e- 1-0-x- t-h-e- w-e-i-g-h-t- o-f- n-e-w-s- s-e-n-t-i-m-e-n-t.- N-e-w-s- s-e-n-t-i-m-e-n-t- a-c-t-s- a-s- a- m-o-d-i-f-i-e-r,- n-o-t- t-h-e- p-r-i-m-a-r-y- s-i-g-n-a-l.-

\*- \*-C-o-m-m-o-n- M-i-s-t-a-k-e-s:-\*- B-e-l-i-e-v-i-n-g- n-e-w-s- s-e-n-t-i-m-e-n-t- i-s- a-n- u-n---g-a-m-e-a-b-l-e- m-e-t-r-i-c.-

\*- \*-B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r:-\*- W-e- i-m-p-l-e-m-e-n-t- a- d-i-s-c-r-e-p-a-n-c-y- a-n-o-m-a-l-y- d-e-t-e-c-t-o-r:- i-f- a- c-o-m-p-a-n-y-'s- p-r-e-d-i-c-t-e-d- r-i-s-k- f-r-o-m- t-a-b-u-l-a-r- d-a-t-a- i-s- h-i-g-h,- b-u-t- n-e-w-s- s-e-n-t-i-m-e-n-t- i-s- h-i-g-h-l-y- p-o-s-i-t-i-v-e,- w-e- f-l-a-g- t-h-e- r-e-c-o-r-d- f-o-r- r-e-v-i-e-w- a-n-d- p-u-l-l- t-h-i-r-d---p-a-r-t-y- e-m-p-l-o-y-e-e- r-e-v-i-e-w-s- (e.-g.-, G-l-a-s-s-d-o-o-r)- t-o- v-e-r-i-f-y.-

\*- \*-F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s:-\*- H-o-w- d-o- y-o-u- m-a-t-h-e-m-a-t-i-c-a-l-l-y- f-o-r-m-a-l-i-z-e- t-h-e- w-e-i-g-h-t- p-e-n-a-l-t-y- f-o-r- n-e-w-s- d-i-s-c-r-e-p-a-n-c-y-?

\*- \*-D-e-e-p- D-i-v-e:-\*- D-e-s-i-g-n- a-n- o-n-l-i-n-e- a-d-v-e-r-s-a-r-i-a-l- t-r-a-i-n-i-n-g- m-e-t-h-o-d- t-o- m-a-k-e- t-h-e- t-e-x-t- c-l-a-s-s-i-f-i-e-r- r-o-b-u-s-t- a-g-a-i-n-s-t- w-o-r-d- s-u-b-s-t-i-t-u-t-i-o-n-s.-

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### -Q-u-e-s-t-i-o-n- 9-8:- S-e-n-i-o-r-/S-t-a-f-f- O-u-t-l-i-e-r- Q-u-e-s-t-i-o-n- 9-8:- W-h-a-t- h-a-p-p-e-n-s- i-f- a-c-o-m-p-a-n-y- d-y-n-a-m-i-c-a-l-l-y- a-l-t-e-r-s- i-t-s- p-u-b-l-i-c- P-R- m-e-s-s-a-g-i-n-g- t-o- t-r-i-c-k- y-o-u-r- s-e-n-t-i-m-e-n-t- m-o-d-e-l?-

\*- \*-R-o-u-n-d:-\*- H-i-r-i-n-g- C-o-m-m-i-t-t-e-e-

\*- \*-D-i-f-f-i-c-u-l-t-y:-\*- H-a-r-d- |- \*-l-m-p-o-r-t-a-n-c-e:-\*- 1-0-/1-0-

\*- \*-C-o-n-c-e-p-t- T-e-s-t-e-d:-\*- A-d-v-e-r-s-a-r-i-a-l- a-t-t-a-c-k-s,- d-a-t-a- m-a-n-i-p-u-l-a-t-i-o-n,- v-a-l-i-d-a-t-i-o-n- c-o-n-s-t-r-a-i-n-t-s.-

\*- \*-W-h-y- l-n-t-e-r-v-i-e-w-e-r- A-s-k-s- l-t:-\*- T-e-s-t-s- a-d-v-e-r-s-a-r-i-a-l- r-o-b-u-s-t-n-e-s-s- a-n-d- s-y-s-t-e-m- b-o-u-n-d-a-r-i-e-s.-

\*- \*-l-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r:-\*- A-d-v-e-r-s-a-r-i-a-l- P-R- -(e.-g.- p-a-c-k-i-n-g- r-e-l-e-a-s-e-s- w-i-t-h- p-o-s-i-t-i-v-e- b-u-z-z-w-o-r-d-s- l-i-k-e- 's-y-n-e-r-g-y'- a-n-d- 'h-y-p-e-r---g-r-o-w-t-h'- t-o- m-a-s-k- l-a-y-o-f-f-s)- w-o-u-l-d- i-n-f-l-a-t-e- t-h-e- s-e-n-t-i-m-e-n-t- s-c-o-r-e.- T-o- c-o-u-n-t-e-r- t-h-i-s,- w-e- a-s-s-i-g-n- d-i-f-f-e-r-e-n-t- w-e-i-g-h-t-s:- d-i-r-e-c-t- D-B- f-i-l-i-n-g-s- (-W-A-R-N- n-o-t-i-c-e-s)- a-n-d- a-c-t-u-a-l- l-a-y-o-f-f- r-e-c-o-r-d-s- h-a-v-e- 1-0-x- t-h-e- w-e-i-g-h-t- o-f- n-e-w-s- s-e-n-t-i-m-e-n-t.- N-e-w-s- s-e-n-t-i-m-e-n-t- a-c-t-s- a-s- a- m-o-d-i-f-i-e-r,- n-o-t- t-h-e- p-r-i-m-a-r-y- s-i-g-n-a-l.-

-\*-\*-\*C-o-m-m-o-n- M-i-s-t-a-k-e-s-\*-\* B-e-l-i-e-v-i-n-g- n-e-w-s- s-e-n-t-i-m-e-n-t- i-s- a-n- u-n---g-a-m-e-a-b-l-e- m-e-t-r-i-c.-

-\*-\*-\*B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r-\*-\* W-e- i-m-p-l-e-m-e-n-t- a- d-i-s-c-r-e-p-a-n-c-y- a-n-o-m-a-l-y- d-e-t-e-c-t-o-r-:- i-f- a- c-o-m-p-a-n-y-'s- p-r-e-d-i-c-t-e-d- r-i-s-k- f-r-o-m- t-a-b-u-l-a-r- d-a-t-a- i-s- h-i-g-h,- b-u-t- n-e-w-s- s-e-n-t-i-m-e-n-t- i-s- h-i-g-h-l-y- p-o-s-i-t-i-v-e,- w-e- f-l-a-g- t-h-e- r-e-c-o-r-d- f-o-r- r-e-v-i-e-w- a-n-d- p-u-l-l- t-h-i-r-d---p-a-r-t-y- e-m-p-l-o-y-e-e- r-e-v-i-e-w-s- (-e-.g.-, -G-l-a-s-s-d-o-o-r)- t-o- v-e-r-i-f-y.-

-\*-\*-\*F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s-\*-\* H-o-w- d-o- y-o-u- m-a-t-h-e-m-a-t-i-c-a-l-l-y- f-o-r-m-a-l-i-z-e- t-h-e- w-e-i-g-h-t- p-e-n-a-l-t-y- f-o-r- n-e-w-s- d-i-s-c-r-e-p-a-n-c-y-?-

-\*-\*-\*D-e-e-p- D-i-v-e-\*-\* D-e-s-i-g-n- a-n- o-n-l-i-n-e- a-d-v-e-r-s-a-r-i-a-l- t-r-a-i-n-i-n-g- m-e-t-h-o-d- t-o- m-a-k-e- t-h-e- t-e-x-t- c-l-a-s-s-i-f-i-e-r- r-o-b-u-s-t- a-g-a-i-n-s-t- w-o-r-d- s-u-b-s-t-i-t-u-t-i-o-n-s.-

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###- Q-u-e-s-t-i-o-n- 9-9:- S-e-n-i-o-r-/S-t-a-f-f- O-u-t-l-i-e-r- Q-u-e-s-t-i-o-n- 9-9:- W-h-a-t- h-a-p-p-e-n-s- i-f- a- c-o-m-p-a-n-y- d-y-n-a-m-i-c-a-l-l-y- a-l-t-e-r-s- i-t-s- p-u-b-l-i-c- P-R- m-e-s-s-a-g-i-n-g- t-o- t-r-i-c-k- y-o-u-r- s-e-n-t-i-m-e-n-t- m-o-d-e-l-?

-\*-\*-\*R-o-u-n-d-\*-\* H-i-r-i-n-g- C-o-m-m-i-t-t-e-e-

-\*-\*-\*D-i-f-f-i-c-u-l-t-y-\*-\* H-a-r-d- |- \*-\*-l-m-p-o-r-t-a-n-c-e-\*-\* -1-0/-1-0-

-\*-\*-\*C-o-n-c-e-p-t- T-e-s-t-e-d-\*-\* A-d-v-e-r-s-a-r-i-a-l- a-t-t-a-c-k-s,- d-a-t-a- m-a-n-i-p-u-l-a-t-i-o-n,- v-a-l-i-d-a-t-i-o-n- c-o-n-s-t-r-a-i-n-t-s.-

-\*-\*-\*W-h-y- I-n-t-e-r-v-i-e-w-e-r- A-s-k-s- I-t-\*-\* T-e-s-t-s- a-d-v-e-r-s-a-r-i-a-l- r-o-b-u-s-t-n-e-s-s- a-n-d- s-y-s-t-e-m- b-o-u-n-d-a-r-i-e-s.-

-\*-\*-\*I-d-e-a-l- F-A-A-N-G- A-n-s-w-e-r-\*-\* A-d-v-e-r-s-a-r-i-a-l- P-R- (-e-.g.- p-a-c-k-i-n-g- r-e-l-e-a-s-e-s- w-i-t-h- p-o-s-i-t-i-v-e- b-u-z-z-w-o-r-d-s- l-i-k-e- 's-y-n-e-r-g-y-' a-n-d- 'h-y-p-e-r---g-r-o-w-t-h-' t-o- m-a-s-k- l-a-y-o-f-f-s)- w-o-u-l-d- i-n-f-l-a-t-e- t-h-e- s-e-n-t-i-m-e-n-t- s-c-o-r-e.- T-o- c-o-u-n-t-e-r- t-h-i-s,- w-e- a-s-s-i-g-n- d-i-f-f-e-r-e-n-t- w-e-i-g-h-t-s:- d-i-r-e-c-t- D-B- f-i-l-i-n-g-s- (-W-A-R-N- n-o-t-i-c-e-s)- a-n-d- a-c-t-u-a-l- l-a-y-o-f-f- r-e-c-o-r-d-s- h-a-v-e- 1-0-x- t-h-e- w-e-i-g-h-t- o-f- n-e-w-s- s-e-n-t-i-m-e-n-t.- N-e-w-s- s-e-n-t-i-m-e-n-t- a-c-t-s- a-s- a- m-o-d-i-f-i-e-r,- n-o-t- t-h-e- p-r-i-m-a-r-y- s-i-g-n-a-l.-

-\*-\*-\*C-o-m-m-o-n- M-i-s-t-a-k-e-s-\*-\* B-e-l-i-e-v-i-n-g- n-e-w-s- s-e-n-t-i-m-e-n-t- i-s- a-n- u-n---g-a-m-e-a-b-l-e- m-e-t-r-i-c.-

-\*-\*-\*B-e-t-t-e-r- P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- A-n-s-w-e-r-\*-\* W-e- i-m-p-l-e-m-e-n-t- a- d-i-s-c-r-e-p-a-n-c-y- a-n-o-m-a-l-y- d-e-t-e-c-t-o-r-:- i-f- a- c-o-m-p-a-n-y-'s- p-r-e-d-i-c-t-e-d- r-i-s-k- f-r-o-m- t-a-b-u-l-a-r- d-a-t-a- i-s- h-i-g-h,- b-u-t- n-e-w-s- s-e-n-t-i-m-e-n-t- i-s- h-i-g-h-l-y- p-o-s-i-t-i-v-e,- w-e- f-l-a-g- t-h-e- r-e-c-o-r-d- f-o-r- r-e-v-i-e-w- a-n-d- p-u-l-l- t-h-i-r-d---p-a-r-t-y- e-m-p-l-o-y-e-e- r-e-v-i-e-w-s- (-e-.g.-, -G-l-a-s-s-d-o-o-r)- t-o- v-e-r-i-f-y.-

-\*-\*-\*F-o-l-l-o-w---u-p- Q-u-e-s-t-i-o-n-s-\*-\* H-o-w- d-o- y-o-u- m-a-t-h-e-m-a-t-i-c-a-l-l-y- f-o-r-m-a-l-i-z-e- t-h-e- w-e-i-g-h-t- p-e-n-a-l-t-y- f-o-r- n-e-w-s- d-i-s-c-r-e-p-a-n-c-y-?-

-\*-\*-\*D-e-e-p- D-i-v-e-\*-\* D-e-s-i-g-n- a-n- o-n-l-i-n-e- a-d-v-e-r-s-a-r-i-a-l- t-r-a-i-n-i-n-g- m-e-t-h-o-d- t-o- m-a-k-e- t-h-e- t-e-x-t- c-l-a-s-s-i-f-i-e-r- r-o-b-u-s-t- a-g-a-i-n-s-t- w-o-r-d- s-u-b-s-t-i-t-u-t-i-o-n-s.-

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###- Q-u-e-s-t-i-o-n- 1-0-0:- S-e-n-i-o-r-/S-t-a-f-f- O-u-t-l-i-e-r- Q-u-e-s-t-i-o-n- 1-0-0:- W-h-a-t- h-a-p-p-e-n-s- i-f- a- c-o-m-p-a-n-y- d-y-n-a-m-i-c-a-l-l-y- a-l-t-e-r-s- i-t-s- p-u-b-l-i-c- P-R- m-e-s-s-a-g-i-n-g- t-o- t-r-i-c-k- y-o-u-r- s-e-n-t-i-m-e-n-t- m-o-d-e-l-?

-\*-\*-\*R-o-u-n-d-\*-\* H-i-r-i-n-g- C-o-m-m-i-t-t-e-e-

-\*-D-i-f-f-i-c-u-l-t-y:-\*- -H-a-r-d- -|-\*-I-m-p-o-r-t-a-n-c-e:-\*- -1-0-/1-0-

-\*-C-o-n-c-e-p-t- -T-e-s-t-e-d:-\*- -A-d-v-e-r-s-a-r-i-a-l- -a-t-t-a-c-k-s,- -d-a-t-a- -m-a-n-i-p-u-l-a-t-i-o-n,- -v-a-l-i-d-a-t-i-o-n-  
-c-o-n-s-t-r-a-i-n-t-s.-

-\*-W-h-y- -I-n-t-e-r-v-i-e-w-e-r- -A-s-k-s- -l-t:-\*- -T-e-s-t-s- -a-d-v-e-r-s-a-r-i-a-l- -r-o-b-u-s-t-n-e-s-s- -a-n-d- -s-y-s-t-e-m-  
-b-o-u-n-d-a-r-i-e-s.-

-\*-I-d-e-a-l- -F-A-A-N-G- -A-n-s-w-e-r:-\*- -A-d-v-e-r-s-a-r-i-a-l- -P-R- -(e.g.- -p-a-c-k-i-n-g- -r-e-l-e-a-s-e-s- -w-i-t-h-  
-p-o-s-i-t-i-v-e- -b-u-z-z-w-o-r-d-s- -l-i-k-e- -'s-y-n-e-r-g-y'- -a-n-d- -'h-y-p-e-r---g-r-o-w-t-h'- -t-o- -m-a-s-k- -l-a-y-o-f-f-s)- -w-o-u-l-d-  
-i-n-f-l-a-t-e- -t-h-e- -s-e-n-t-i-m-e-n-t- -s-c-o-r-e,- -T-o- -c-o-u-n-t-e-r- -t-h-i-s,- -w-e- -a-s-s-i-g-n- -d-i-f-f-e-r-e-n-t- -w-e-i-g-h-t-s:-  
-d-i-r-e-c-t- -D-B- -f-i-l-i-n-g-s- -(W-A-R-N- -n-o-t-i-c-e-s)- -a-n-d- -a-c-t-u-a-l- -l-a-y-o-f-f- -r-e-c-o-r-d-s- -h-a-v-e- -1-0-x- -t-h-e-  
-w-e-i-g-h-t- -o-f- -n-e-w-s- -s-e-n-t-i-m-e-n-t,- -N-e-w-s- -s-e-n-t-i-m-e-n-t- -a-c-t-s- -a-s- -a- -m-o-d-i-f-i-e-r,- -n-o-t- -t-h-e-  
-p-r-i-m-a-r-y- -s-i-g-n-a-l.-

-\*-C-o-m-m-o-n- -M-i-s-t-a-k-e-s:-\*- -B-e-l-i-e-v-i-n-g- -n-e-w-s- -s-e-n-t-i-m-e-n-t- -i-s- -a-n- -u-n---g-a-m-e-a-b-l-e- -m-e-t-r-i-c.-

-\*-B-e-t-t-e-r- -P-r-o-d-u-c-t-i-o-n---G-r-a-d-e- -A-n-s-w-e-r:-\*- -W-e- -i-m-p-l-e-m-e-n-t- -a- -d-i-s-c-r-e-p-a-n-c-y- -a-n-o-m-a-l-y-  
-d-e-t-e-c-t-o-r:- -i-f- -a- -c-o-m-p-a-n-y-'s- -p-r-e-d-i-c-t-e-d- -r-i-s-k- -f-r-o-m- -t-a-b-u-l-a-r- -d-a-t-a- -i-s- -h-i-g-h,- -b-u-t- -n-e-w-s-  
-s-e-n-t-i-m-e-n-t- -i-s- -h-i-g-h-l-y- -p-o-s-i-t-i-v-e,- -w-e- -f-l-a-g- -t-h-e- -r-e-c-o-r-d- -f-o-r- -r-e-v-i-e-w- -a-n-d- -p-u-l-l-  
-t-h-i-r-d---p-a-r-t-y- -e-m-p-l-o-y-e-e- -r-e-v-i-e-w-s- -(e.g.-, -G-l-a-s-s-d-o-o-r)- -t-o- -v-e-r-i-f-y.-

-\*-F-o-l-l-o-w---u-p- -Q-u-e-s-t-i-o-n-s:-\*- -H-o-w- -d-o- -y-o-u- -m-a-t-h-e-m-a-t-i-c-a-l-l-y- -f-o-r-m-a-l-i-z-e- -t-h-e- -w-e-i-g-h-t-  
-p-e-n-a-l-t-y- -f-o-r- -n-e-w-s- -d-i-s-c-r-e-p-a-n-c-y-?

-\*-D-e-e-p- -D-i-v-e:-\*- -D-e-s-i-g-n- -a-n- -o-n-l-i-n-e- -a-d-v-e-r-s-a-r-i-a-l- -t-r-a-i-n-i-n-g- -m-e-t-h-o-d- -t-o- -m-a-k-e- -t-h-e-  
-t-e-x-t- -c-l-a-s-s-i-f-i-e-r- -r-o-b-u-s-t- -a-g-a-i-n-s-t- -w-o-r-d- -s-u-b-s-t-i-t-u-t-i-o-n-s.-

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-#- -S-e-c-t-i-o-n- -1-7,- -E-v-a-l-u-a-t-i-o-n- -M-e-t-r-i-c-s- -&- -M-o-c-k- -I-n-t-e-r-v-i-e-w- -P-l-a-n-

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-\*-O-v-e-r-a-l-l- -E-n-g-i-n-e-e-r-i-n-g- -M-a-t-u-r-i-t-y:-\*- \*-8-8-/1-0-0\*-\*

- -\*-S-t-r-e-n-g-t-h-s:-\*- -C-l-e-a-n- -s-e-p-a-r-a-t-i-o-n- -o-f- -t-r-a-i-n-i-n-g- -a-n-d- -i-n-f-e-r-e-n-c-e,- -a-u-t-o-m-a-t-e-d-  
-f-a-l-l-b-a-c-k- -s-y-s-t-e-m-s,- -r-o-b-u-s-t- -d-a-t-a- -p-i-p-e-l-i-n-e-s.-

- -\*-W-e-a-k-n-e-s-s-e-s:-\*- -L-a-c-k- -o-f- -a-u-t-o-m-a-t-e-d- -m-o-d-e-l- -d-r-i-f-t- -m-o-n-i-t-o-r-i-n-g,- -d-e-p-e-n-d-e-n-c-y-  
-o-n- -p-i-c-k-l-e- -s-e-r-i-a-l-i-z-a-t-i-o-n- -i-n-s-t-e-a-d- -o-f- -O-N-N-X.-

-\*-M-a-c-h-i-n-e- -L-e-a-r-n-i-n-g- -M-a-t-u-r-i-t-y:-\*- \*-8-5-/1-0-0\*-\*

- -\*-S-t-r-e-n-g-t-h-s:-\*- -C-a-l-i-b-r-a-t-e-d- -p-r-o-b-a-b-i-l-i-t-i-e-s,- -B-a-y-e-s-i-a-n- -o-p-t-i-m-i-z-a-t-i-o-n,- -l-o-c-a-l-  
-e-x-p-l-a-n-a-t-i-o-n- -i-n-t-e-g-r-a-t-i-o-n.-

- -\*-W-e-a-k-n-e-s-s-e-s:-\*- -K-e-y-w-o-r-d---b-a-s-e-d- -s-e-n-t-i-m-e-n-t- -h-e-u-r-i-s-t-i-c- -i-n-s-t-e-a-d- -o-f-  
-t-r-a-n-s-f-o-r-m-e-r- -N-L-P- -e-m-b-e-d-d-i-n-g-s.-

-\*-P-r-o-d-u-c-t-i-o-n- -R-e-a-d-i-n-e-s-s:-\*- \*-8-2-/1-0-0\*-\*

- -\*-S-t-r-e-n-g-t-h-s:-\*- -D-y-n-a-m-i-c- -v-e-n-v- -p-a-t-h- -c-o-r-r-e-c-t-i-o-n,- -t-r-a-n-s-a-c-t-i-o-n-a-l- -s-a-f-e-t-y,- -c-a-c-h-e-d-  
-r-e-s-o-u-r-c-e-s.-

- - \*- \*-W-e-a-k-n-e-s-s-e-s:-\*- -L-a-c-k-s- -c-o-n-t-a-i-n-e-r-i-z-a-t-i-o-n- -(D-o-c-k-e-r)- -c-o-n-f-i-g-u-r-a-t-i-o-n-s- -i-n- -t-h-e-  
-r-e-p-o- -r-o-o-t.-

-\*- \*-S-y-s-t-e-m- -D-e-s-i-g-n- -S-c-o-r-e:-\*- \*-8-7-/1-0-0-\*.--

- - \*- \*-S-t-r-e-n-g-t-h-s:-\*- -H-i-g-h-l-y- -d-e-c-o-u-p-l-e-d- -s-t-o-r-a-g-e- -a-n-d- -i-n-f-e-r-e-n-c-e- -l-a-y-e-r-s.-

- - \*- \*-W-e-a-k-n-e-s-s-e-s:-\*- -B-a-t-c-h- -p-r-o-c-e-s-s-i-n-g- -a-r-c-h-i-t-e-c-t-u-r-e-s- -n-o-t- -f-u-l-l-y- -d-e-p-l-o-y-e-d.-

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-#-#- -2.- -M-o-c-k- -l-n-t-e-r-v-i-e-w- -S-t-r-a-t-e-g-y-

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-T-o- -d-e-f-e-n-d- -t-h-i-s- -p-r-o-j-e-c-t- -i-n- -a- -S-t-a-f-f-/S-e-n-i-o-r- -l-o-o-p- -(M-e-t-a- -E-5- -/- -G-o-o-g-l-e- -L-5):-

-1.- \*-F-o-c-u-s- -o-n- -t-h-e- -c-a-l-i-b-r-a-t-i-o-n- -s-t-e-p:-\*- -E-x-p-e-c-t- -i-n-t-e-r-v-i-e-w-e-r-s- -t-o- -p-r-e-s-s- -o-n- -w-h-y-  
-y-o-u- -o-p-t-i-m-i-z-e-d- -t-h-e- -t-h-r-e-s-h-o-l-d- -t-o- -0.-2-0`- -a-n-d- -h-o-w- -P-l-a-t-t- -s-c-a-l-i-n-g- -w-o-r-k-s.-

-2.- \*-D-e-f-e-n-d- -t-h-e- -t-e-m-p-o-r-a-l- -v-a-l-i-d-a-t-i-o-n- -s-p-l-i-t:-\*- -E-x-p-l-a-i-n- -w-h-y- -r-a-n-d-o-m-  
-c-r-o-s-s--v-a-l-i-d-a-t-i-o-n- -f-a-i-l-s- -i-n- -f-o-r-e-c-a-s-t-i-n-g- -t-a-s-k-s- -d-u-e- -t-o- -c-h-r-o-n-o-l-o-g-i-c-a-l- -d-a-t-a-  
-l-e-a-k-a-g-e.-

-3.- \*-B-e- -r-e-a-d-y- -f-o-r- -s-y-s-t-e-m- -d-e-s-i-g-n- -s-c-a-l-i-n-g- -q-u-e-s-t-i-o-n-s:-\*- -E-x-p-l-a-i-n- -t-h-e- -p-a-r-t-i-t-i-o-n-  
-a-n-d- -c-a-c-h-i-n-g- -c-h-o-i-c-e-s- -i-f- -t-h-e- -d-a-t-a-s-e-t- -e-x-p-a-n-d-s- -b-y- -1-0-0-x.-

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